



Covariance Matrix Adaptation Evolution Strategy (CMA-ES): A Comprehensive Survey of Variants and Hybridizations

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Abstract

The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is a benchmark in evolutionary computation, widely used for continuous black-box optimization, with strong empirical performance in many benchmark settings. Despite its impact, its practical performance remains highly dependent on specific structural configurations and parameter settings. To address these challenges, a vast ecosystem of modifications has been developed. Following a systematic review of over 170 peer-reviewed publications—selected from high-impact databases based on their citations and relevance—this survey introduces a novel hierarchical taxonomy of CMA-ES modifications and hybridizations. Beyond a descriptive review, this work provides a critical synthesis of the algorithm’s evolution, offering a structured framework to analyze performance trade-offs and structural synergies. The analysis reveals that while recent hybrids have been reported to improve convergence speed and/or exploration in specific regimes and benchmarks, significant gaps remain in scalability and automated parameter tuning. Across the surveyed literature, a consistent trend is that mechanism-preserving refinements tend to be the most transferable across benchmarks, whereas regime-specific extensions (e.g. surrogate assistance, uncertainty handling, and constraint management) yield the largest gains mainly under expensive, noisy, or feasibility-limited settings. This approach maps current trends and supports the development of future variants, highlighting scalable covariance adaptation and automated parameter tuning, while providing a practitioner-oriented decision guide for navigating the CMA-ES literature.

1 Introduction

Evolutionary Strategies (ES) are a class of evolutionary algorithms [1], such as Genetic Algorithms [2, 3], in which its focus within computational intelligence is based on applying mechanisms of regulation and adaptation based on

Darwinian evolution to result in individuals that improve their strategies to survive their environment [4]. Due to their ease of application and strong empirical robustness across a range of problem classes, they have significantly impacted metaheuristic computation. The family of ES algorithms has found applications in various fields of science, such as biology, medicine, engineering, and computing, thanks to their ability to solve complex optimization problems [5, 6].

The first version of the ES was the (1+1)ES algorithm, which consisted of a parent solution that mutated through a Gaussian function to generate a child solution. If the generated solution improved concerning the parent solution, then the child solution became the new parent solution. If the solution did not improve, it was discarded [7]. Several versions of the (1+1)ES were developed from this simple concept, such as the adaptive (1+1)ES, which implemented mechanisms to modify the variance of the Gaussian function and thus improve its performance. The $(\mu+\lambda)$ ES algorithm emerged as a variant of the original (1+1)ES as it was an algorithm that implemented the concept already seen but with μ parent solutions that were recombined, resulting

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in λ child solutions, being a novel algorithm that began to address complex optimization problems by having a varied population in a multidimensional environment [8, 9].

The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is an algorithm belonging to the family of evolutionary strategies that consists of a distribution-based search, where new solutions are generated based on a Multivariate Normal Distribution (MND) [9]. At the same time, the mean and covariance of the MND are updated from the evaluation of these solutions. The key concept by which the CMA-ES has competitive results in optimization problems lies in the population covariance matrix, with which the variance of the MND is updated, and the mean is a weighting of the average of μ carefully selected solutions [10]. Meanwhile, step size is dynamically adjusted, effectively controlling the dispersion of the next generation of solutions. This algorithm follows in the footsteps of the ES by choosing the best solutions to be the new center from which new solutions will be generated. Hence, its implementation makes it a reference in terms of evolutionary algorithms.

In the context of metaheuristic algorithms, the CMA-ES algorithm has become one of the most influential Evolutionary Strategies, implementing novel concepts that, at the time, placed it as a state-of-the-art high-performance algorithm [11]. However, several advances in evolutionary algorithms were made before the final version of the CMA-ES was developed. The CMA-ES was published by N. Hansen in 2001 at the Technical University of Berlin [10]. Nevertheless, the tools necessary for creating the final version of the algorithm emerged from his previous work. One of the first steps for the development of the CMA-ES was the Derandomized Mutative Step-Size Control by A. Ostermeier in 1994 [12], with which slight modifications in the step-size mutation could reduce random fluctuations and better-defined steps, naming this process as “derandomization,” being later improved to a version that also contemplates

information from past solution generations [13]. Another critical step for the CMA-ES was implementing covariance matrix fitting for the first time to adapt the arbitrary normal distribution of mutations with mean zero to fit hyper-ellipsoids with high axis ratios reliably [14]. Additional advances were required in the development of the CMA-ES operators; however, since its publication, the CMA-ES has been one of the key algorithms in evolutionary computation, being subjected to exhaustive tests and validations with different test sets, and has shown strong empirical performance across several test suites, particularly in ill-conditioned or non-separable landscapes. Its remarkable ability to solve complex problems has led the CMA-ES to be implemented in applications in engineering, medicine, machine learning, and many other fields of science. Despite all the advantages the CMA-ES offers in an optimization environment, several factors weaken its performance and make its implementation impossible, such as its high computational cost, high use of memory resources, and susceptibility to noisy functions [15].

Despite the maturity of CMA-ES, several research gaps remain unaddressed in the current literature:

- **Scalability in Non-Separable Problems:** Most efficiency-oriented variants, e.g., sep-CMA-ES, sacrifice the ability to learn dependencies.
- **Lack of Unified Benchmarking:** Hybrid variants are often tested across disparate suites, such as CEC and BBOB, making direct comparisons difficult.
- **Automated Hyperparameter Tuning:** There is a lack of self-adaptive mechanisms that handle the trade-off between exploration and exploitation in highly noisy environments.

The objective of this article is to make an exhaustive compilation of the modifications that the CMA-ES has had throughout its history to analyze the main aspects in which this algorithm has been modified and improved, as well as to observe the impact that its hybridization with other algorithms in the literature has had. This review bases its references on prestigious research from publishers such as IEEE, Springer, ACM, Elsevier, and MIT Press, among others. Figure 1 shows the number of publications included in this exhaustive analysis in terms of the publisher related to modifications, hybridizations, and applications of the CMA-ES since its initial development.

This paper is organized as follows: The CMA-ES algorithm is analyzed in detail with a theoretical foundation in Sect. 2. The different tests to which the CMA-ES algorithm has been subjected will be discussed, and the areas of improvement that have been worked on will be identified in Sect. 3. Section 4 will analyze the different modifications

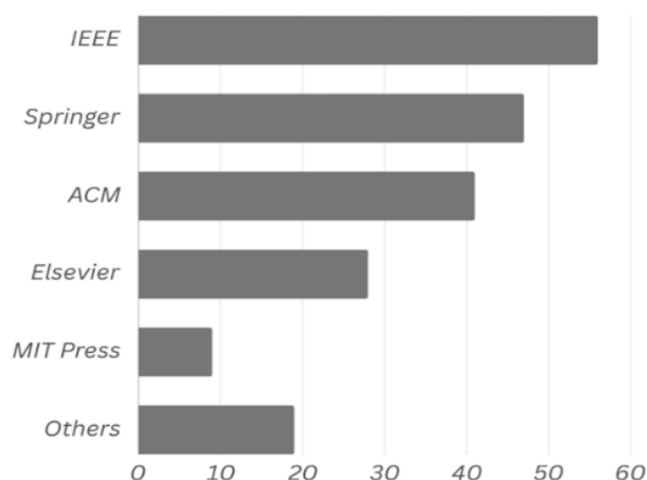


Fig. 1 Number of publications of CMA-ES per publisher

that the CMA-ES has had throughout its history; this is classified according to the aspect of the algorithm it intends to improve. Section 5 discusses the hybridizations of the CMA-ES and its versatility when combined with other methodologies in literature. Finally, Sect. 6 gives conclusions about the literature related to CMA-ES and an analysis of the algorithm's future and the direction in which its development could take.

2 The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) Algorithm

2.1 Statistics and Publications of the CMA-ES

Since the publication of the CMA-ES algorithm in 2001, hundreds of articles have been published on modifications, hybridizations, and applications, and over 200 publications with the most significant impact and novelty have been selected for the present analysis.

The CMA-ES is an algorithm that has had high exposure in congresses worldwide, since of the 200 publications analyzed, 118 are conference papers, being the Genetic and Evolutionary Computation Conference (GECCO) and the IEEE Congress on Evolutionary Computation (CEC) the congresses where the CMA-ES algorithm has had a greater presence. Therefore, 82 of the publications analyzed have been in international impact journals with JCR indexing, such as IEEE Transactions on Evolutionary Computation, Swarm and Evolutionary Computation, Expert Systems with Applications, Applied Intelligence, Soft Computing, Evolutionary Computation, Neural Computing and Applications, among others.

In Fig. 2, it's possible to see how the CMA-ES algorithm has captured the interest of researchers more and more over time since its creation, maintaining a relevant impact with a constant and considerable number of publications more than two decades after its development.

2.2 Structure of the Original CMA-ES Algorithm

The process must be divided into four fundamental stages to explain how the CMA-ES algorithm works adequately: Sampling, Moving the Mean, Adapting the Covariance Matrix, and Step-Size Control [16].

The optimization process performed by the CMA-ES is detailed in Fig. 3, where, from initialization in a specific area, the candidate solutions will move through the search space according to the defined operators to progressively improve candidate solutions toward an optimum.

2.2.1 Sampling

A new population of search points is generated by sampling a multivariate normal distribution using the following equation:

$$\mathbf{x}_k^{(g+1)} \sim \mathbf{m}^{(g)} + \sigma^{(g)} \mathcal{N}(\mathbf{0}, \mathbf{C}^{(g)}) \quad \text{for } k = 1, \dots, \lambda \quad (1)$$

where $\sim$ denotes the same distribution on the left and right sides.

$\mathcal{N}(\mathbf{0}, \mathbf{C}^{(g)})$ is a multivariate normal distribution with zero mean and covariance matrix $\mathbf{C}^{(g)}$.

$\mathbf{x}_k^{(g+1)} \in \mathbb{R}^n$, k -th offspring from generation $g+1$.

Fig. 2 Number of publications analyzed for this CMA-ES study by year



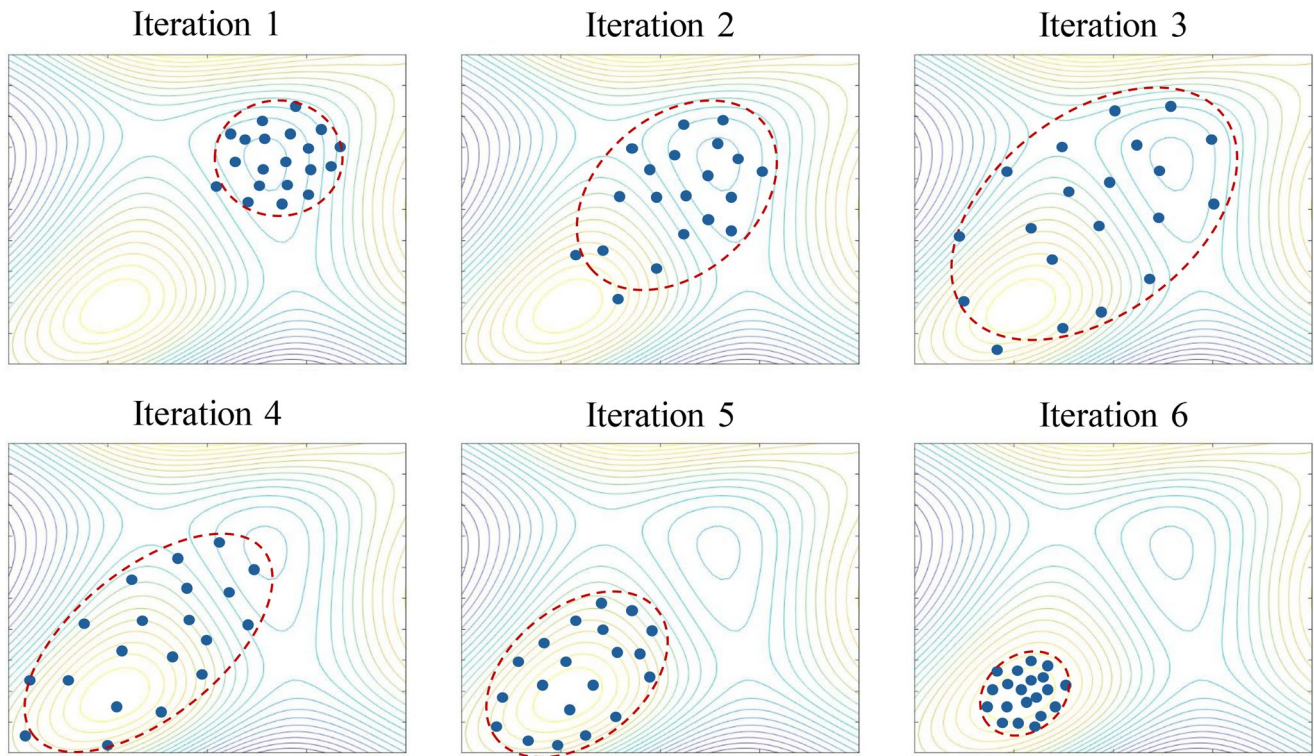


Fig. 3 Optimization of a search space with the CMA-ES algorithm

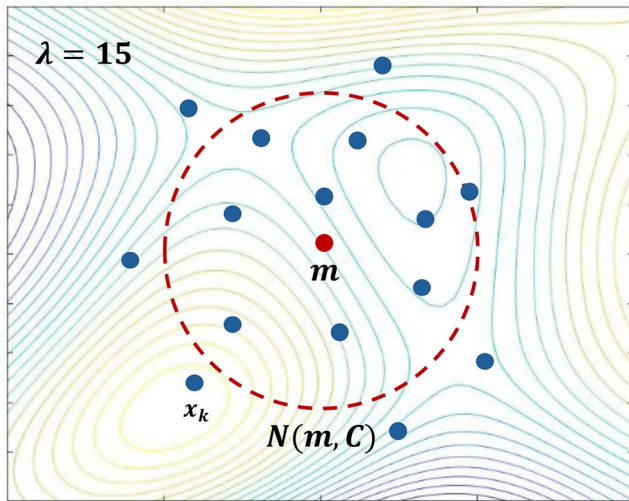


Fig. 4 Sampling in the CMA-ES algorithm

$\mathbf{m}^{(g)} \in \mathbb{R}^n$, the mean value of the search distribution at generation g .

$\sigma^{(g)} \in \mathbb{R}_{>0}$, “overall” standard deviation, step-size, at generation g .

$\mathbf{C}^{(g)} \in \mathbb{R}^{n \times n}$, the covariance matrix of the search distribution at generation g .

$\lambda \geq 2$, population size, sample size, number of offspring.

The features described above are shown on a search map in Fig. 4.

2.2.2 Moving the Mean

The new mean $\mathbf{m}^{(g+1)}$ of the search distribution is a weighted average of μ selected points from the sample $\mathbf{x}_1^{(g+1)}, \dots, \mathbf{x}_\lambda^{(g+1)}$:

$$\mathbf{m}^{(g)} = \sum_{i=1}^{\mu} w_i \mathbf{x}_{i:\lambda}^{(g+1)} \quad (2)$$

$$\sum_{i=1}^{\mu} w_i = 1, w_1 \geq w_2 \geq \dots \geq w_\mu > 0 \quad (3)$$

where $\mu \leq \lambda$ is the parent population size, i.e., the number of selected points.

$w_{i=1 \dots \mu} \in \mathbb{R}_{>0}$, the positive weight coefficients for recombination.

$\mathbf{x}_{i:\lambda}^{(g+1)}$, i -th best individual out of $\mathbf{x}_1^{(g+1)}, \dots, \mathbf{x}_\lambda^{(g+1)}$ from (1). The index $i : \lambda$ denotes the index of the i -th ranked individual.

Equation (2) implements truncation selection by choosing $\mu < \lambda$ out of λ offspring points. Assigning different weights w_i should also be interpreted as a selection mechanism. Equation (2) implements weighted intermediate recombination by taking $\mu > 1$ individuals into account for a weighted average.

The measure

$$\mu_{eff} = \left(\frac{\|w\|_1}{\|w\|_2} \right)^2 = \frac{\|w\|_1^2}{\|w\|_2^2} = \frac{\left(\sum_{i=1}^{\mu} |w_i|^2 \right)^2}{\sum_{i=1}^{\mu} w_i^2} = \frac{1}{\sum_{i=1}^{\mu} w_i^2} \quad (4)$$

can be paraphrased as the *effective sample size* of the selected samples or *variance effective selection mass*.

The final equation rewrites (2) as an update of $\mathbf{m}$,

$$\mathbf{m}^{(g+1)} = \mathbf{m}^{(g)} + c_m \sum_{i=1}^{\mu} w_i \left(\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)} \right) \quad (5)$$

where c_m is a learning rate, usually set to 1. This operator is graphically represented in Fig. 5.

2.2.3 Adapting the Covariance Matrix

Assuming the population contains enough information to estimate a covariance matrix from the population reliably, we can re-estimate the original covariance matrix $\mathbf{C}^{(g)}$ using the sampled population from Eq. (1) via the empirical covariance matrix

$$\mathbf{C}_{emp}^{(g+1)} = \frac{1}{\lambda-1} \sum_{i=1}^{\lambda} \left(\mathbf{x}_i^{(g+1)} - \frac{1}{\lambda} \sum_{j=1}^{\lambda} \mathbf{x}_j^{(g+1)} \right) \left(\mathbf{x}_i^{(g+1)} - \frac{1}{\lambda} \sum_{j=1}^{\lambda} \mathbf{x}_j^{(g+1)} \right)^T \quad (6)$$

The empirical covariance matrix $\mathbf{C}_{emp}^{(g+1)}$ is an unbiased estimator of $\mathbf{C}^{(g)}$. Now, a slightly different approach to get an estimator for $\mathbf{C}^{(g)}$ is proposed:

$$\mathbf{C}_{\lambda}^{(g+1)} = \frac{1}{\lambda} \sum_{i=1}^{\lambda} \left(\mathbf{x}_i^{(g+1)} - \mathbf{m}^{(g)} \right) \left(\mathbf{x}_i^{(g+1)} - \mathbf{m}^{(g)} \right)^T \quad (7)$$

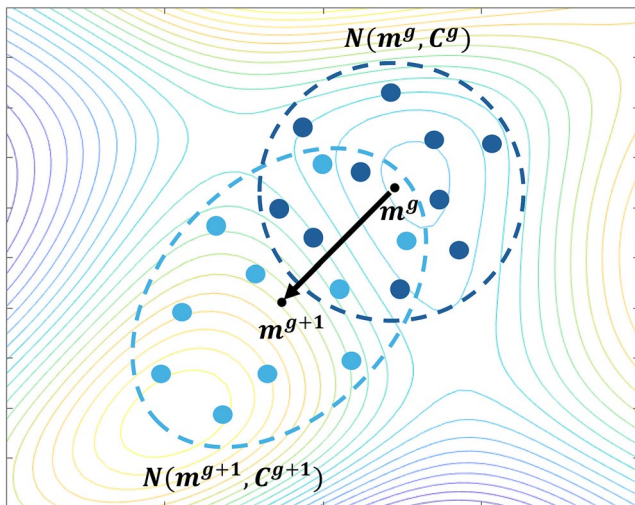


Fig. 5 Movement of the mean as the candidate solutions progress

Also, the matrix $\mathbf{C}_{\lambda}^{(g+1)}$ is an unbiased estimator of $\mathbf{C}^{(g)}$ but with the remarkable difference in the reference mean value. For $\mathbf{C}_{emp}^{(g+1)}$ it is the mean of the *actually realized* sample and for $\mathbf{C}_{\lambda}^{(g+1)}$ it is the *true* mean value, $\mathbf{m}^{(g)}$, of the sampled distribution.

$$\mathbf{C}_{\mu}^{(g+1)} = \sum_{i=1}^{\mu} w_i \left(\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)} \right) \left(\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)} \right)^T \quad (8)$$

The matrix $\mathbf{C}_{\mu}^{(g+1)}$ seen in Eq. (8) is an estimator for the distribution of *selected steps*, just as $\mathbf{C}_{\lambda}^{(g+1)}$ is an estimator of the original distribution of steps before selection. Sampling from $\mathbf{C}_{\mu}^{(g+1)}$ tends to reproduce selected successful steps, justifying what a “better” covariance matrix means.

To achieve a *fast* search, the population size λ must be small; however, getting a reliable estimator for a good covariance matrix from Eq. (8) is impossible. As a remedy, information from previous generations is used additionally.

$$\mathbf{C}^{(g+1)} = \frac{1}{g+1} \sum_{i=0}^g \frac{1}{\sigma^{(i)^2}} \mathbf{C}_{\mu}^{(i+1)} \quad (9)$$

After several generations, Eq. (9) becomes a reliable estimator for the selected steps. To make $\mathbf{C}_{\mu}^{(g)}$ from different generations comparable, the different $\sigma^{(i)}$ are incorporated. All generation steps have the same weight, whereby exponential smoothing is introduced to assign recent generations a higher weight. Choosing $\mathbf{C}^{(0)} = \mathbf{I}$ to be the unity matrix and a learning rate $0 < c_{\mu} \leq 1$, then $\mathbf{C}^{(g+1)}$ reads:

$$\begin{aligned} \mathbf{C}^{(g+1)} &= (1 - c_{\mu}) \mathbf{C}^{(g)} + c_{\mu} \frac{1}{\sigma^{(g)^2}} \mathbf{C}_{\mu}^{(g+1)} \\ &= (1 - c_{\mu}) \mathbf{C}^{(g)} + c_{\mu} \sum_{i=1}^{\mu} w_i \mathbf{y}_{i:\lambda}^{(g+1)} \mathbf{y}_{i:\lambda}^{(g+1)^T}, \end{aligned} \quad (10)$$

where $c_{\mu} \leq 1$ is the learning rate for updating the covariance matrix.

$w_{1 \dots \mu} \in \mathbb{R}$ such that $w_1 \geq \dots \geq w_{\mu} > 0$ and $\sum_i w_i = 1$.

$\mathbf{y}_{i:\lambda}^{(g+1)} = \left(\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)} \right) / \sigma^{(g)}$.

$\mathbf{z}_{i:\lambda}^{(g+1)} = \mathbf{C}^{(g)^{-1/2}} \mathbf{y}_{i:\lambda}^{(g+1)}$ is the mutation vector expressed in the unique coordinate system where the sampling is isotropic, and the respective coordinate system transformation does not rotate the original principal axes of the distribution.

This covariance matrix update is called rank- μ -update because the sum of outer products in Eq. (10) is of rank $\min(\mu, n)$ with probability one. This approach is generalized

to λ weight values, which need neither sum to 1 nor be non-negative anymore,

$$\begin{aligned} C^{(g+1)} &= \left(1 - c_\mu \sum_{i=1}^{\lambda} w_i\right) C^{(g)} + c_\mu \sum_{i=1}^{\lambda} w_i \mathbf{y}_{i:\lambda}^{(g+1)} \mathbf{y}_{i:\lambda}^{(g+1)T} \\ &= C^{(g)1/2} \left(I + c_\mu \sum_{i=1}^{\lambda} w_i \left(\mathbf{z}_{i:\lambda}^{(g+1)} \mathbf{z}_{i:\lambda}^{(g+1)T} - I \right) \right) C^{(g)1/2}, \end{aligned} \quad (11)$$

where $w_{1\dots\mu} \in \mathbb{R}$ such that $w_1 \geq \dots \geq w_\mu > 0 \geq w_{\mu+1} \geq w_\lambda$, and usually $\sum_{i=1}^{\mu} w_i = 1$ and $\sum_{i=1}^{\lambda} w_i \approx 0$.

The second line of Eq. (11) expresses the update in the natural coordinate system. The identity covariance matrix is updated, and a coordinate system transformation is applied afterward by multiplication with $C^{(g)1/2}$ on both sides. The number $1/c_\mu$ is the *backward time horizon*, which contributes roughly 63% of the overall information.

Taking the opposite viewpoint, the covariance matrix will be updated in the generation sequence using a *single selected step* only. A specific method to produce n -dimensional normal distributions with zero mean is considered to achieve this. Let the vector $\mathbf{y}_1, \dots, \mathbf{y}_{g_0} \in \mathbb{R}^n$, $g_0 \geq n$, span $\mathbb{R}^n$, and let $\mathcal{N}(0, 1)$ denote independent (0,1)-normally distributed random numbers, then:

$$\mathcal{N}(0, 1) \mathbf{y}_1 + \dots + \mathcal{N}(0, 1) \mathbf{y}_{g_0} \sim \mathcal{N}\left(0, \sum_{i=1}^{g_0} \mathbf{y}_i \mathbf{y}_i^T\right) \quad (12)$$

is a normally distributed random vector with zero mean and covariance matrix $\sum_{i=1}^{g_0} \mathbf{y}_i \mathbf{y}_i^T$. The random vector (12) is generated by adding “line-distributions” $\mathcal{N}(0, 1) \mathbf{y}_1$.

Considering Eq. (12) and a slight simplification of Eq. (11), we try to gain insight into the adaptation rule for the covariance matrix. Let the sum in Eq. (11) consist of a single

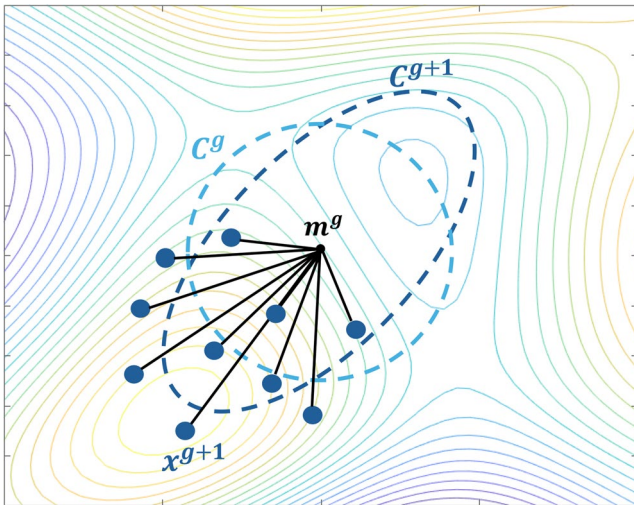


Fig. 6 Adaptation of the covariance matrix

summand only, and let $\mathbf{y}_{g+1} = \frac{\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)}}{\sigma^{(g)}}$, then, the rank-one update for the covariance matrix reads:

$$C^{(g+1)} = (1 - c_\mu) C^{(g)} + c_1 \mathbf{y}_{g+1} \mathbf{y}_{g+1}^T. \quad (13)$$

The right summand in Eq. (13) is of degree one and adds the maximum likelihood term for $\mathbf{y}_{g+1}$ into the covariance matrix $C^{(g)}$. Therefore, the probability to generate $\mathbf{y}_{g+1}$ in the next generation increases.

The selected steps $\mathbf{y}_{i:\lambda}^{(g+1)} = (\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)}) / \sigma^{(g)}$ are used to update the covariance matrix in (11) and (13). Because $\mathbf{y} \mathbf{y}^T = -\mathbf{y}(-\mathbf{y})^T$, the sign of the steps is lost when calculating $C^{(g+1)}$. An evolution path is constructed to reintroduce the sign information, thus naming a sequence of successive steps that the strategy takes over some generations. It can be expressed by a sum of consecutive steps, called cumulation. To construct an evolution path, the step-size σ is disregarded.

Exponential smoothing is used when an evolution path, $\mathbf{p}_c \in \mathbb{R}^n$, is constructed, starting with $\mathbf{p}_c^{(0)} = 0$.

$$\mathbf{p}_c^{(g+1)} = (1 - c_c) \mathbf{p}_c^{(g)} + \sqrt{c_c(2 - c_c) \mu_{eff}} \frac{\mathbf{m}^{(g+1)} - \mathbf{m}^{(g)}}{c_m \sigma^g} \quad (14)$$

where $\mathbf{p}_c^{(g)} \in \mathbb{R}^n$, evolution path at generation g and $c_c \leq 1$. The factor $\sqrt{c_c(2 - c_c) \mu_{eff}}$ is a normalization constant for $\mathbf{p}_c$.

The rank-one update of the covariance matrix $C^{(g)}$ via the evolution path $\mathbf{p}_c^{(g+1)}$ reads:

$$C^{(g+1)} = (1 - c_1) C^{(g)} + c_1 \mathbf{p}_c^{(g+1)} \mathbf{p}_c^{(g+1)T}. \quad (15)$$

Using the evolution path for the update of C is a significant improvement of (11) for small μ_{eff} because correlations between consecutive steps are heavily exploited.

The final update of the covariance matrix combines (11) and (15), being described visually in Fig. 6 and giving rise to the following equation:

$$\begin{aligned} C^{(g+1)} &= \left(1 - c_1 - c_\mu \sum_{i=1}^{\lambda} w_i\right) \\ &\quad C^{(g)} + c_1 \mathbf{p}_c^{(g+1)} \mathbf{p}_c^{(g+1)T} \\ &\quad + c_\mu \sum_{i=1}^{\lambda} w_i \mathbf{y}_{i:\lambda}^{(g+1)} \left(\mathbf{y}_{i:\lambda}^{(g+1)}\right)^T \end{aligned} \quad (16)$$

where $c_1 \approx 2/n^2$.

$$c_\mu \approx \min\left(\frac{\mu_{eff}}{n^2}, 1 - c_1\right).$$

$$\mathbf{y}_{i:\lambda}^{(g+1)} = \left(\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)} \right) / \sigma^{(g)}.$$

Equation (16) combines the advantages of Eqs. (11) and (15). On the one hand, information on correlations between generations is exploited using the evolution path for the rank-one update. The former is essential in large populations and is particularly important in small populations.

2.2.4 Step-Size Control

The covariance matrix adaptation increases or decreases the scale only in a *single direction* for each selected step. It also reduces the scale by fading old information by a given non-adaptive factor. Also, exist two reasons to introduce a step-size control in addition to the adaptation rule (16) for $\mathbf{C}^{(g+1)}$:

1. The optimal overall step length cannot be well approximated by (16) if μ_{eff} is chosen larger than one.
2. The largest reliable learning rate for the covariance matrix update in (16) is too slow to achieve competitive change rates for the overall step length.

To control the step size $\sigma^{(g)}$ an evolution path is utilized. The method can be applied independently of the covariance matrix update and is denoted as *cumulative step length adaptation* (CSA).

To decide whether the evolution path is “long” or “short,” the length of the path is compared with its expected length under random selection, where consecutive steps are independent and, therefore, uncorrelated.

In contrast to Eq. (14), a conjugate evolution path is constructed because the expected length of the evolution path $\mathbf{p}_c$ from (14) depends on its direction. Initialized with $\mathbf{p}_\sigma^{(0)} = \mathbf{0}$, the conjugate evolution path reads

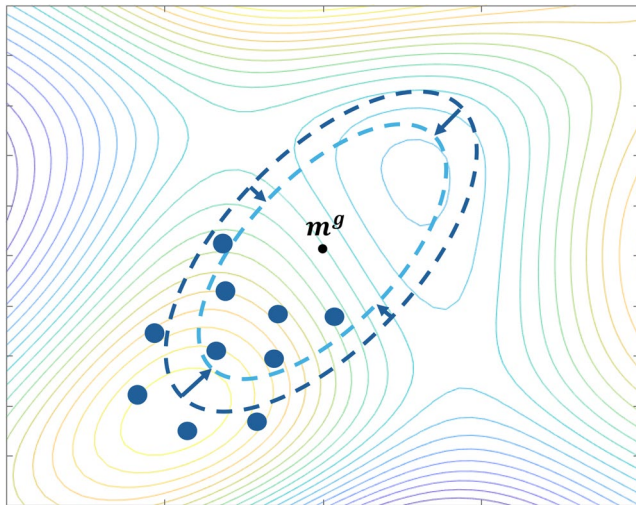


Fig. 7 Adjusting the step size

$$\mathbf{p}_\sigma^{(g+1)} = (1 - c_\sigma) \mathbf{p}_\sigma^{(g)} + \sqrt{c_\sigma (2 - c_\sigma) \mu_{eff}} \mathbf{C}^{(g)^{-1/2}} \frac{\mathbf{m}^{(g+1)} - \mathbf{m}^{(g)}}{c_m \sigma^{(g)}} \quad (17)$$

where $\mathbf{p}_\sigma^{(g)} \in \mathbb{R}^n$ is the conjugate evolution path at generation g , $c_\sigma < 1$ and $\sqrt{c_\sigma (2 - c_\sigma) \mu_{eff}}$ is a normalization constant.

$\mathbf{C}^{(g)^{-1/2}} \stackrel{\text{def}}{=} \mathbf{B}^{(g)} \mathbf{D}^{(g)^{-1}} \mathbf{B}^{(g)^T}$, where $\mathbf{C}^{(g)} = \mathbf{B}^{(g)} (\mathbf{D}^{(g)})^2 \mathbf{B}^{(g)^T}$ is a eigendecomposition of $\mathbf{C}^{(g)}$, where $\mathbf{B}^{(g)}$ is an orthonormal basis of eigenvectors, and the diagonal elements of the diagonal matrix $\mathbf{D}^{(g)}$ are square roots of the corresponding positive eigenvalues.

For $\mathbf{C}^{(g)} = \mathbf{I}$, we have $\mathbf{C}^{(g)^{-1/2}} = \mathbf{I}$ and Eq. (17) replicates Eq. (14). The transformation $\mathbf{C}^{(g)^{-1/2}}$ re-scales the step $\mathbf{m}^{(g+1)} - \mathbf{m}^{(g)}$ within the coordinate system given by $\mathbf{B}^{(g)}$. Consequently, the transformation $\mathbf{C}^{(g)^{-1/2}}$ makes the expected length of $\mathbf{p}_\sigma^{(g+1)}$ independent of its direction and for any sequence of realized covariance matrices $\mathbf{C}_{g=0,1,2,\dots}^{(g)}$ it has under the random selection $\mathbf{p}_\sigma^{(g+1)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, given $\mathbf{p}_\sigma^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

To update $\sigma^{(g)}$, we compare $\|\mathbf{p}_\sigma^{(g+1)}\|$ with its expected length $E\|\mathcal{N}(\mathbf{0}, \mathbf{I})\|$, that is

$$\ln \sigma^{(g+1)} = \ln \sigma^{(g)} + \frac{c_\sigma}{d_\sigma} \left(\frac{\|\mathbf{p}_\sigma^{(g+1)}\|}{E\|\mathcal{N}(\mathbf{0}, \mathbf{I})\|} - 1 \right), \quad (18)$$

where $d_\sigma \approx 1$, damping parameter, scales the change magnitude of $\ln \sigma^{(g)}$, and $E\|\mathcal{N}(\mathbf{0}, \mathbf{I})\|$ is the expectation of the Euclidean norm of a $\mathcal{N}(\mathbf{0}, \mathbf{I})$ distributed random vector.

The step-size change is unbiased on the log scale because $E[\ln \sigma^{(g+1)} | \sigma^{(g)}] = \ln \sigma^{(g)}$ for $\mathbf{p}_\sigma^{(g+1)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

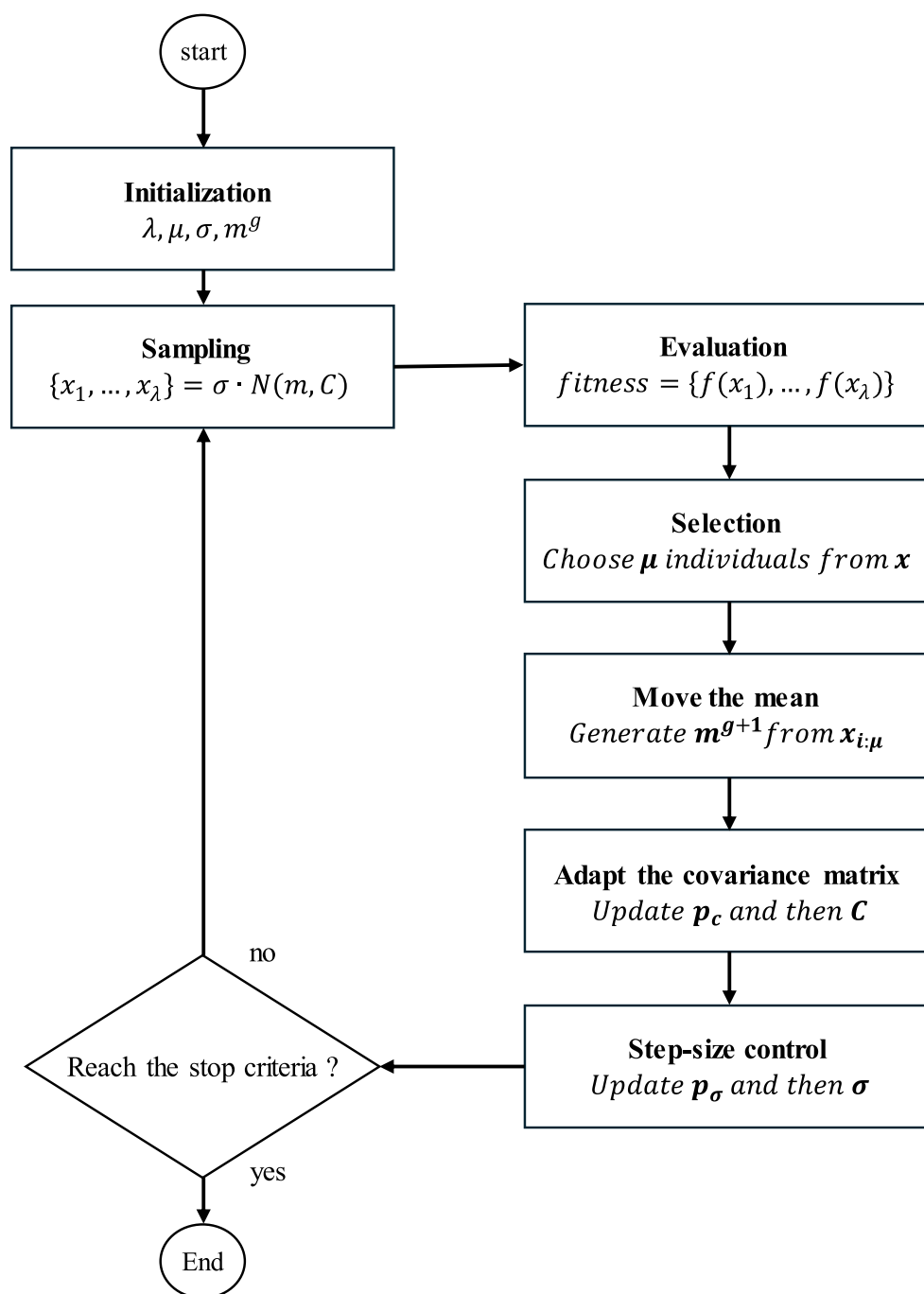
Because $\sigma^{(g)} > 0$, Eq. (18) is equivalent to:

$$\sigma^{(g+1)} = \sigma^{(g)} \exp \left(\frac{c_\sigma}{d_\sigma} \left(\frac{\|\mathbf{p}_\sigma^{(g+1)}\|}{E\|\mathcal{N}(\mathbf{0}, \mathbf{I})\|} - 1 \right) \right) \quad (19)$$

The process of adjusting the step size of the CMA-ES algorithm is shown visually in Fig. 7.

2.2.5 CMA-ES Flowchart

All the processes that allow the CMA-ES algorithm to work are summarized in Fig. 8.

Fig. 8 CMA-ES algorithm flowchart

3 Studies on State-of-the-Art CMA-ES and Its Research Areas

3.1 Basic Design Principles of CMA-ES

The CMA-ES algorithm can overcome the most common problems encountered in evolutionary computation, such as low performance on poorly scaled and poorly separable objective functions, the need for large population size, or premature system convergence [17]. To overcome all these

conflicts, the CMA-ES uses some basic design principles listed below:

- **Change Rates:** A rate change is defined as the change in the anticipated parameter for each selected search point, depending on a particular choice situation. Fine-tuning the possible rates of change of the adaptive parameters is essential to achieve competitive performance on various objective functions. The CMA-ES independently controls the rates of change of the mean value of the

distribution m , the covariance matrix C , and the step size σ .

- **Invariance:** The invariance characteristics of a search algorithm indicate equal behavior on a set or category of objective functions. Invariance is a crucial feature of CMA-ES. Translational invariance is considered a requirement in continuous domain optimization. Other invariance, such as invariance to certain linear transformations in the search space, are highly desirable as they ensure consistent performance across different classes of functions and thus facilitate extrapolation of empirical results. The CMA-ES have other invariances besides translation invariance, like invariance to order-preserving transformations of the objective function value and invariance to angle-preserving transformations of the search space if the initial search point is transformed accordingly, among others.
- **Unbiasedness:** The unbiasedness of variations of object and strategy parameters is a crucial design criterion for a randomized search procedure [18]. For step sizes, a bias toward increasing or decreasing could result in premature divergence or convergence, respectively, in case the selection pressure is low or no improvement is observed. In noisy objective functions, it may be appropriate to have a bias toward a properly controlled increase. It has the disadvantage that the termination decision becomes even more difficult.

3.2 Metric Studies at CMA-ES

Several studies have been conducted to test the elementary design features and principles of the CMA-ES, dealing directly with its behavior according to the population size λ , analyzing the dispersion metric in functions, comparing the impact of search invariance, and reviewing the results on large-scale optimization problems.

3.2.1 Effect of Population Size on the Performance

The population size is one of the most influential and directly controllable parameters affecting performance that the CMA-ES will have. The optimal population size that the CMA-ES can take a wide range of values, as it depends mainly on two key factors: the type of objective function and the size of the problem. It has been shown empirically that the optimal population size scales sub-linearly concerning the problem size [19]. Similarly, the performance of the CMA-ES when trying to find the optimal solution to an optimization problem scales between linear and cubic behavior with the problem size.

Generally, if the population size is increased, the CMA-ES has a better perspective of the function by detecting

the problem's topology. However, despite obtaining better results with a larger population, this is not always a rule since there are functions with particular topologies that make it difficult to search for a larger population, besides increasing the accesses to the function and the computational cost linearly. With this in mind, searching for an ideal population size for a wide range of optimization problems is a recurring theme when discussing improvements to the CMA-ES.

3.2.2 Effect of the Dispersion of a Function on the CMA-ES

The dispersion of a function is a concept introduced by Lunacek et al. [20] to classify objective functions based on their global topology, i.e., how close a sample of points is to each other, where the sample is constructed based on a threshold that is expressed as an objective percentile of the search space.

It has been shown that the CMA-ES obtains better results in functions categorized as “low dispersion,” receiving a reduced performance in “high dispersion” functions. This phenomenon is because the cumulation strategy does not work correctly in regions with high sparsity, so the CMA-ES can spend a large amount of computational resources on populations that can get stuck in a local optima. In this way, the CMA-ES is inefficient in multimodal functions where the underlying structure of the objective function is unpredictable, which has become a fruitful field of research by implementing improvements to the CMA-ES to work with noisy and unexpected functions.

3.2.3 CMA-ES on Large-Scale Optimization Problems

The CMA-ES algorithm performs well in an assumable number of dimensions. However, like many optimization algorithms, its performance is reduced when the problem's dimensionality increases. The course of dimensionality is a problem that generally affects evolutionary algorithms, but it is mainly reflected in the CMA-ES [21]. It is possible to relate the dimensionality problem with its high computational cost, which increases exponentially, thus limiting its capabilities when implemented in real-life applications with many design variables.

The problem of the CMA-ES when dealing with high dimensionalities lies in the eigen-decomposition of the covariance matrix, which is a vital part of the CMA-ES performance. Despite this, there are several ways to deal with CMA-ES at high dimensionalities, such as reducing its computational cost or using different techniques to scale population algorithms and obtain competitive results.

3.2.4 Impact of Invariance on Search

As discussed above, invariance is the ability of an algorithm to obtain the same performance regardless of the environment under which the experiments are performed.

When we evaluate the ability of CMA-ES to work in environments where there are rotations of the coordinate system, we see that CMA-ES is invariant to orthogonal transformations by design, which often yields more stable behavior under rotated landscapes and has been reported to outperform several baselines on rotated benchmark functions, for example, Particle Swarm Optimization (PSO) [22] in functions with this type of transformations [23].

3.3 Research Areas

Since its development, CMA-ES has had several versions and modifications to meet different optimization paradigms, remaining at the edge of contemporary algorithms regarding their versatility.

In addition to its areas of improvement, such as reducing the computational cost, improving its performance in high dimensionality, and dynamically modifying the population size, versions of CMA-ES have been developed for tasks such as multiobjective optimization [24], mixed-variable optimization [25], constrained optimization [26], parallel optimization [27], as well as multiple multi-start proposals [28].

4 CMA-ES Modifications

The various modifications of the CMA-ES that have been included in this survey have been classified into different categories based on the type of improvement sought (such as improving the overall performance of the algorithm or reducing the computational cost) or based on the kind of optimization problem to be solved (such as multiobjective functions or constrained optimization).

The following sections will include a brief summary of each article proposing modifications and hybridizations of CMA-ES, including representative equations and flowcharts for the modifications with the greatest academic impact in each subsection. However, all the algorithms proposed in all the articles will be condensed at the end of each subsection into a comparative table, which will summarize the main modifications of the proposal with respect to the original CMA-ES algorithm, including the advantages and disadvantages that each method offers.

To provide an actionable complement to our hierarchical taxonomy, Fig. 9 introduces a regime-driven decision flow that summarizes when and why major CMA-ES variant

families are typically preferred. The flowchart maps common problem characteristics—variable type, multiobjective structure, constraints/safety, noise, evaluation cost, dimensionality, and multimodality—to the corresponding CMA-ES families discussed throughout Sects. 4 and 5.

4.1 Reducing Computational Complexity

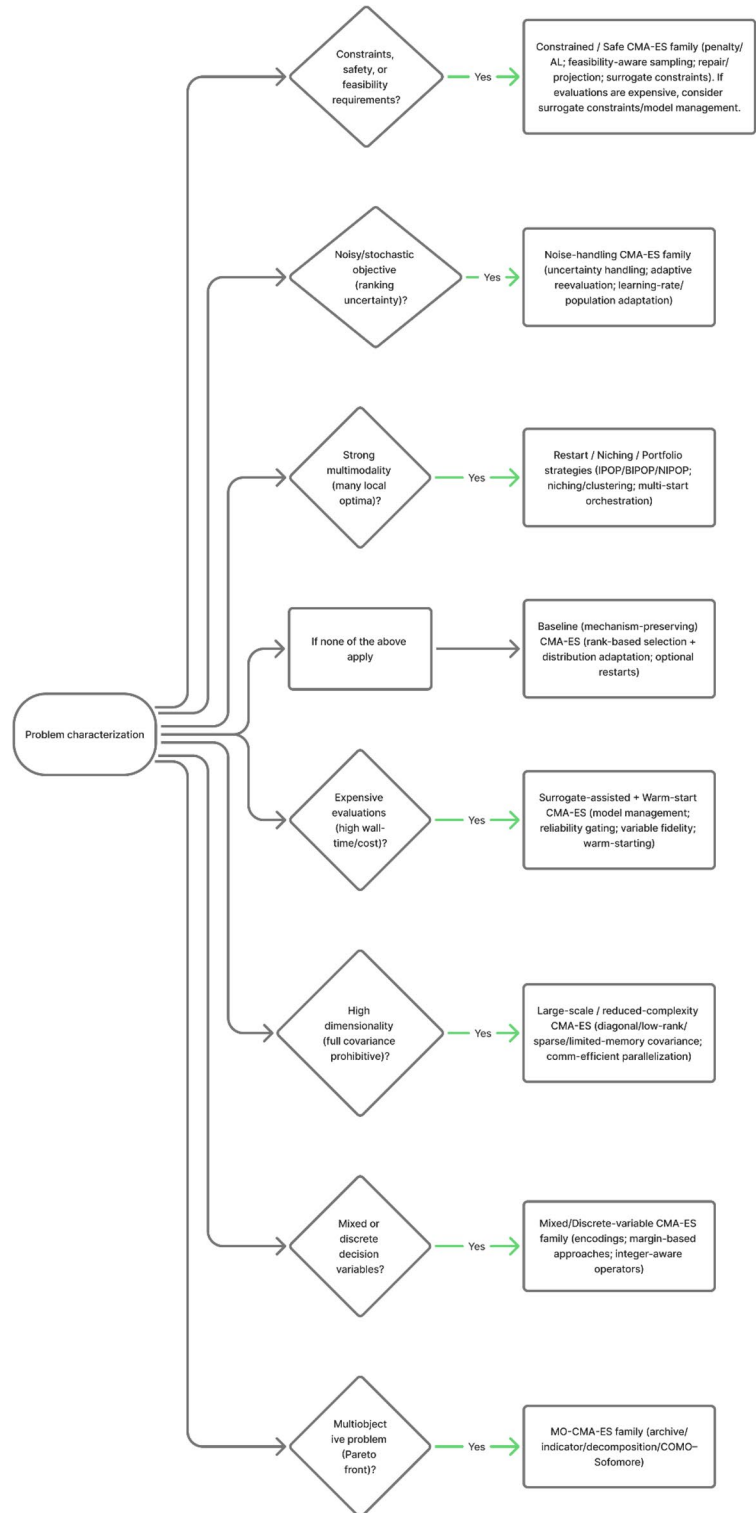
Reducing computational complexity is a fundamental concept for reducing an algorithm's resources and implementing more efficient strategies, as demonstrated in [29], where a variant called (1+1)-Cholesky-CMA-ES is proposed. This variant makes use of the covariance matrix adaptation in the (1+1)ES algorithm, using an incremental Cholesky update to replace the decomposition of the covariance matrix, obtaining a computationally efficient process that reduces the computational complexity from $O(n^3)$ to $O(n^2)$. The update of the Cholesky factors A (where $C = AA^T$) for a mutation vector $v = Az$ is:

$$A^{(g+1)} = \sqrt{\alpha}A^{(g)} + \frac{\sqrt{\alpha}}{\|z\|^2} \left(\sqrt{1 + \frac{\beta}{\alpha}\|z\|^2} - 1 \right) (A^{(g)}z) z^T \quad (20)$$

where $\alpha = 1 - C_{cov}$, $\beta = C_{cov}$, and C_{cov} is the learning rate. Figure 10 graphically shows the sequence of Cholesky-CMA-ES operators.

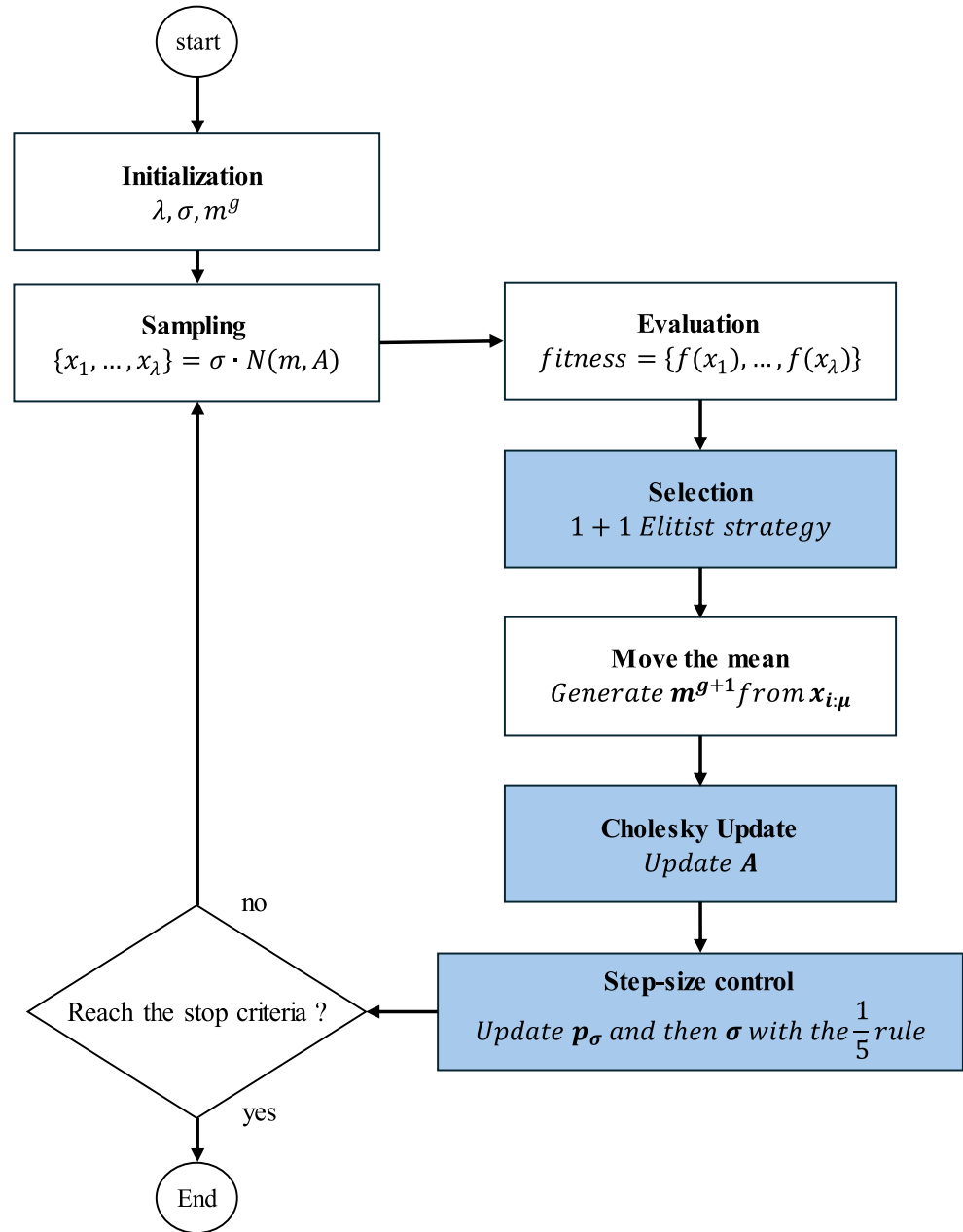
In [30], a modification of the CMA-ES is proposed that reduces the computational complexity from $O(n^3)$ to $O(n^2)$, employing an update mechanism based on Cholesky factorization that replaces the decomposition of the covariance matrix, making it a more robust and faster method in high dimensions. This methodology maintains characteristics similar to the previous proposal, with the difference that it preserves the evolution path, retains the information, and enhances the adaptation of the search metric.

When talking about reducing computational complexity, it is possible to refer to two fundamental concepts in optimization: Space complexity and time complexity, which refer to the amount of memory and execution time used by an algorithm. In this sense, modifications are based on optimizing one of the two parameters or finding a balance between them, as in [15], a modification proposed by the author of the original CMA-ES. This proposal reduces the number of generations by utilizing information from large-scale populations, thereby reducing the time complexity from $O(n^2)$ to $O(n)$. Similar to this, in [31], a variant named Limited Memory Evolution Strategy with Covariance Matrix Adaptation (L-CMA-ES) is proposed, which reduces the time and space complexity by reducing the dimension of the mutation distribution and avoiding the computation and storage of the covariance matrix, resulting in a space complexity of $O(nm)$ and time complexity of $O(nm^2)$. Instead of

Fig. 9 Regime-driven decision flow for CMA-ES variant selection

explicitly calculating C , the dominant eigenvalues $\hat{B}$ and $\hat{D}$ are updated directly by a low-rank spectral decomposition. The key equation is an adaptation of the rank-one update from the original CMA-ES, preserving the structure of C without storing it.

On the other hand, Ros and Hansen [32] proposed a simple modification to the CMA-ES where the covariance matrix is restricted to be diagonal, and the learning rate is increased, naming it Separable Covariance Matrix Adaptation Evolution Strategy (sep-CMA-ES), reducing the time and space complexity to a linear behavior under certain

Fig. 10 CMA-ES with Cholesky update flowchart

conditions. With this approach, the covariance matrix C is restricted to be diagonal, as shown in Eq. 22, eliminating the fitting of correlations between variables. This simplifies the model but speeds up fitting in problems with low interdependence between variables.

Akimoto and Hansen [33] developed a variant of CMA-ES that reduces the covariance matrix parameters, naming this variant as Restricted Covariance Matrix Adaptation (VkD-CMA). This approach yields similar results to the original CMA-ES or sep-CMA-ES under certain conditions, achieving superior performance in terms of function accesses, thereby reducing internal complexity to a linear

behavior rather than a quadratic behavior. The covariance matrix is parameterized as:

$$C = D(I + VV^T)D \quad (21)$$

where D is a diagonal matrix that scales the variables, and V is a matrix that captures dependencies between variables. A projection-based method is derived to update D and V without explicitly calculating the full covariance matrix, reducing computational complexity. Despite this, in [34], the same authors develop an improved version of VkD-CMA, removing the need for a pre-selection of the restricted covariance matrix and thus achieving better results than its predecessor.

A simplification of the CMA-ES is described in [35], where processes such as the evolutive path or the calculation of the covariance matrix are discarded, achieving similar results in terms of performance but with a notably lower computational burden. This modification, called Matrix Adaptation Evolution Strategy (MA-ES), can obtain these results by transforming the covariance learning to direct learning of the matrix square root of the covariance matrix, performing at the level of the original CMA-ES in the BBOB COCO test [36]. This proposal replaces C with a matrix M that adapts directly, avoiding complex numerical operations. It maintains a competitive performance with the original CMA-ES in continuous optimization problems but with lower computational costs.

Mao proposed an efficient conjugate gradient-based Cholesky CMA-ES (AM-CG Cholesky CMA) [37], which was developed from the need to implement the CMA-ES in an environment of parameter estimation of nonlinear systems with colored noise, where the computational complexity of the covariance matrix is imposed as a constraint. This variant of the Cholesky CMA-ES uses the Cholesky factor instead of computing the covariance matrix, updating the search direction and the mean distribution employing the conjugate gradient method (CG), and performing two simulations in parameter optimization environments to demonstrate its effectiveness. Instead of updating and decomposing the covariance matrix, the algorithm directly updates its Cholesky factor L .

Finally, one recent modification to reduce the computational complexity of the CMA-ES is the Correlated Evolution Path CMA-ES (cCMA-ES) [38]. Its principle of operation is based on the similarity of the evolution path p_σ concerning the evolution path p_c of the original CMA-ES,

combining them to obtain a single evolutionary path, reducing the computational cost and preserving the original structure of the algorithm without losing its vital characteristics, as other modifications of the CMA-ES dedicated to saving computational resources do. This proposal was tested on 30 functions of the IEEE CEC 2024 benchmark suite, compared with five other modifications of the CMA-ES, achieving outstanding performance in most of the tests.

Table 1 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection in terms of reducing computational complexity.

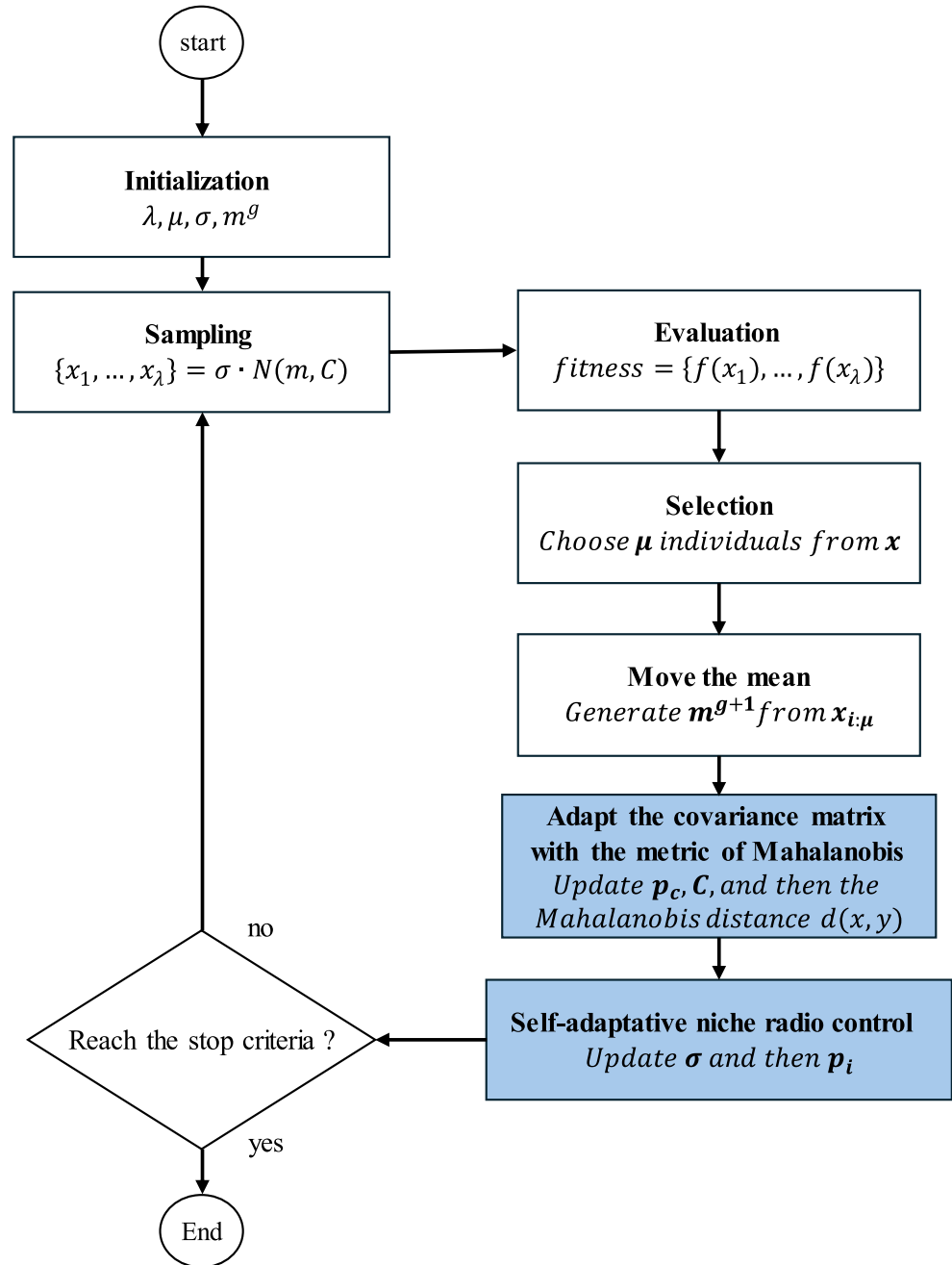
4.2 Large-Scale Optimization

High-dimensional optimization problems have been a research area in evolutionary computation, involving concepts such as niching methods. Dynamic niching is a concept introduced to evolutionary algorithms in [39] and applied to CMA-ES in [40], resulting in the CMA-ES Dynamic Niching (CMA-Nich), where different fitness peaks are identified dynamically using Dynamic Peak Identification (DPI), classifying the solutions based on a niche radius, thus producing recombinations only in the solutions within the niches. This proposal did not improve its quality in large-scale problems, so the development of a Niche Radius Adaptation method in the CMA-ES Niching algorithm [41, 42] improved the performance of the CMA-ES in high dimensions, introducing the concept of an individual niche radius that adapts the global step size during the course of the algorithm. It proposes a self-adaptive niche radius mechanism for each individual, coupled with the step size of the CMA-ES, as can be seen in Fig. 11. This allows niches to adjust dynamically according to the landscape topology, avoiding specifying a

Table 1 Summary of reducing computational complexity CMA-ES modifications

Algorithm	Key Modification	Effect on CMA-ES	Complexity	Strengths	Weaknesses
sep-CMA-ES [32]	Diagonal covariance only	Removes correlation learning	$O(n)$	Very fast in high-dim.	Not rotation-invariant
L-CMA-ES [31]	Low-rank eigenbasis (m components)	Stores/updates only m eigenpairs	$O(nm^2)$	Captures main search directions	Performance depends on m
VkD-CMA-ES [33]	Restricted covariance model	Updates $diag(D) + k$ direction vectors	$O(nk)$	Flexible model complexity	Sensitive to choice of k
VkD-CMA-ES (Online) [34]	Adaptive number of directions k	Dynamically adjusts model size	$O(nk)$	No manual tuning	k may grow large
(1+1) Cholesky-CMA-ES [29]	Rank-one Cholesky update	Removes eigen decomposition	$O(n^2)$	Simple and efficient	Still quadratic cost
Cholesky-CMA-ES (2009) [30]	Incremental Cholesky factor update	Direct factor update instead of EVD	$O(n^2)$	Equivalent behavior to CMA-ES	Eigenvalues unavailable directly
MA-ES [35]	Matrix square-root adaptation	Removes covariance matrix and one path	$O(n^2)$	Fewer components, simpler	Less accurate for large λ
AM-CG CMA-ES [37]	Conjugate-gradient enhanced updates	CG modifies mean and path updates	$O(n^2)$	Fast and accurate estimation	Domain-specific
cCMA-ES [38]	Single correlated evolution path	Merges step-size and covariance paths	$O(n^2)$	Slightly reduced overhead	Small complexity gain

Fig. 11 Self-adaptive niche radius CMA-ES flowchart



fixed radius. Furthermore, this proposal utilizes the covariance matrix of the CMA-ES to define niches with ellipsoidal shapes, employing the Mahalanobis distance, which better captures the geometry of the basins of attraction. The update of the radius ρ for individual i in generation $g+1$ is defined in the following equation:

$$\rho_i^{(g+1)} = (1 - c_i^{(g+1)}) \cdot \rho_{parent}^{(g)} + c_i^{(g+1)} \cdot \sqrt{n} \cdot \sigma_{parent}^{(g+1)} \quad (22)$$

where $c_i^{(g+1)}$ is an adaptive learning coefficient based on the change in step-size, and $\sigma_{parent}^{(g+1)}$ is the step size of the

parent. This proposal was tested on 8 multimodal benchmark functions, achieving notable improvements in 7 of the 8 functions.

In [43] a variant of the CMA-ES using Cooperative Coevolution (CC-CMA-ES), where this proposal adopts a strategy where the population of solutions is not subdivided, nor does it perform mutations or recombinations, is presented. Within the adaptive decomposition method of cooperative coevolution, the probability of selecting a strategy i is calculated as:

$$probability_i = \frac{e^{\theta_i}}{\sum_{p=1}^3 e^{\theta_p}}, \text{ where } \theta_i = \sum_{j=1}^n \delta_{ij} \quad (23)$$

δ_{ij} measure the fitness improvement of strategy i in cycle j . However, instead, it obtains a local Gaussian distribution from the global Gaussian distribution to generate samples in a subspace, in addition to two decomposition strategies based on the diagonal of the covariance matrix are proposed to maintain a correct balance between exploration and exploitation, resulting in superior performance on 11 out of 15 multimodal functions in 1000 dimensions against its original counterpart.

Loshchilov [44] proposed a variant of the CMA-ES inspired by the Limited Memory Broyden-Fletcher-Goldfarb-Shanno (LM-BFGS) [45] numerical method called Limited Memory CMA-ES (LM-CMA-ES), dedicated to optimizing large-scale problems. This strategy is based on memory limitation by storing solution information generating samples according to a reproduced covariance matrix of m direction vectors selected in the optimization process. This algorithm has been tested against variants such as sep-CMA-ES in 100,000 dimensions on ellipsoidal functions, obtaining better results as iterations progress while its competitors remain stagnant. However, Loshchilov [46] proposed an improvement to this methodology, adding a sampling procedure from a family of search representations defined by Cholesky factors according to a Rademacher distribution. Instead of storing the covariance matrix C , the LM-CMA stores m pairs of direction vectors and dynamically reconstructs the Cholesky factor A using linear operations, as shown in the following equation:

$$A^{(g+1)} = \sqrt{1 - c_1} A^{(g)} + \frac{\sqrt{1 - c_1}}{\|v^g\|^2} \left(\sqrt{1 + \frac{c_1}{1 - c_1} \|v^g\|^2} - 1 \right) p_c^g (v^g)^T \quad (24)$$

where $v^g = A^{-g} p_c^g$, and p_c^g is the evolution path. The proposed improvements enabled the development of a more robust algorithm at higher dimensionalities, yielding better results with fewer function accesses than the original LM-CMA-ES.

In [47], a methodology is proposed in which the CMA-ES stores the historical information of all generations of solutions to evaluate the parameters of the next-generation distribution. This algorithm, called Gradient Information Evolutionary Strategy (GI-ES), replaces the adaptation of the covariance matrix with the expected fitness of the candidate solution, simulating the gradient through all the information of the individuals, thereby defining a search direction based on this data.

The use of approximate gradient information enables the evolution to be guided more efficiently, without relying on first-order details, which is particularly crucial in

high-dimensional problems. GI-ES outperforms CMA-ES, SDA-ES, DECC-G, and MOS in terms of convergence and runtime on test functions in 1,000 dimensions, excelling on both rotated and unrotated functions (Ellipsoid, Cigar, Ras-trigin) with a lower average runtime.

Li [48] proposed a variant where the CMA-ES optimizes large-scale black-box problems by using a low-rank matrix approximation of the covariance matrix and using two of its vectors to generate the new candidate solutions. Fast CMA-ES approximates the covariance matrix using a low-rank model that uses only m evolution vectors (paths), where each new solution is generated using only two evolution vectors. Fast CMA-ES is n times faster than CMA-ES in high-dimensional problems, and performs similarly to LM-CMA-ES in optimization problems for up to 8,192 dimensions. Nevertheless, this algorithm is highly competitive in the CEC 2010 large-scale global optimization benchmark, achieving a remarkable performance.

Finally, Shimizu [49] developed a system that can be embedded in algorithms dedicated to optimizing high-dimensional problems called Coordinate Selection CMA-ES (CS-CMA-ES). The CS-CMA-ES is a methodology that vectorizes the step size, adapting each vector component to the scale of the coordinates selected in an objective function. This simple adaptation enables reducing the number of function accesses required to reach the global optimum in test functions ranging from 10,000 to 100,000 dimensions, making it a modification that can be implemented in other variants of the CMA-ES, such as the sep-CMA-ES and the VD-CMA-ES. CS-CMA-ES defines an index vector t and an endpoint e , where each iteration selects s consecutive coordinates from e , then samples individuals in the selected subspace, and estimates the curvature for intelligent coordinate selection.

High-dimensional optimization problems have a property where the design variables contain redundant information that practically does not affect the value of the objective function. This phenomenon, known as Low Effective Dimensionality (LED), is addressed by the sep-CMA-ES-LED [50] algorithm, an enhanced version of sep-CMA-ES designed explicitly for high dimensionality, which utilizes hyperparameter adaptation and a refined version of the step-size adaptation update rule. Sep-CMA-ES-LED estimates the effectiveness of each dimension using the signal-to-noise ratio (SNR) of the update directions of the distribution parameters and adapts the hyperparameters and step-size adaptation rules accordingly. Sep-CMA-ES-LED achieves a more efficient search step since the modifications in CSA and TPA ignore redundant dimensions when calculating norms and evolution directions, improving convergence. Due to its various adaptive operators, this algorithm outperforms sep-CMA-ES and other variants on benchmark functions of up

to 1024 dimensions, resulting in notable improvements in functions such as Sphere, Ellipsoid, and Ackley.

Several algorithms have been developed to avoid the curse of dimensionality, including variants of the CMA-ES, such as [51], which implement correlation coefficient grouping (CCG) and Model Complexity Control (MCC) to form the MCC-CCG-CMAES and the CC-CCG-CMAES variants. This algorithm combines these techniques, using the CCG to dynamically divide the variables into groups based on the information obtained from their correlation matrix, where the covariance matrix C and the correlation coefficients r_{ij} are calculated as follows:

$$C = \sum_{k=1}^{\mu} w_k (x_{k:\lambda} - m)(x_{k:\lambda} - m)^T, r_{ij} = \frac{C_{ij}}{\sqrt{C_{ii}C_{jj}}} \quad (25)$$

At the same time, the MCC framework is implemented to reduce the degree of freedom of the Gaussian model used in the original CMA-ES. These modifications effectively reduce the computational cost of the system, yielding a competitive algorithm that can produce robust results in the CEC 2010 Large-Scale Global Optimization Benchmark.

Table 2 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on large-scale optimization.

4.3 Reducing Function Accesses

Within optimization schemes, it is essential to maintain strict control over the function accesses performed by an algorithm, as efficiency depends on achieving high-quality solutions under a given evaluation budget [52]. Although reducing the computational complexity of an algorithm, such as the CMA-ES, can help obtain results similar to the original method with fewer logical and/or arithmetic operations, the objective of reducing evaluations in the objective function allows us to improve the relationship between the resources used and the final result obtained.

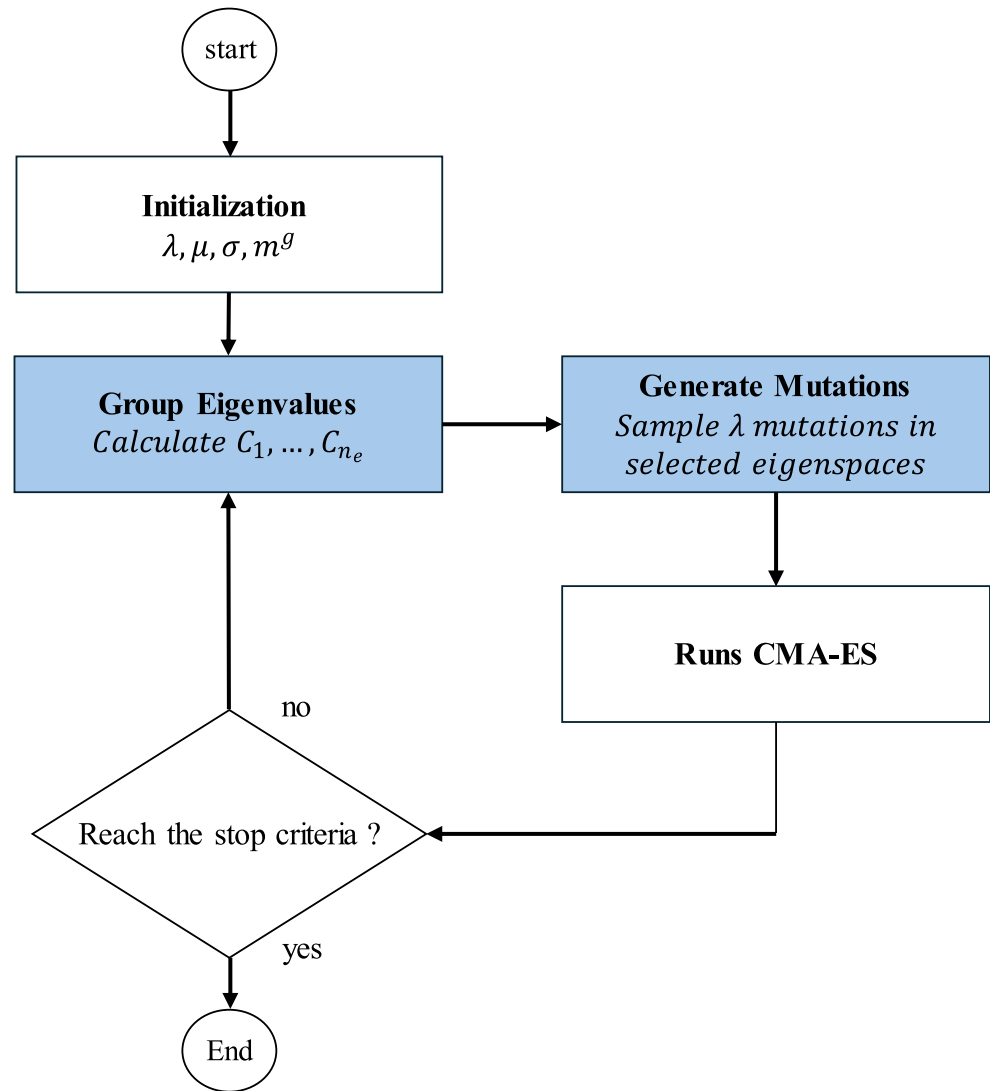
One of the first proposals to reduce function accesses is the ϵ -CMA-ES [53], which achieves its objective by reducing the number of evaluations of the objective function necessary to adapt the covariance matrix in cases where the Hessian matrix of the objective function has repeated or clustered eigenvalues. The main idea is to randomly evaluate the minor eigenspace of the covariance matrix, which is associated with the smallest eigenvalues, as can be seen in Fig. 12. To accelerate the adaptation of the covariance matrix in problems where there are dominant and minor eigenspaces, the eigenvalues of the covariance matrix C are grouped into C_i clusters, so that for each direction j in the minor eigenspace:

$$z_{i,j} \sim \begin{cases} \mathcal{N}(0, 1) & \text{with probability } p_m \\ 0 & \text{otherwise} \end{cases} \quad (26)$$

Table 2 Summary of large scale optimization CMA-ES modifications

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
Adaptive-Radius CMA-ES Niching [40]	Niching mechanism	Individual niche radius self-adapts via step-size coupling	Reduces sensitivity to fixed niche radius; improves robustness in multimodal search	Additional niching parameters; increased algorithmic complexity
Step-Size Coupled Niching CMA-ES [42]	Niching + CSA	Niche radius dynamically coupled to global step size	Automatic radius adaptation; better scalability across landscapes	Assumes meaningful relation between step size and niche structure
Mahalanobis-Niching CMA-ES [42]	Niching + Covariance	Ellipsoidal niches defined via Mahalanobis distance	Rotation-aware niche shapes; improved handling of anisotropic basins	Extra distance computations; relies on accurate covariance estimation
LM-CMA-ES [44]	Covariance adaptation	Covariance reconstructed from limited direction vectors	Scales to very high dimensions; reduced memory footprint	Approximate covariance may miss weak dependencies
Improved LM-CMA [46]	Covariance adaptation	Refined limited-memory sampling and update rules	Competitive performance with gradient-free efficiency	Sensitive to memory size selection
fast-CMA-ES [48]	Covariance model	Dominant eigen-subspace approximation of covariance	Preserves main correlations with reduced cost	Loss of full covariance information
Improved sep-CMA-ES [50]	Diagonal covariance	Enhanced diagonal adaptation exploiting low effective dimensionality	Very fast; effective when many variables are redundant	Loses rotation invariance; limited on non-separable problems
MCC-CCG-CMA-ES [51]	Covariance + grouping	Correlation-based grouping with MCC framework	Strong performance on LSGO; reduced model complexity	Grouping overhead; additional hyperparameters
CC-CMA-ES [43]	Search decomposition	CMA-ES embedded in cooperative coevolution framework	Enables scalability via divide-and-conquer	Performance depends heavily on decomposition quality

Fig. 12 E-CMA-ES flowchart



with $z_{i,j} \sim \mathcal{N}(0, 1)$ for the other eigenspaces. ϵ -CMA-ES significantly reduces the number of objective function evaluations on benchmark functions such as Cigar, Tablet, and Twoaxes, where the Hessian matrix has two well-defined eigenvalue clusters. On these functions, the ϵ -CMA-ES reduces the number of evaluations by 23% to 40% compared to the traditional CMA-ES.

In [54], an implementation of the CMA-ES in combination with Kriging metamodels is presented. Integrating Kriging helps the CMA-ES approximate the robustness of solutions efficiently by reducing the number of objective function evaluations. In addition, the use of the CMA-ES covariance matrix is explored to enhance the accuracy of Kriging metamodel predictions by introducing a method for selecting points from an archive to construct local metamodels, thereby reducing the need for additional objective function evaluations. Although the Kriging-based approach is initially slower due to the need to build the file, it outperforms

traditional methods after approximately 3,000 function evaluations. Experiments show that this approach outperforms traditional methods, such as SEM (Single Evaluation Method) and MEM (Multi Evaluation Method), in terms of the quality of the final solutions, especially when the evaluation budget is limited. However, given the effectiveness of the Kriging model in [55], it is possible to see an implementation of these models in the CMA-ES from a different perspective, adding an approximate ranking process to estimate the ranking of the current population. In addition, a training set selection method based on confidence intervals is introduced to construct the surrogate model. The proposed algorithm significantly reduces the number of objective function evaluations compared to the standard CMA-ES and other Kriging-assisted versions.

Despite the good results that Kriging metamodels offer us, there are more variants of the CMA-ES with integrated metamodels, such as the variant proposed in [56], where the

main focus is to reduce the number of evaluations of the objective function using local quadratic metamodels, which saves time in optimization problems where each evaluation can take minutes or even hours. This proposal analyzes how the Imm-CMA algorithm scales with population size more significant than the default and proposes a new variant of the algorithm, called nlmm-CMA, which relaxes the acceptance criteria of the metamodel to improve its performance on large populations. Instead of requiring that the exact ranking of the best individuals be maintained, the proposed variant only requires that the set of best individuals and the best individuals remain unchanged. This change allows the algorithm to be more efficient with large populations while maintaining a significant improvement in convergence speed compared to standard CMA-ES. Nlmm-CMA outperforms Imm-CMA in most test functions, especially in multimodal functions such as Ackley. In contrast, on functions such as Rastrigin, the speedup of nlmm-CMA reaches between 2.5 and 5, while that of Imm-CMA is between 1.5 and 3.

Pitra [57] proposed his variant of the CMA-ES called S-CMA-ES (Surrogate Covariance Matrix Adaptation Evolution Strategy), which uses surrogate models to optimize computationally expensive objective functions. However, the main contribution is the introduction of a Doubly Trained Evolution Control, which uses not only the predictions of the surrogate model but also the variances of those predictions to improve the algorithm's efficiency, thus reducing the number of costly evaluations of the objective function. The DTS-CMA-ES significantly improves convergence speed compared to the original S-CMA-ES and other surrogate model-based algorithms. In particular, DTS-CMA-ES is more effective in the middle phase of the optimization process (between 30 and 80 evaluations per dimension).

Table 3 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection in terms of reducing function accesses.

Table 3 Summary of reducing function accesses CMA-ES modifications

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
DTS-CMA-ES [57]	Surrogate-assisted evaluation control	Surrogate-assisted CMA-ES with doubly trained evolution control; uses uncertainty/utility criteria to decide true vs surrogate evaluations.	Reduces expensive objective evaluations while limiting surrogate over-trust.	Extra surrogate-training overhead; benefit depends on model quality and evaluation cost.
ε -CMA-ES [53]	Sampling/exploration control	Detect eigenvalue clusters; with probability p_m perform random evaluation in the minor eigenspace to boost exploration of low-variance directions.	Improves exploration along neglected directions; can help on ill-conditioned problems.	Adds hyperparameters; may waste evaluations on simple/separable landscapes.
nlmm-CMA [56]	Surrogate (local meta-model) acceptance rule	Relax acceptance: preserve the μ -best set and best individual (not full ranking) to maintain speedup for large λ .	Sustains speedups at larger population sizes; better scaling than Imm-CMA.	Can accept imperfect rankings; requires sensible initialization of surrogate parameters to avoid early model trust.
KA-CMA-ES (PS-Interval) [55]	Surrogate-assisted selection (Kriging+pre-selection)	Kriging-assisted CMA-ES with pre-selection; training set updated at predefined intervals.	Reduces true evaluations via surrogate ranking; update frequency is controllable.	If update interval is too large, surrogate drift can mis-rank candidates.
KA-CMA-ES (FGC) [55]	Surrogate refresh control	Fixed generation-based control: periodically perform true evaluations to refresh the Kriging model.	Simple safeguard against stale surrogates; easy to implement.	Refresh schedule is problem-dependent; may spend extra evaluations when not needed.
KA-CMA-ES (ARP-Mean) [55]	Approximate ranking (surrogate-based)	Approximate Ranking Procedure using Kriging mean prediction to rank candidates.	More adaptive than fixed schedules; can improve efficiency under tight budgets.	Sensitive to surrogate bias; may mis-rank when uncertainty is high.
Kriging-assisted robust CMA-ES (archive-based, normal kernel) [54]	Robustness handling via surrogate	Archive-based local Kriging model approximates robust objective within CMA-ES.	Effective under limited budgets; enables robustness-driven search with fewer true evaluations.	Archive/model management overhead; robustness approximation may fail under non-stationary noise.
Kriging-assisted robust CMA-ES (Mahalanobis kernel using C) [54]	Robustness handling via surrogate	Uses CMA-ES covariance to define a Mahalanobis-distance kernel for Kriging.	Aligns surrogate locality with CMA-ES anisotropy; can improve prediction accuracy.	Depends on covariance quality; adds modeling choices/hyperparameters.

4.4 Constrained Optimization

Constrained optimization is an area of mathematical optimization that focuses on finding the optimal value of an objective function subject to a set of constraints [58]. These constraints can be equalities or inequalities that limit the possible solutions to the problem. These problems represent a significant challenge for metaheuristic algorithms due to several reasons inherent in these algorithms and the complexity of constrained problems [59]. Metaheuristic algorithms are designed to explore substantial and complex search spaces. However, they are not specifically tailored to handle constraints, as their primary focus is on exploring and exploiting the solution space without guaranteeing that the generated solutions meet the problem's requirements. To overcome these challenges, metaheuristic algorithms often incorporate specific techniques for handling constraints and develop novel techniques.

The CMA-ES has received several modifications to improve its performance in constrained optimization environments, one of the first being the approach developed by Kramer [60], where the constraints are not available explicitly but are assumed as “black boxes” that return a numerical or Boolean value when evaluating a solution vector. According to Fig. 13, the main contribution of this methodology is introducing a surrogate model to approximate the constraint functions, which allows for improved constraint handling in CMA-ES. This surrogate model is used for three primary purposes: covariance matrix rotation, feasibility verification of solutions utilizing the surrogate model, and repair of infeasible solutions. To estimate the constraint hyperplane h_0 , N points $s_1, \dots, s_n$ are obtained by binary search between a feasible point x_f and infeasible points ξ_i sampled on a hypersphere, as shown in the following equation:

$$(\xi)_i = \frac{1}{\sqrt{(\xi_1)^2 + \dots + (\xi_N)^2}} \cdot ((\xi_1)_i, \dots, (\xi_N)_i)^T \quad (27)$$

where the normal vector of h_0 is then calculated using Gram-Schmidt orthogonalization.

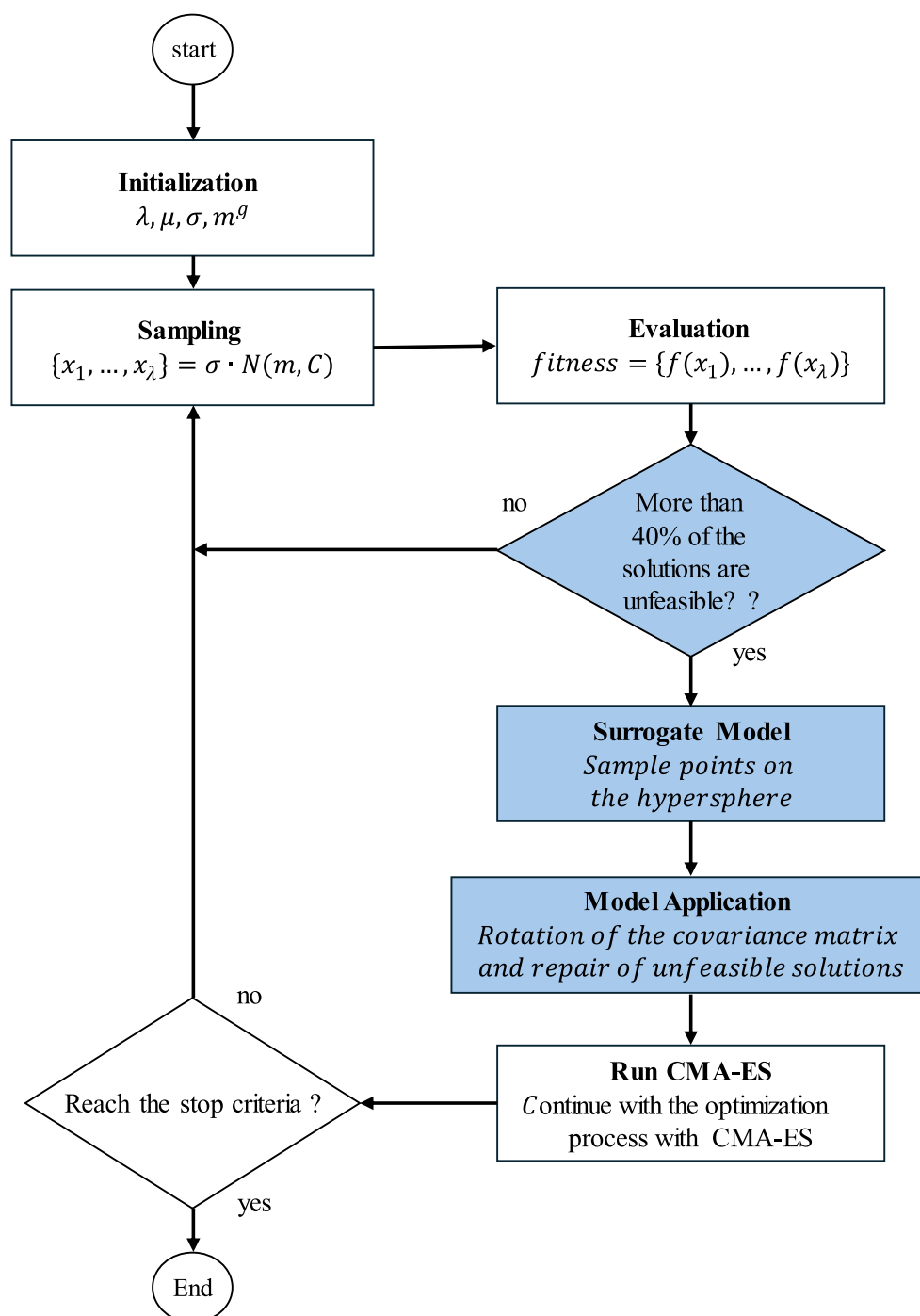
The feasibility check based on the surrogate model reduces the number of constraint evaluations, especially in problems where constraint evaluations are costly, an example being the TR2 problem, where the number of objective function evaluations was reduced from 6,754 to 3,249, and constraint evaluations from 12,019 to 3,650 compared to the original CMA-ES.

Arnold [61] proposed a new approach in which the CMA-ES handles the constraints, specifically in the (1+1)-CMA-ES variant. This method proposes to approximate the directions of the normal vectors of the constraint boundaries

by accumulating steps that violate the constraints. Then, the variances of the mutation distribution in those directions are reduced, allowing the algorithm to approach the boundary of the feasible region without losing the ability to search in directions tangential to the constraints. In problems with multiple active constraints, the proposed method demonstrates superior efficiency in reducing the number of constraint evaluations while maintaining high accuracy in the search for optimal solutions, and it proves to be feasible for constrained optimization problems in practical applications.

De Melo has proposed two different CMA-ES approaches for optimizing constrained environments, the first publication being a combination of the CMA-ES algorithm with an adaptive penalty function that dynamically handles the constraints during the optimization process [62]. This approach is based on adjusting the penalty according to constraint violations and the quality of feasible solutions found, which enables exploring infeasible regions close to the feasible region without discarding them altogether. Its performance in constrained environments, such as engineering problems, was remarkable. In designing a welded beam and a speed reducer, the modified CMA-ES was reported to reach near-optimal/feasible solutions on the welded-beam and speed-reducer problems with fewer objective evaluations than the compared baselines. Based on this model, the methodology presented in [63] was developed, which proposes a two-stage framework where the CMA-ES serves as a global search engine. The best result found by CMA-ES is used as a starting point for a Local Optimizer (LO), testing three classical local optimizers: Simplex, BOBYQA, and L-BFGS-B. This proposal also implements the adaptive penalty function of [62], which allows infeasible solutions with minor constraint violations to be considered if they have a better fitness value than feasible solutions. This proposal was evaluated on nine constrained optimization problems, including four benchmark functions and five engineering design problems, of which, in five of the nine problems, the combination of CMA-ES with the Simplex local optimizer showed significant improvements in solution quality.

The ε CMAgES [64] is a variant of the CMA-ES designed to tackle real-world constrained optimization problems, incorporating improvements in the sampling scheme and stopping criteria, in addition to a ε -constrained constraint-based technique and a gradient-based repair method to handle nonlinear constraints. The performance of the ε CMAgES algorithm was evaluated on 57 optimization problems with real-world constraints. Compared with state-of-the-art algorithms, ε CMAgES obtained the best overall ranking, outperforming methods such as IUDE, ε MAgES, and ILSHADE ε .

Fig. 13 Surrogate constraint CMA-ES flowchart

Within the variants of the CMA-ES to solve problems with constraints, the adaptive augmented Lagrangian method has been implemented on several occasions, as it is possible to see in [65], where the minimization of a function $f(x)$ subject to a single linear inequality constraint $g(x) \leq 0$ in a black-box environment is studied. Within this proposal, a penalty method based on the augmented Lagrangian is incorporated to transform the constrained problem into an unconstrained one, in addition to implementing a theoretical approach that

allows adapting unconstrained optimization algorithms for constrained problems as well as a scheme to dynamically update the penalty factors and Lagrange multipliers, avoiding the need to set values manually. This variant was tested on functions such as sphere, ellipsoid, and Rosenbrock, demonstrating robustness even in ill-conditioned cases compared to other constraint handling methods, showing that the variant with CMA-ES is more efficient under challenging problems. Considering the promising results seen in

[65], the augmented Lagrangian was implemented in a new variant of the CMA-ES, named AL-CMA-ES [66], differing from the previous method by making use of Constraint Modeling by Substitution (CMS). CMS is a technique based on linear models to approximate the constraints, allowing a more stable optimization under challenging problems and the ability to identify failures of the Augmented Lagrangian in non-differentiable constraints and propose solutions. The CMA-ES with AL and constraint modeling is an option that combines robustness with efficiency in problems with multiple constraints, in addition to significantly improving convergence compared to the original CMA-ES.

As noted, different techniques for solving constrained problems are based on CMA-ES. Biedrzycki [67] performed a comprehensive analysis of 22 boundary constraint handling methods, evaluating their impact on the performance of CMA-ES on unimodal and multimodal optimization problems. The choice of boundary constraint handling method is crucial for CMA-ES performance. In contrast, Darwinian reflection and resampling methods are the most recommended, as they significantly improve performance on boundary-constrained problems. Furthermore, it is shown that projection repair methods, although fast and straightforward, can harm the performance of CMA-ES. This research concludes by mentioning that future research could improve CMA-ES performance by incorporating more advanced boundary constraint management methods.

Table 4 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on constrained optimization.

4.5 Multi-Objective Optimization

Multi-objective optimization is an area of mathematical optimization that seeks to find solutions to problems involving multiple objectives that are generally in conflict with each other [68]. The main difference concerning single-objective optimization is that, where a single optimal solution is sought, no single solution optimizes all objectives simultaneously in multiobjective optimization. Instead, a set of solutions representing optimal trade-offs among the goals, known as a Pareto front, is sought. Metaheuristic algorithms are widely used in multiobjective optimization due to their ability to explore complex and nonlinear search spaces. However, they face several challenges, such as avoiding premature convergence to a small subset of solutions, balancing exploration and exploitation, and handling problems with many objectives (many-objective optimization), among others [69].

To transform the CMA-ES into a multi-objective tool, in 2007, the authors of the CMA-ES published the Multiobjective CMA-ES (MO-CMA-ES) [70] variant, combining

the adaptation of the covariance matrix with multi-objective selection strategies based on the concept of Pareto dominance. The MO-CMA-ES is based on an elitist version of the single-target CMA-ES, which uses $(1+\lambda)$ selection and step size control based on a success rule. Then, this version is extended to the multi-target domain, where a population of individuals is maintained that adapts its search strategy, similar to the elitist CMA-ES, but is subject to multi-target selection. The selection is based on non-dominated ordering, using crowding distance or contributory hypervolume as a secondary criterion, as can be seen in Fig. 14. Multi-objective selection can use the crowding-distance criterion as shown in Eq. 28, or the contributing hypervolume criterion as detailed in Eq. 29. Therefore, the crowding-distance criterion is calculated as shown below:

$$c(a, A') = \sum_{m=1}^M \frac{c_m(a, A')}{f_m^{max} - f_m^{min}} \quad (28)$$

where M is the number of objectives in the optimization problem, A' is the set of non-diminished solutions to which individual a belongs, f_m^{max} is the maximum value of objective function m in set A' , f_m^{min} is the minimum value of objective function m found in A' , and $c_m(a, A')$ is the individual contribution of objective m to the total distance.

The contributing hypervolume criterion is calculated as shown below:

$$\Delta s(a, A') = S_{a_{ref}}(A') - S_{a_{ref}}(A' \setminus \{a\}) \quad (29)$$

where a_{ref} is a reference point in the target space and $s_{a_{ref}}(A')$ is the total hypervolume of set A' , representing the volume of the union of all hyperorthants defined between each point on the Pareto front and the reference point.

The MO-CMA-ES inherits important invariance properties from the original CMA-ES, such as invariance against search space rotations, making it robust against affine transformations of the search space.

The MO-CMA-ES, especially the variant using the contributory hypervolume as a secondary selection criterion (s-MO-CMA), performed superiorly on most test functions compared to NSGA-II and NSDE. In particular, the s-MO-CMA was significantly better on problems where the search space was rotated or not separable, contributing considerably to multi-objective optimization in complex environments.

The MO-CMA-ES algorithm has undergone numerous modifications to enhance its performance in multi-target environments. One of the first, as noted in [71], was based on MO-CMA-ES and introduced a comparison of different

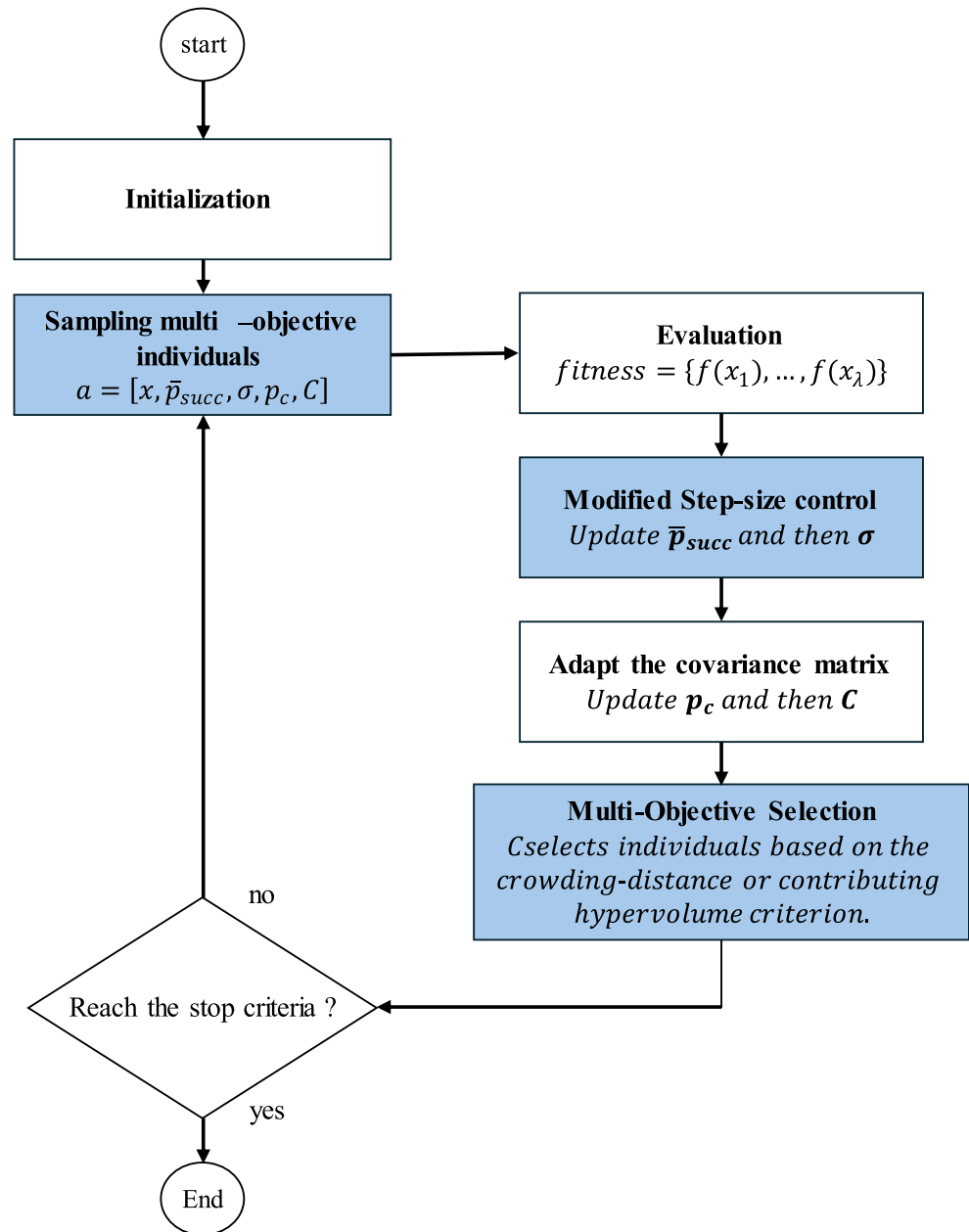
Table 4 Summary of constrained optimization CMA-ES modifications

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
MM-AL-($\mu/\mu_w, \lambda$)-CMA-ES [66]	Constraint handling+Surrogate modeling (constraints)	Train surrogate constraint functions; use them inside AL to reduce constraint calls.	Can reduce expensive constraint evaluations; maintains AL framework.	Surrogate inaccuracies can mislead feasibility search; extra model training cost.
(1+1)-CMA-ES with feasibility-only, active covariance adaptation [61]	Constraint handling (covariance adaptation)	Estimate constraint-normal directions from infeasible steps; reduce variance along violation directions; use only feasibility.	No need objective values for infeasible samples; preserves invariance properties; can closely follow feasibility boundary.	Needs enough constraint hits to learn normals; may degrade with many/complex constraints or disconnected feasible sets.
AL-($\mu/\mu_w, \lambda$)-MSR-CMA-ES [65]	Constraint handling (Augmented Lagrangian)+step-size control	CMA-ES with Median Success Rule (MSR) and adaptive augmented Lagrangian for one linear inequality.	Linear convergence observed on linearly constrained convex/ill-conditioned tests; principled adaptive penalties.	Scope limited (single linear constraint); extension to multiple/nonlinear constraints is nontrivial.
CMA-ES+BCHMs [67]	Bound-constraint handling	Systematic benchmarking of 22 BCHMs on CEC2017 & BBOB; analyzes interactions with restarts.	Provides practical guidance on BCHM choice; broad empirical evidence.	No new CMA-ES variant; conclusions can be benchmark/setting dependent; focuses on box constraints only.
Modified CMA-ES (adaptive penalty+restart) [62]	Constraint handling (penalty)+restart	Adaptive penalty function for constraints; restart increases λ and resets σ /mean on stop criteria.	Improves feasibility handling; restarts help escape stagnation; competitive on benchmarks+engineering cases.	Penalty scaling can distort ranking; additional control rules; restarts may increase evaluation budget.
SCF-CMA-ES (covariance rotation) [60]	Constraint handling (surrogate constraints)	Learn constraint surrogate; rotate mutation ellipsoid near boundary to align with feasible region.	Uses boundary information to improve feasibility; can reduce wasted infeasible samples.	Relies on surrogate quality; may be unstable with noisy/complex constraints.
SCF-CMA-ES (feasibility check) [60]	Constraint handling (surrogate constraints)	Use surrogate to predict feasibility of candidates (screening) before expensive evaluations.	Potentially reduces constraint evaluations; speeds up constrained search.	False positives/negatives harm progress; adds modeling overhead.
SCF-CMA-ES (repair via projection) [60]	Constraint handling (surrogate constraints)	Project infeasible mutations onto estimated boundary using surrogate model.	Can increase feasible-sample rate; effective near smooth boundaries.	Projection may bias search; requires reasonably smooth/learnable constraint surface.
esCMaGES [64]	Constraint handling+sampling strategy	Hybrid ES with ε -constraint ranking, tailored sampling, repair operators, and restart/stagnation checks.	Strong results on real-world constrained benchmarks; practical operators for nonconvex/nonlinear constraints.	Less pure CMA-ES (reduced invariances); several ad-hoc components; performance may depend on constraint geometry.
2-stage memetic CMA-ES+Nelder-Mead Simplex [63]	Constraint handling (penalty)+local search	Stage 1: modified CMA-ES (adaptive penalty+restart); Stage 2: Simplex refinement from best feasible.	Better final accuracy via deterministic local refinement; good for smooth feasible basins.	Local search can get trapped; extra evaluations; sensitive to noisy objectives.
2-stage memetic CMA-ES+BOBYQA [63]	Constraint handling (penalty)+local search	Same Stage 1; Stage 2: BOBYQA derivative-free trust-region refinement.	Strong local refinement for expensive black-box problems; good exploitation.	More parameters/overhead; may fail on highly multimodal landscapes.
2-stage memetic CMA-ES+L-BFGS-B [63]	Constraint handling (penalty)+local search	Same Stage 1; Stage 2: L-BFGS-B (box-constrained quasi-Newton) for refinement.	Fast convergence when gradients are approximable/smooth; efficient near optimum.	Assumes smoothness; limited to box constraints for stage-2; can struggle with nonlinear constraints.

selection schemes, including both generational and steady-state approaches. In addition, a version with a Cholesky update is introduced, which reduces the computation time by avoiding the decomposition of the covariance matrix. Thanks to this research, it was concluded that steady-state algorithms perform better on unimodal problems. In contrast, generational algorithms are superior on multimodal problems and recommend using the $(\mu+1)$ -MO-CMA-ES due to its trade-off between performance and efficiency.

From diverse ideas, the MO-CMA-ES algorithm has evolved into multiple variants to encompass different optimization techniques, such as the $(\mu_{MO}+\mu_{MO})$ -MO-CMA-ES [72], which introduces a covariance matrix update scheme that allows the recombination of information between nearby individuals in the search space. The $(\mu_{MO}+\mu_{MO})$ -MO-CMA-ES approach seeks to accelerate covariance matrix adaptation by incorporating information from multiple genealogies rather than being limited to information

Fig. 14 MO-CMA-ES flowchart



from a single individual and their offspring. This algorithm was evaluated on a diverse set of test functions, including constrained and unconstrained problems as well as rotated functions, where, in all cases, the $(\mu_{MO}+\mu_{MO})$ -MO-CMA-ES outperformed the original MO-CMA-ES, suggesting that the information recombination approach is robust and effective in a variety of optimization scenarios. However, taking into account the good results of $(\mu_{MO}+\mu_{MO})$ -MO-CMA-ES, the same authors developed years later a variant that improves MO-CMA-ES from a different approach, enhancing the adaptation of step size at the individual level, proposing a new criterion to determine if a mutation is successful [73]. In the original version of MO-CMA-ES, a mutation was

considered successful if the offspring was better than its parent in terms of Pareto dominance. However, in this proposal, a mutation is considered successful if the offspring are selected for the next parent population. This criterion is easier to implement and reduces computational complexity, especially in the steady-state variant of MO-CMA-ES. This step-size adaptation scheme improves the convergence speed of MO-CMA-ES. It reduces the risk of convergence to suboptimal local optima, where experiments show that this approach is efficient on multimodal functions, whereas the original MO-CMA-ES tends to get trapped in local optima.

Krimpmann et al. [74] proposed a new evolutionary strategy for multi-objective optimization called

active- $(\mu+\lambda)$ -MO-CMA-ES. This strategy focuses on optimization problems with a limited budget for fitness function evaluations, as is often the case in hardware optimizations, where time and resources are scarce. The proposed algorithm extends existing approaches by utilizing both successful and failed mutations to adjust the covariance matrix, thereby enabling faster convergence to optimal solutions. The active- $(\mu+\lambda)$ -MO-CMA-ES shows faster convergence than most of its comparative competitors, especially on problems with a limited budget of fitness evaluations. Moreover, on tests with more evaluations (50,000), the active- $(\mu+\lambda)$ -MO-CMA-ES performs similarly to NSGA-II, demonstrating that it is competitive even on long-term optimizations.

In [75], it is possible to analyze a variant of the multi-objective CMA-ES called “Population-based MO-CMA-ES.” This variant incorporates a population associated with each Gaussian distribution, driving the optimization and representing a significant improvement over previous versions of MO-CMA-ES, which used a population size of one. This population-based variant proved more robust than earlier versions of MO-CMA-ES, especially in constrained problems. In unconstrained problems, such as BD_2 , this variant could find both ends of the Pareto front, while previous versions only found one of the ends.

Different MO-CMA-ES methodologies have been developed to improve their performance from various perspectives. However, the study conducted in [76] shows the impact of other strategies to increase the population size in the algorithm, determining which modulation methodology gives us the best benefits. The MO-CMA-ES requires an initial warm-up phase to adapt its internal parameters, which may affect its performance in the first generations. This research proposes and evaluates two static and three dynamic strategies to increase the population size in MO-CMA-ES. The static strategy increases the population size at fixed intervals, while the dynamic strategies adjust the population size based on the algorithm’s behavior during optimization. The study determined that dynamic strategies are more effective than static and fixed population strategies in most test problems. However, in highly multimodal problems, a dynamic strategy may not be suitable, as it tends to converge quickly to poor local optima.

A framework called Sofomore for building multi-objective optimization algorithms from single-objective optimization algorithms is presented in [77], which addresses the problem of finding p solutions in an n -dimensional search space by maximizing an indicator through dynamic optimization of subspaces. This strategy is instantiated in the CMA-ES, resulting in the Comma-Selection Multiobjective CMA-ES (COMO-CMA-ES) algorithm. The COMO-CMA-ES exhibits linear convergence to the optimal

distribution of p solutions in convex-quadratic bi-objective problems. This was experimentally demonstrated on several test functions, including sphere, ellipsoid, and cigtab problems. COMO-CMA-ES outperforms or equals the performance of MO-CMA-ES, NSGA-II, and SMS-EMOA regarding the convergence gap. In particular, COMO-CMA-ES shows faster and more robust convergence on problems with independent rotations of Hessian matrices.

Among the most recent variants of MO-CMA-ES, it is possible to find the Scalable Small Subpopulations CMA-ES algorithm (S^3 -CMA-ES) [78], which is dedicated to solving large-scale multi-objective optimization problems (LSMaOPs). The S^3 -CMA-ES algorithm uses a series of small subpopulations, each converging to a solution instead of a single large population. The comparison of S^3 -CMA-ES was carried out together with five representative algorithms (LMEA, MOEA/DVA, NSGA-III, RVEA, and BiGE) on 36 test instances with 5 to 15 objectives and 500 to 1500 decision variables. Experimental results showed that S^3 -CMA-ES outperforms the other algorithms in terms of convergence and diversity in most cases.

In [79], the MO-CMA-ES is taken as a basis, adapting it to maximize the hypervolume covered by all evaluated points without limiting the population size. This improved version of the CMA-ES, named Unbounded Population Multi-Objective CMA-ES (UP-MO-CMA-ES), combines a parallel exploration strategy and a recombination operator that adapts the covariance matrices to align them with the Pareto front, plus the parent selection is based on the contribution to the dominated hypervolume, allowing a more effective exploration of the search space. Although the initial parallel exploration strategy allowed finding better values than the benchmark on weakly structured problems, UP-MO-CMA-ES showed inferior performance on multimodal problems, such as Rastrigin and Schaffer-F7 functions, especially in higher dimensions.

Given the remarkable effectiveness of the MO-CMA-ES model for multiobjective space optimization, this variant continues to receive modifications and improvements to this day, one of the most recent being the variant called COMO-lq-CMA-ES [80], which uses a linear-quadratic surrogate model to approximate the Uncongested Hypervolume Improvement (UHVI), which is a dynamic function that changes with the set of non-dominated solutions, being a key metric in multiobjective optimization. The proposed algorithm handles this dynamic by recalculating and adjusting the surrogate model at each iteration. The COMO-lq-CMA-ES algorithm shows linear convergence on the dual-sphere function, with a convergence rate 6 times faster than COMO-CMA-ES with a predetermined population size and 20 times faster with a population size 4 times larger. Moreover, compared to the NSGA-III algorithm,

COMO-lq-CMA-ES shows superior performance, especially on higher-dimensional problems with a larger number of evaluations.

Table 5 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on multi-objective optimization.

4.6 Mixed-Variable Optimization

Mixed Variable Optimization refers to optimization problems in which the decision variables can differ, such as continuous, discrete, categorical, or even binary [81]. This type of problem is common in real-world applications, where systems or processes involve multiple interacting variables of different types. Metaheuristic algorithms face several challenges when addressing problems with mixed variables, such as the heterogeneity of the search space, where variables of other types require specific search strategies [82]. For example, continuous variables can be optimized using gradient methods, while categorical variables require combinatorial approaches. In addition, variables of different types may interact in complex ways, making it challenging to explore the search space [83]. For example, a change in one variable can significantly affect the performance of continuous variables.

To design a strategy that would be able to tackle mixed variable optimization problems, the author of the CMA-ES, Nikolaus Hansen, proposes a relatively straightforward adaptation of the CMA-ES to handle integer variables in optimization problems, where some components of the solution vector can only take integer values [84]. This proposal describes two key modifications to adapt CMA-ES to problems with mixed variables: The Integer Mutation and the Integer Variable Mask, as can be seen in Fig. 15. The integer mutation occurs when the variance of an integer variable is too slight; an additional mutation is introduced into that variable. This mutation is used to update the mean of the distribution. However, it is not considered in the update of the covariance matrix or the step size. At the same time, the Integer Variable Mask is used for integer variables with slight variance from the global update of the step size, which avoids random fluctuations in the step size. These simple improvements proved to be quite effective, as they significantly increase the probability of finding the optimal solution in problems where some variables are integers. For instance, in certain cases, the probability of convergence to the optimum increased from less than 1% to 100%.

In [85] presents an investigation on solving satisfiability problems in fuzzy logics (SAT_∞) using an extension of the CMA-ES algorithm called M-CMA-ES (Mixing CMA-ES). The satisfiability problem in fuzzy logic (SAT_∞) is an extension of the classical satisfiability problem (SAT) in

propositional logic, where variables can take truth values in the continuous interval $[0, 1]$. The M-CMA-ES proposes an optimization-based approach to solving SAT_∞ , using an objective function that measures the degree of satisfiability of a set of fuzzy logic formulas. The M-CMA-ES is an extension of CMA-ES that incorporates a population mixing mechanism to improve performance on problems with large plateaus. Tests show that M-CMA-ES outperforms standard CMA-ES on problems with high granularity (L100) but not on problems with low granularity (L4). Using multiple independent populations also improves performance, but the blending mechanism in M-CMA-ES provides an additional benefit.

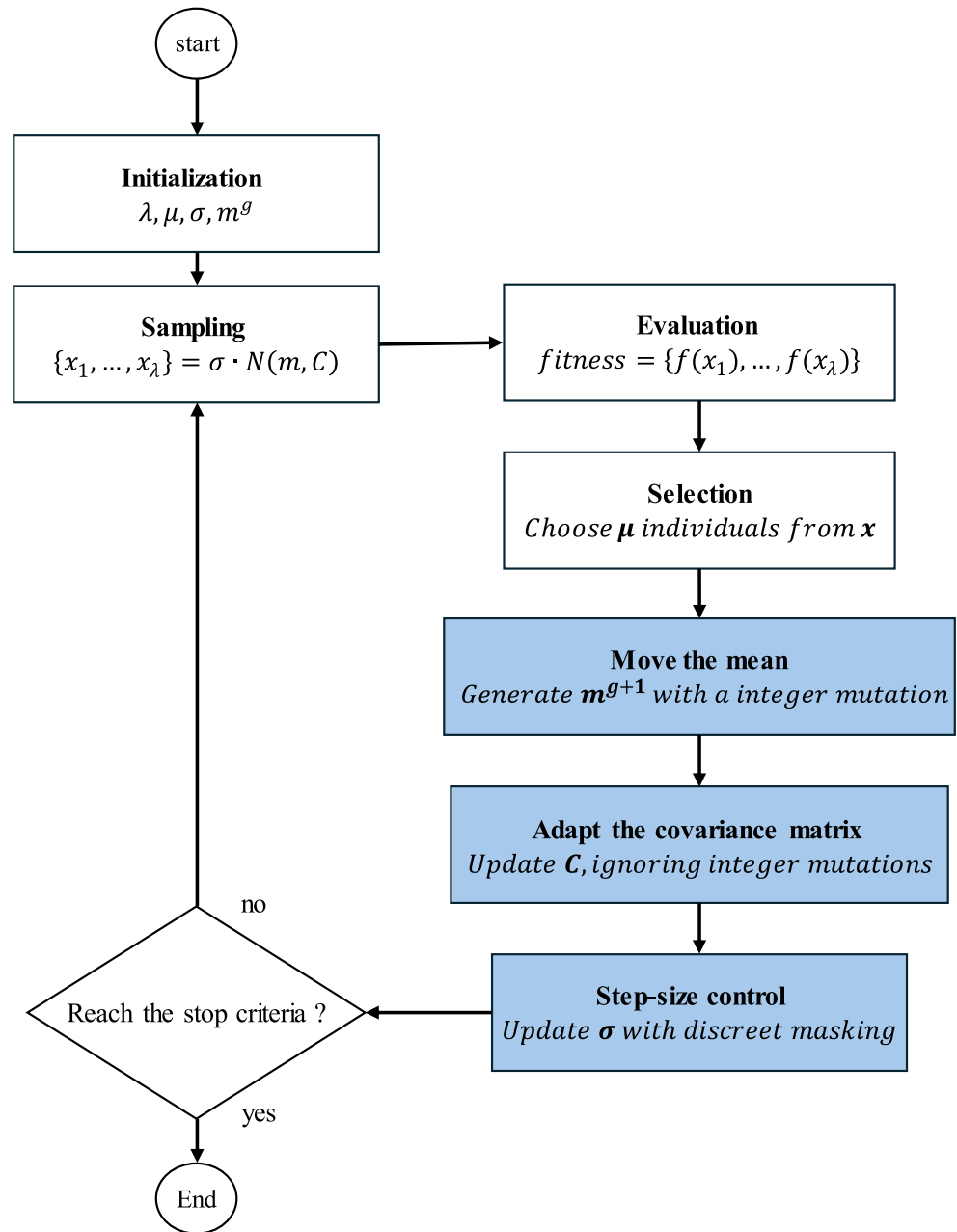
A variant of CMA-ES called CMA-ES for One-Class Constraint Synthesis (CMAESOCCS) [86] was developed as a strategy that synthesizes Mixed Integer Linear Programming (MILP) models from feasible example solutions using the evolutionary CMA-ES strategy. The method is beneficial for single-class classification problems, where only feasible example solutions are available, and infeasible examples are unavailable. This algorithm is compared with the results of an earlier method, ESOCSS, showing that CMAESOCCS produces more accurate and better-performing models, outperforming it on synthetic problems.

Proposals have been developed, such as the one seen in [87], where the optimization problem with mixed variables is approached from a black box perspective since the CMA-ES is an efficient algorithm for continuous optimization, but when applied to MI-BBO problems with a direct discretization of the integer variables, it can suffer from stagnation because the variance of the integer variables is reduced too much before reaching the optimal solution. This proposal defines a simple but effective modification of CMA-ES, called CMA-ES with Margin, which introduces a lower bound on the marginal probability of generating integer values. This modification prevents integer variables from being prematurely fixed to a single value, allowing more effective search space exploration. CMA-ES with Margin is robust and efficient on MI-BBO problems with increasing dimensions, maintaining high success rates and requiring fewer evaluations than the compared methods.

One of the most recent modifications of CMA-ES to operate with various types of variables is the CMA-ES on Sets of Points (CMA-ES-SoP) [88], which is a variant designed to address problems where acceptable solutions are selected from disjoint subspaces containing possible points. The CMA-ES-SoP focuses on optimizing problems where solutions are chosen from sets of points rather than continuous or integer spaces, as the traditional CMA-ES is not suitable for these problems due to its tendency to converge to sub-optimal points prematurely. Through various tests on mixed optimization problems, the CMA-ES-SoP outperformed the

Table 5 Summary of multi-objective CMA-ES modifications

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
Improved-Step MO-CMA-ES [73]	Step-size (σ)	MO-specific success criterion for σ adaptation.	More stable σ in MO runs; preserves MO-CMA-ES structure.	Depends on selection/indicator; extra bookkeeping.
Recombination MO-CMA-ES [72]	Strategy-parameter learning (C , σ)	Lateral transfer/recombination of successful strategy steps across neighbors.	Faster learning via continuity; better sample efficiency.	Negative transfer possible; more hyperparameters.
Elitist CMA-ES (+ selection) [70]	Selection + step-size (σ)	+ selection with success-rule σ control.	Often faster on unimodal problems; simple elitist scheme.	Higher risk of premature convergence.
MO-CMA-ES [70]	MO selection/archiving + per-individual adaptation	Population of individuals each learning its Gaussian mutation distribution under MO selection.	Strong invariance; competitive MO performance.	Can be costly; sensitive to archive/selection design.
COMO-CMA-ES (Sofomore) [77]	Indicator (UHVI) + multi-kernel population	Multiple CMA-ES kernels optimize Uncrowded Hypervolume Improvement.	Modular MO construction; good convergence-diversity trade-off.	Kernel/indicator overhead; scaling to many objectives harder.
lq-CMA-ES (kernel) [80]	Surrogate-assisted selection/evaluation	Linear-quadratic surrogate replaces part of expensive evaluations.	Fewer true evaluations; faster early progress.	Surrogate bias risk; training overhead.
COMO-lq-CMA-ES [80]	UHVI + surrogate-assisted kernels	COMO-CMA-ES enhanced with linear-quadratic surrogate in UHVI optimization.	Higher evaluation efficiency; keeps UHVI diversity pressure.	Complex pipeline; sensitive to surrogate quality.
UP-MO-CMA-ES [79]	Population model + selection	Unbounded parent population + strategy-parameter recombination.	High-precision front filling; avoids fixed- $\hat{\mu}$ limits.	Memory grows; runtime rises with archive size.
Active MO-CMA-ES [74]	Covariance update (C)	Active covariance adaptation uses information from unsuccessful steps.	Better learning under tight budgets.	May destabilize C if noisy; tuning needed.
S3-CMA-ES [78]	Population structure + diversity	Scalable small subpopulations + diversity-improvement strategy.	Scales to large n and many objectives; better diversity.	Coordination choices; weaker global covariance coupling.
MO-CMA-ES with increasing μ/λ [76]	Population size schedule (μ , λ)	Static/dynamic (incl. success-driven) population-size growth strategies.	Improves coverage/robustness; easy wrapper.	Higher evals per generation; schedule is problem-dependent.
Population-based MO-CMA-ES [75]	Population/selection + constraint handling	Clustered MO population with SO-pressure clusters; evaluates multiple constraint-handling schemes.	More robust than size-one MO-CMA-ES; maintains extremes.	Cluster mean shifts can be erratic; more complexity.
MO-CMA-ES ($\mu+\mu$) [71]	Selection (generational) + MO ranking/indicator	Generational MO-CMA: each of μ parents generates 1 offspring; select best μ from 2μ via non-dominated sorting + hypervolume contribution.	Strong on multi-modal problems vs greedy steady-state; standard baseline for MO-CMA-ES variants.	More selection/indicator work per generation; can be slower on unimodal cases than steady-state.
MO-CMA-ES ($\mu+\mu$)chol [71]	Covariance adaptation/sampling (Cholesky factor A)	Store/update Cholesky factor A directly (rank-one Cholesky update) to avoid per-generation covariance decomposition.	Reduces rank-one covariance adaptation cost from $O(n^3)$ to $O(n^2)$; simpler implementation without hidden decompositions.	Trade-off: slower adaptation rates in terms of generations; may degrade when fast metric learning is needed.
MO-CMA-ES ($\mu+1$) [71]	Selection (steady-state, $\lambda=1$)	Steady-state MO-CMA: generate 1 offspring; parent chosen uniformly from population; offspring replaces worst if better.	Recommended by authors as robust; not significantly worse than generational while benefiting on unimodal problems.	Can be inferior to generational variants on multi-modal landscapes.
MO-CMA-ES ($\mu<+1$) [71]	Selection (greedy steady-state; ndom-based parent choice)	Steady-state with parent drawn from non-dominated set.	Often best on unimodal problems (fast convergence).	Very poor on multi-modal problems; explicitly not recommended as alternative despite unimodal gains.

Fig. 15 CMA-ES for mixed-integer optimization flowchart

CMA-ES, especially on high dimensionality problems, and proved effective in optimizing test functions on discrete point sets.

Table 6 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on mixed-variable optimization.

4.7 Parallel Optimization

Parallel optimization is a technique that seeks to accelerate the process of solving optimization problems by simultaneously executing multiple tasks or solution evaluations [89]. This is achieved using parallel computing architectures,

such as multicore processors, clusters, or graphics processing units. In the context of metaheuristic algorithms, parallel optimization allows exploring the search space more efficiently, reducing execution time and improving the quality of solutions [90]. Although parallel optimization offers excellent advantages, it presents significant challenges, such as interprocessor communication, process load balancing, and implementation complexity in different environments.

Müller et al. [91] presented a modification of CMA-ES intending to improve its performance in sequential and parallel environments, especially when using large populations. The original CMA-ES is known for its ability to adapt the covariance matrix in small populations, but its performance

Table 6 Summary of mixed-variable optimization CMA-ES modifications

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
CMA-ES with Margin [87]	Sampling/encoding + mean update (discretization-aware affine transform)	Applies a margin-based affine transform so each discrete variable keeps a minimum marginal; extends to integer variables.	Mitigates premature convergence on binary/integer vars; retains CMA-ES affine invariance; strong robustness on mixed-integer benchmarks.	Adds margin hyperparameters/extra steps; may distort search if margin is too aggressive; representation assumes meaningful marginal probabilities.
CMA-ES-SoP [88]	Sampling/encoding + adaptation of discrete-variable marginals (margin correction/adaptation)	Adds sample encoding for sets-of-points, margin correction, and adaptive target-margin update to avoid premature convergence in discrete choices.	Enables CMA-ES on set-structured/discrete+continuous problems; adaptive margin reduces premature convergence vs naive CMA-ES.	Requires problem-specific encoding; extra hyperparameters/overhead; performance depends on how well the encoding captures neighborhood structure.
Mixing CMA-ES (M-CMA-ES) [85]	Offspring generation/recombination (mixing operator across runs)	Introduces a mixing operator that exchanges/combines variable values between elite solutions to escape plateaus in fuzzy SAT landscapes.	Better plateau escape than plain CMA-ES on SAT ∞ ; leverages CMA-ES local search while adding global jumps via mixing.	Usually needs multiple instances and tuning of mixing frequency; mixing can be disruptive when variables are strongly dependent.
CMAES-OCCS [86]	Fitness design/constraint handling (CMA-ES used as optimizer; core update unchanged)	Uses CMA-ES to learn linear constraints (LP/MILP) from feasible examples, with x-means partitioning and constraint selection to build a compact model.	Turns constraint/model acquisition into black-box optimization; can synthesize interpretable linear constraints from data.	Not a structural CMA-ES variant; depends on data coverage & clustering quality; scalability can be challenging for many variables/constraints.
Mixed-integer CMA-ES (integermutation + masking) [84]	Sampling + CSA step-size adaptation (masking) + mean update	Rounds/truncates integer coordinates for evaluation; when variance is below granularity, injects integer mutation and masks those coordinates in step-size adaptation.	Simple retrofit to handle granular/integer vars; prevents stagnation when variance collapses in integer coordinates; minimal changes to CMA-ES.	Integer mutation is independent (limited covariance learning in discrete space); effectiveness depends on variance thresholds and granularity choice.

degrades when the population size is increased. The proposed modification, called Hybr-CMA, accelerates covariance matrix adaptation by taking advantage of the increased amount of information available in large populations. The modification makes it possible to make better use of the parallelization of the algorithm, especially when a large number of processors are available. In parallel environments, the Hybr-CMA modification shows a remarkable improvement in the number of generations needed to converge, especially when the population size is proportional to the problem size.

Another proposal created to operate effectively under a parallel processing scheme is the one presented in [92], where several parallel versions of the CMA-ES algorithm are shown, including a basic version called BasicpCMAES and a version optimized to reduce communication called LowCommpCMAES. The LowCommpCMAES algorithm minimizes the amount of data that needs to be communicated between processors in each generation, which improves performance in parallel environments, showing a significant improvement in execution time compared to the serial and basic parallel versions, especially on large-dimension problems. In addition, this method also shows performance improvement when more processing nodes are used, although the speedup is not linear.

Table 7 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on parallel optimization.

4.8 Performance Improvements

So far, variants of the CMA-ES designed to attack a specific problem have been analyzed; however, there are modifications to improve the algorithm's performance in general or special environments. Under this scheme, we find that one of the first modifications to the CMA-ES is the LS-CMA-ES algorithm [93], which uses a second-order approximation to adapt the covariance matrix in the Gaussian mutation, combining the covariance matrix adaptation (CMA) with the least squares (LS) approximation to estimate the local curvature of the objective function, as can be seen in Fig. 16. This modification allows a more accurate adaptation of the covariance matrix in parameter optimization problems. CMA-ES updates the covariance matrix through rank-one and rank- μ corrections derived solely from successful step directions, effectively using only local first-order information. In contrast, LS-CMA-ES reconstructs a local quadratic model by solving the least-squares problem, using approximately n^2 archived points (X) to estimate both gradient ($\hat{g}$) and Hessian ($\hat{H}$). Moreover, the algorithm introduces a dynamic quality criterion Q , given in Eq. 30, which

Table 7 Summary of parallel CMA-ES modifications

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
NaivepC-MAES [92]	Parallelization (multi-start/independent runs)	Run multiple serial CMA-ES instances in parallel with different initial guesses; communicate only the best result from each processor at the end.	Embarrassingly parallel; minimal synchronization; improves robustness via multi-start.	Does not speed up a single CMA-ES run; redundant work if processors converge to the same basin.
BasicpC-MAES [92]	Parallelization (objective evaluation per generation)	Split λ evaluations across P processors each generation; communicate all (v, q) pairs to every processor so all perform the same selection and CMA-ES updates.	Reduces wall-time when the objective is expensive; preserves the standard CMA-ES update logic.	High communication overhead each generation; scalability limited by exchanging all candidates.
LowCompCMAES [92]	Parallelization (communication-efficient selection)	Each processor locally sorts candidates and sends only its top $(\mu/P + y)$ candidates (instead of all λ/P), aiming to cover the global top- μ while reducing communication.	Cuts communication significantly; improves runtime over Serial/Naive/Basic variants; best runtime in their study.	Approximate global selection (may miss true top- μ); requires choosing y ; speedup not necessarily linear with more nodes.
New-CMA ($\alpha_{\text{cov}}=0$) [91]	Covariance matrix adaptation (C update)	Replace the rank-one-only update with a blend of rank-one path term and a rank- μ term Z built from selected z -vectors. Increase c_{cov} for faster learning in large populations.	Uses more information from large λ/μ per generation; faster covariance adaptation (fewer generations) when many processors are available.	Introduces α_{cov} choice; may trade off stability vs speed if mis-set; per-generation eigendecomposition cost remains.
Hybr-CMA ($\alpha_{\text{cov}}=1/\mu$) [91]	Covariance matrix adaptation (C update)	Same modified covariance update as above, with α_{cov} set to $1/\mu$ to interpolate between Orig-CMA and New-CMA for reliability vs adaptation speed.	Practical compromise between rank-one (path) and rank- μ information; targets improved serial & parallel performance for large populations.	Still requires selecting α_{cov} (function of μ); benefits depend on using sufficiently large populations.

determines whether the second-order approximation is reliable; if not, the algorithm automatically reverts to the standard CMA update.

$$Q = \frac{1}{N} \sum_{k=1}^N \left(\frac{f(X_k) - f(X_0) - (X_k - X_0)^T \hat{g} - \frac{1}{2} (X_k - X_0)^T \hat{H} (X_k - X_0)}{f(X_k) - f(X_0) - (X_k - X_0)^T \hat{g}} \right)^2 \quad (30)$$

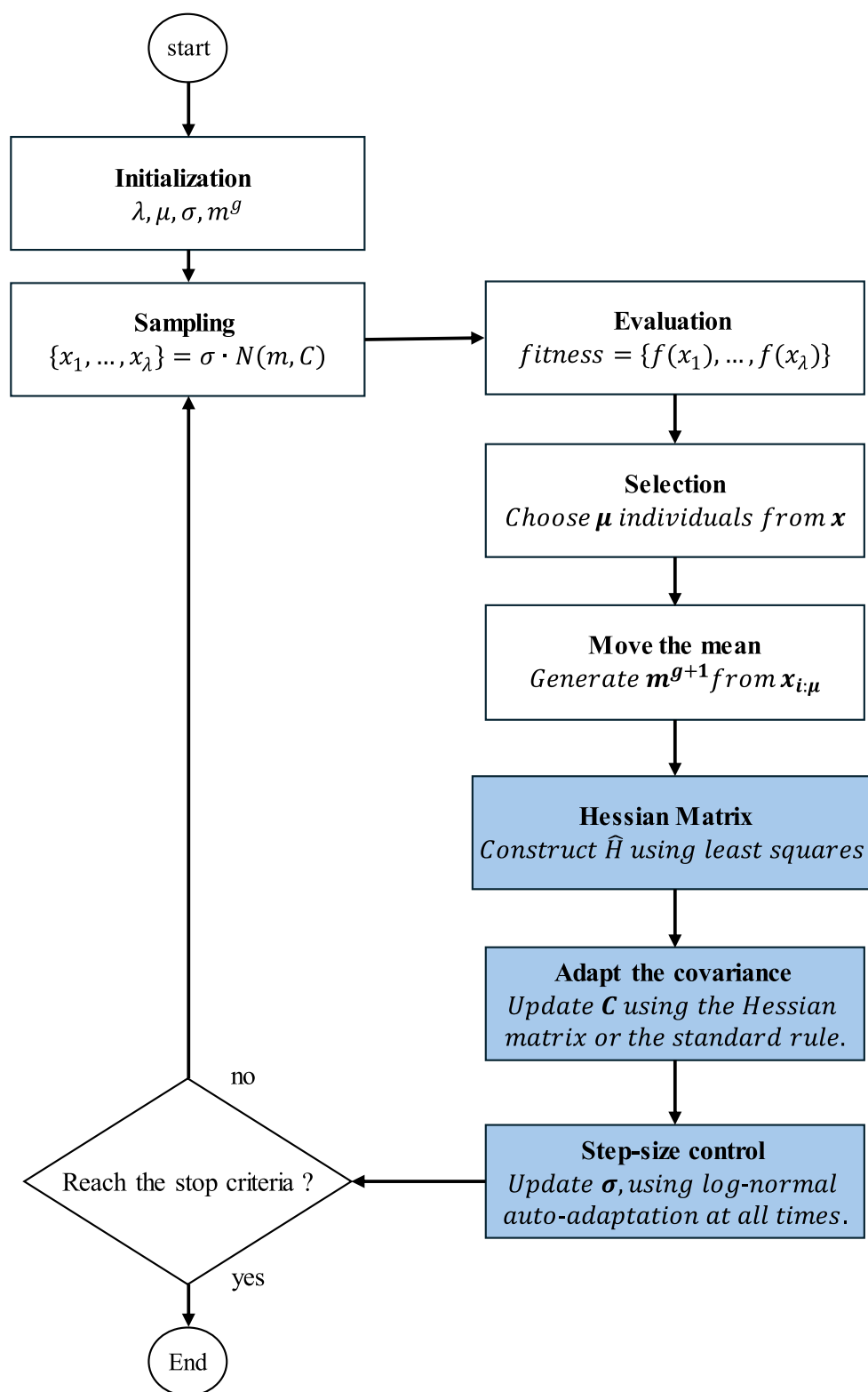
where if Q is small, the model is reliable and the algorithm enters LS mode, and if Q is large, the model is unreliable and returns to CMA mode. Experimental evaluations show that LS-CMA-ES consistently outperforms the classical CMA-ES on elliptic and quasi-quadratic functions, where the local curvature structure is well captured by the least-squares Hessian approximation. In these settings, LS-CMA-ES converges markedly faster, often requiring only a fraction of the evaluations needed by the original algorithm, and exhibiting a notably steeper descent during the early stages of optimization.

The algorithm presented in [94] the Active-CMA-ES, shows a different perspective of implementing the original algorithm since its main mechanism is to use information from successful candidates and unsuccessful ones to adapt the covariance matrix actively. This strategy seeks to actively reduce variances in search directions that have proven to be unpromising rather than allowing this information to decay over time passively. When subjected to performance testing, Active-CMA-ES consistently outperforms Original-CMA-ES and Hybrid-CMA-ES in all test functions, especially those with an eigenvalue spectrum dominated by a much

larger value than the others. Furthermore, in some functions, Active-CMA-ES reduces the number of generations required to reach the target value in certain target functions by more than 40% compared to Hybrid-CMA-ES. In addition to Active-CMA-ES, more modifications have been proposed that actively adapt the covariance matrix, as does the algorithm proposed in [95], which is based on the (1+1)-CMA-ES variant. At first glance this proposal may look very similar to Active CMA-ES; however, this variant of (1+1)-CMA-ES avoids expensive matrix decompositions and maintains quadratic complexity, while Active-CMA-ES introduces a β parameter to control active adaptation in large populations, in addition to having completely different contexts in population management, since Active-CMA-ES was designed for implementation with large populations, while the (1+1)-CMA-ES variant is intended to work in more individualistic environments.

In [96], an improved niching technique for the CMA-ES is presented. The main approach is the introduction of the Mahalanobis metric into the niching mechanism instead of the traditional Euclidean metric. The Mahalanobis metric allows a more accurate spatial classification based on the ellipsoids of the distribution, which is particularly useful in high-dimensional landscapes with non-isotropic basins of attraction. This approach is compared to two existing niching frameworks: one with a fixed niche radius and one with a self-adaptive niche radius based on coupling to step size. Studies on variant performance on different types of problems show that although the Mahalanobis metric does not

Fig. 16 LS-CMA-ES flowchart



significantly improve the localization of the global optimum in all cases, it does improve the quality of the sub-optimal found, which is crucial in multimodal problems. However, the niching technique results in an interesting proposal, so

its implementation in the CMA-ES has generated multiple variants, including the one in [97], where premature convergence in multimodal environments is addressed. This research proposes a niche technique based on Nearest-Better

Clustering (NBC). This technique allows the identification and exploration of multiple promising regions of the search space in parallel, avoiding premature convergence to a single local optimum by combining the CMA-ES with a high-level control mechanism that manages the exploration of multiple niches identified by the NBC. When tested, this variant of the CMA-ES shows fast progress at search initiation and competitive performance on multimodal features such as BBOB set features 17, 18, 21, 22, and 23.

It is possible to find another variant of CMA-ES by applying a niche technique called Attraction Basin Sphere Estimation (ABSE), where the main goal is to improve the ability of CMA-ES to find multiple solutions to multimodal problems [98]. This approach, called ABSE-CMA-ES, uses ABSE to estimate the regions of attraction (attraction spheres) around the optima, which allows the algorithm to maintain diversity and avoid premature convergence to a single optimum. Although ABSE-CMA-ES requires more evaluations of the objective function due to the estimation of the spheres of attraction, its overall efficiency is comparable or superior to other niche algorithms.

Stagnation at local minima is one of the significant challenges facing the CMA-ES due to the high exploitation rate that characterizes it. However, several methodologies have been developed to explore CMA-ES within a search space, such as the CMA-ES-DIG (CMA-ES with Directed Individual Generation) variant [99]. The CMA-ES-DIG algorithm incorporates a directional feature in the generation of individuals, using information from previous generations to retain the influence of the last point recombination. This allows for better exploration of the search space, combining the efficiency of CMA-ES with increased exploration capability. In addition, an adaptive scaling factor is introduced for the mutation operator, which varies with generations, improving diversity and exploration in the early stages of the algorithm. The CMA-ES-DIG algorithm was tested on 28 IEEE CEC 2013 benchmark functions in 10, 30, and 50 dimensions. The results showed that CMA-ES-DIG outperformed traditional CMA-ES in most cases, obtaining lower errors and faster convergence.

In [100], a variant of CMA-ES is presented that implements the idea of generating a high-performance individual called a “super-fit” that is then used as a starting point for a RIS-enhanced local search. This approach, called CMA-ES-RIS, seeks to balance exploration and exploitation of the search space, avoiding premature convergence to local optima by introducing a new local search mechanism that combines partial gene inheritance of the best individual with an exploitative search. The CMA-ES-RIS was tested on 28 optimization functions of the CEC 2013 benchmark in dimensions 10, 30, and 50. The results showed that the algorithm can solve all unimodal functions and obtain good

results on multimodal and compositional functions, demonstrate excellent scalability, and maintain good performance as the problem’s dimensionality increases.

A vital part of the correct functioning of an algorithm is the balance in the adjustment of its parameters, so in [101] it is proposed the self-CMA-ES algorithm, which uses a second instance of CMA-ES to dynamically adapt the hyperparameters of the main instance of CMA-ES, to improve its overall performance. The approach is based on maximizing the likelihood of generating the best solutions so far, which allows adjusting the hyperparameters to favor the generation of high-quality individuals. By subjecting self-CMA-ES to various test functions, this algorithm achieves comparable or superior performance to CMA-ES, with a speedup of up to 1.5 times on functions such as Sharp Ridge. This demonstrates that online adaptation of hyperparameters can improve the algorithm’s performance.

Exponential parameterization is a technique that has served as the basis for some optimization algorithms, such as the Exponential Natural Evolution Strategy (xNES) [102]. Its operator has been compatible with developing more complex functions within the CMA-ES, as with the xCMA-ES algorithm [103]. Applying the exponential parameterization within the covariance matrix allows a multiplicative update instead of an additive one. This leads to the xCMA-ES showing a clear advantage in constrained optimization problems, where active updates with negative weights are beneficial. In particular, the algorithm is more stable and converges faster than traditional CMA-ES on problems with multiple constraints.

In [104], a study on parameter tuning of the CMA-ES to improve its performance on expensive CEC’15 optimization problems is presented. This research uses a bilevel optimization approach to find optimal parameter values, both specialized and generalized. The optimized parameters significantly improve the performance of the CMA-ES; in particular, they improve the performance by 81% for the median and 128% for the best result compared to the generalized parameters. However, even with the results obtained by configuring the parameters of the original CMA-ES, a variant emerged with the premise of modeling the distribution of the quality of the solutions and the conditional distribution of the hyperparameter configurations given the solution quality [105]. This TPE-CMA-ES proposal owes its acronym to the Tree Structure Parzen Estimators (TPE). Using TPE, this approach proposes two stages to fit the c_0 , c_1 , and c_μ hyperparameters of CMA-ES. In the first stage, the hyperparameters are fitted to a set of training functions, and in the second stage, the fitted CMA-ES is applied to a set of test functions. In most evaluated test problems, the TPE-CMA-ES converges faster to the optimal solutions than the default CMA-ES and self-CMA-ES. This is evidenced by

a convergence analysis and the number of function evaluations required to reach the termination criterion.

Quasi-random numbers are designed to have a more uniform distribution than traditional random numbers, which can improve efficiency in the search for optimal solutions. Implementing these numbers has given interesting results in implementing optimization algorithms, like the Derandomized Covariance Matrix Adaptation (DCMA) algorithm [106]. The DCMA proposes to replace the pseudo-random numbers in the CMA-ES mutations with quasi-random numbers, which improves the uniformity of the distribution of the search points. This algorithm was tested on the BBOB framework problems, where the DCMA version outperforms the standard CMA-ES in terms of runtime and accuracy on 22 of the 24 functions evaluated, as well as showing a particularly noticeable improvement on multimodal problems, suggesting that quasi-random mutations are especially effective on this type of problem.

Population size modulation is a popular paradigm within optimization due to the good results presented by the algorithms that implement it, influencing researchers to develop a version of the CMA-ES that adapts the population size. The strategy presented in [107] is based on the interpretation of the CMA-ES as a stochastic natural gradient approach in the parameter space of the sampling distribution, where a measure of the accuracy of the natural gradient estimation is introduced, and a mechanism to adapt the population size according to this measure is proposed. The population size is adapted in noisy functions according to the signal-to-noise ratio. This proposed strategy can handle both roughness and uncertainty in the functions, which makes it more robust compared to approaches that require a fixed population size.

A modification of CMA-ES based on a Monte Carlo Tree Search (MCTS) variant for optimizing continuous functions of real parameters can be seen in [108]. The proposed approach, called Hierarchical CMA-ES Solvers (CMA-MCTS), decomposes the search space into smaller regions and uses CMA-ES solvers at each node of the MCTS tree to generate adaptive solutions. The CMA-MCTS algorithm outperforms other MCTS variants and the standard CMA-ES in terms of the quality of the solutions found, especially on multimodal and noisy functions, since, unlike the standard CMA-ES, which requires large populations to be efficient, CMA-MCTS works well with small populations and bounded search spaces.

Trust regions based on information geometry are a statistical concept constructed using the information geometry metric related to the curvature of the parameter space defined by the likelihood function or objective function. This tool is embedded in the CMA-ES, resulting in the Trust-Region Covariance Matrix Adaptation Evolution Strategy

(TR-CMA-ES) [109]. The TR-CMA-ES introduces a bound on the Kullback-Leibler (KL) divergence to define a trust region that constrains the change in the search distribution at each update, helping to avoid premature convergence and improving the algorithm's stability. TR-CMA-ES was evaluated on 8 standard benchmark functions and showed superior performance in most cases compared to CMA-ES, especially regarding the number of evaluations required to reach a target fitness value.

In [110], an investigation into how to improve the performance of the CMA-ES algorithm by evolving selection methods for updating the mean is presented. A hyperheuristic is used to explore the space of selection algorithms that choose individuals from the population to update the mean of the CMA-ES, where each selection algorithm is represented by a genetic programming tree that computes the “desirability” of an individual and a selection method based on that desirability. In several classes of benchmark functions, the modified CMA-ES with the evolved selection functions outperformed the standard CMA-ES. In particular, significant improvements were observed in different function classes where the modified CMA-ES solved instances at least 20% more often than the standard CMA-ES in at least one-dimensionality.

To provide a sound theoretical justification for parameter updates in CMA-ES, a variant based on an approach using Bayesian conjugate distributions, called BCMA-ES, was developed [111]. The proposed approach assumes that the parameters of the multivariate normal distribution follow a conjugate distribution, allowing optimal updates of these parameters at each iteration. The proposed approach assumes that the parameters of the multivariate normal distribution follow a conjugate distribution, which allows optimal updates of these parameters at each iteration. Experiments show that BCMA-ES converges quickly on convex and non-convex functions, such as Cone and Rastrigin, respectively, and, in particular, shows similar or superior performance to standard CMA-ES in terms of convergence.

The Graphical Lasso CMA-ES (gl-CMA-ES) [112] algorithm was developed to improve the efficiency of CMA-ES in optimization problems with sparse Hessian structures, which manages to reduce the covariance matrix adaptation time without compromising the algorithm's performance. The gl-CMA-ES combines the CMA-ES approach with the Graphical Lasso technique, which allows for the estimation of sparse precision matrices through penalization. This is particularly useful in optimization problems where the Hessian matrix of the objective function is sparse, reducing the number of free parameters in the search model and speeding up the fitting of the covariance matrix. In problems with sparse Hessian structures, gl-CMA-ES significantly improves convergence speed compared to standard

CMA-ES. For example, in the Rosenbrock problem, whose Hessian is tridiagonal, gl-CMA-ES is more than 3 times faster in high dimensions.

Akimoto and Hansen proposed in [113] a new perspective of CMA-ES where adaptive diagonal decoding is applied to speed uCMA-ES performance in continuous black-box optimization problems. The proposed algorithm, named dd-CMA, suggests an adaptive damping mechanism to control the learning rate of the diagonal matrix based on the condition number of the underlying correlation matrix, combining the advantages of standard CMA-ES and separable CMA-ES, allowing faster adaptation on problems with separable structures without compromising performance on non-separable problems. In numerical experiments, the dd-CMA-ES algorithm outperforms standard CMA-ES and separable CMA-ES on problems with dimensions up to 5120, also showing significant improvements on problems with poor coordinate conditioning.

In [114], the proposed PPO-CMA method incorporates a CMA-ES-inspired mechanism to adapt and contract the exploration variance. PPO-CMA outperforms PPO and CMA-ES in several Roboschool continuous control environments, showing faster convergence and better final performance, particularly efficient in complex tasks such as controlling a humanoid in the MuJoCo Humanoid-v2 environment.

The Covariance Matrix Adaptation MAP-Elites (CMA-ME) [115] is an algorithm developed to find diverse, high-quality solutions in complex continuous domains. It uses emitters or modified instances of CMA-ES to explore and optimize the search space, maintaining an archive of elite behavior-based solutions. In addition, it has adaptive exploration abilities by dynamically modulating the exploration variance, allowing for more efficient exploration of the behavior space. The CMA-ME finds strategies of higher quality and diversity than MAP-Elites and CMA-ES, demonstrating a greater ability to explore and optimize the behavior space.

Although various modifications to the CMA-ES operators have been studied for higher performance, some modifications with the same objective focus on predicting CMA-ES operator configurations. Research by Friess et al. [116] identifies that CMA-ES operator configurations, such as the step size σ and the covariance matrix C , can act as inductive biases, allowing for more effective algorithm initialization on specific problems. Under this scheme, a method is proposed to predict these operator configurations during the algorithm's execution. It uses a multilayer neural network, allowing it to dynamically adjust the CMA-ES to improve its performance on specific problems. The approach proposed by Friess is most effective when applied

to early generations of the CMA-ES, suggesting that early operator prediction can significantly improve convergence.

Table 8 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection in terms of overall performance improvement.

4.8.1 Step-Size Adaptations

The step size is one of the fundamental parts of the original CMA-ES operation, so, in addition to the simplicity of its operation, it is an operator with the ability to be improved with different techniques. An example of the above-mentioned is the Functional Specialization CMA-ES (FS-CMA-ES) algorithm [117], which proposes to separate the adaptation functions of the covariance matrix and the step size, allowing a better adaptation of the global step size and a reduction in the number of generations needed to converge. In addition, two methods of normalizing the covariance matrix after each update are introduced: normalization by determinant and normalization by trace, as can be seen in Fig. 17. This prevents the CMA from adjusting the overall variance, allowing the SSA to take care of this aspect exclusively.

The experimental results demonstrate that FS-CMA-ES achieves consistently improved performance over the standard CMA-ES, particularly when large population sizes are used. By separating the roles of covariance-matrix shape adaptation and global step-size control, the algorithm exhibits greater stability, more reliable step-size behavior, and faster convergence across a variety of unimodal and multimodal benchmark functions. In high-dimensional or ill-conditioned landscapes, FS-CMA-ES outperforms CMA-ES in terms of both convergence speed and robustness, while maintaining comparable performance on simpler problems.

On the other hand, it is possible to find more variants that operate on the step size, as we can see in [118], where an improved CMA-ES approach is presented by using reinforcement learning (RL) to adjust the step size parameter dynamically. The main objective is to automate the dynamic setting of this parameter, which is traditionally tuned by manual heuristics such as cumulative step size adaptation (CSA). The authors propose to use guided policy search (GPS), an RL technique, to learn policies that adjust the step size more efficiently than existing heuristics. The learned policies outperform CSA on most of the evaluated functions, showing significant improvement in final performance and convergence speed, an example being the Rastrigin function, where the learned policy achieves much better performance than CSA after only 15 training iterations. Although these versions of the CMA-ES achieve excellent results on different types of problems, the strategy presented in [119] shows a modification based on the

Table 8 Summary of performance modifications of CMA-ES

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
gl-CMA-ES [112]	Learns a sparse *precision* via Graphical Lasso; adaptive learning rates depend on sparsity; τ controls regularization.	Covariance/precision matrix adaptation (structure sparsification)	Targets partially separable structure; can reduce model degrees-of-freedom; coincides with CMA-ES when $\tau=0$; keeps monotonic-invariance.	Affine-invariance breaks when $\tau>0$; needs regularization/threshold tuning; extra cost for Graphical Lasso solve.
gl-CMA-ES (weighted penalties) [112]	Extends Graphical Lasso with non-equal/weighted l_1 penalties to bias sparsity pattern learning.	Covariance/precision matrix adaptation (weighted sparsification)	More flexible sparsity induction than uniform penalties; may better match heterogeneous variable couplings.	More hyperparameters (weights); increased risk of misspecified sparsity; extra tuning effort.
self-CMA-ES [101]	Online maximum-likelihood adaptation of CMA-ES hyperparameters (beyond just λ) during the run.	Hyper-parameter control (learning rates, damping, etc.)	Reduces manual tuning; particularly targets non-default (larger) population settings.	Added adaptation logic/overhead; potential instability if ML updates are noisy; harder theoretical analysis.
BCMA-ES (Normal-Wishart prior) [111]	Bayesian update of Gaussian parameters using a conjugate prior on (μ, Σ) with a Normal-Wishart formulation.	Mean and covariance update (Bayesian derivation)	Provides probabilistic justification; yields closed-form updates; reported fast convergence in experiments.	Prior/strength selection required; can add computational overhead; behavior may depend on modeling assumptions.
BCMA-ES (Normal-Inverse-Wishart prior) [111]	Alternative Bayesian update using a conjugate prior parametrized by the *precision* (Σ^{-1}).	Mean and precision update (Bayesian derivation)	Natural fit when working with precision matrices; offers an explicit Bayesian interpretation.	Same prior-sensitivity caveats; additional book-keeping; may be less plug-and-play than standard CMA-ES.
TR-CMA-ES [109]	Re-derives CMA-ES updates under an information-geometric trust-region (KL) constraint; improves step-size update.	Step-size adaptation (and update coefficients via KL bound)	More principled foundation; adaptive coefficients per iteration; improved step-size rule vs standard CSA.	Requires solving trust-region/dual parameters; extra computational steps; introduces trust-region tuning.
dd-CMA (diagonal decoding CMA) [113]	Adds an adaptive diagonal decoding matrix D (DCD form) to accelerate learning of coordinate-wise scaling.	Covariance parameterization/sampling (diagonal acceleration)	Bridges default and separable CMA-ES advantages; improves performance on classes with strong coordinate scaling.	Extra state/updates; benefits may diminish on fully rotated interactions not captured by diagonal scaling.
CMAES-D (default) [104]	Baseline CMA-ES with default parameter settings (used for comparison against tuned settings).	Baseline reference	No tuning overhead; robust default behavior.	Can be far from optimal for expensive problems/population settings.
CMAES-G (generalized tuned) [104]	Single tuned parameter set optimized to perform well across the full expensive-problem suite.	Hyper-parameter tuning	Better average performance than defaults; transferable across the considered suite.	Still problem-dependent; tuning cost needed up front.
CMAES-S (specialized tuned) [104]	Per-function (specialized) tuned parameters, allowing each problem to use its own best settings.	Hyper-parameter tuning	Can yield the strongest per-problem results; highlights sensitivity of CMA-ES parameters.	Poor transferability; requires repeated tuning for new problems.
CMAES-R (replicated tuned best set) [104]	Best tuned parameter set replicated on new starting positions to validate generalization.	Hyper-parameter tuning/validation	Checks robustness of tuned settings; reduces overfitting to starting points.	Still inherits tuning overhead; may not generalize beyond the suite.
Active (1+1)-CMA-ES [95]	Updates mutation distribution using both successful and unsuccessful steps (active covariance adaptation).	Covariance update (using negative information)	Can speed up convergence (reported up to $\sim 2x$ on some tests); keeps invariance properties; $O(n^2)$ like base (1+1)-CMA-ES.	May be less robust on noisy/unstable signals; extra care needed to keep covariance well-conditioned.
LS-CMA-ES [93]	Uses least-squares quadratic approximation of f to drive a second-order covariance update; falls back to CMA when approximation is poor.	Covariance update (second-order/quadratic model)	Can exploit near-quadratic structure; adaptive fallback preserves robustness when model is inaccurate.	Extra evaluations/compute for model fitting; less effective on highly non-quadratic/noisy landscapes.

Table 8 (continued)

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
CMA-ES super-fit scheme (for RIS) [100]	Runs CMA-ES to obtain a high-quality super-fit individual used to seed/boost the Re-sampled Inheritance Search.	Initialization/hybridization (CMA-ES as pre-processing)	Simple hybrid that can boost downstream local search; leverages CMA-ES global search to produce strong seeds.	Extra CMA-ES budget required; improvement depends on available evaluations and problem structure.
Hierarchical CMA-ES (MCTS-structured solvers) [108]	Organizes multiple CMA-ES solvers in a Monte Carlo Tree Search structure to balance exploration/exploitation over regions.	Restart/portfolio orchestration (multi-solver control)	Empirically outperforms standard CMA-ES and an MCTS adaptation on several benchmarks; explicit exploration-exploitation control.	Higher algorithmic complexity and overhead; more design/tuning choices (tree policy, solver allocation).
DCMA (quasi-random CMA-ES sampling) [106]	Replace pseudo-random Gaussian mutations with quasi-random sequences (low-discrepancy) mapped to Gaussian steps.	Sampling/randomness	More regular space-filling sampling; often improves early progress/robustness on BBOB.	Less stochasticity; benefits may be problem-dependent; may break assumptions of standard analyses.
CMA-ME (optimizing emitter) [115]	MAP-Elites archive + multiple CMA-ES-like emitters; optimizing emitter CMA-ES with restarts seeded from archive elites.	Restart/selection (archive seeding), covariance adaptation	Fast local optimization per niche; leverages CMA-ES adaptation while maintaining QD archive.	Needs behavior descriptors + archive; emitter scheduling/parameters add design choices.
CMA-ME (random-direction emitter) [115]	Emitter ranks/updates to push solutions along a chosen behavior-space direction, encouraging coverage.	Selection/ranking rules (covariance update driver)	Strong diversity/coverage when behavior-direction is meaningful.	Requires defining a direction in behavior space; may reduce solution quality vs improvement emitters.
CMA-ME (improvement emitter) [115]	Ranks candidates by archive-improvement (replacing/adding elites), updating distribution to maximize improvement rate.	Selection/ranking rules (covariance update driver)	Good quality-diversity trade-off; improves MAP-Elites coverage and quality.	Still needs archive + descriptors; may be heavier than single-objective CMA-ES.
Meta-initialized CMA-ES (operator/parameter prediction) [116]	Predict CMA-ES operator choices/initial settings as inductive biases for new (related) shape-optimization instances.	Strategy-parameter setting/initialization	Can reduce manual tuning and speed convergence on a target problem family.	Requires training data and generalization may fail outside the trained domain.
PPO-CMA [114]	Inject CMA-ES-style rank- μ and evolution-path-like variance adaptation into PPO.	Covariance/variance adaptation; evolution path heuristic	Mitigates premature variance collapse; improves robustness and reduces hyperparameter sensitivity in tested tasks.	RL-specific; adds buffer size H and architectural choices; not a drop-in for generic black-box optimization.
Active covariance update (active (1+1)-CMA-ES) [94]	Use information from bad solutions via a 'negative' covariance update to reduce variance along detrimental directions (while maintaining SPD).	Covariance matrix update	Faster adaptation of the covariance; can improve progress, especially in ill-conditioned settings.	Needs careful safeguarding to keep covariance valid; can be unstable if misweighted.
xCMA-ES [103]	Replace additive covariance updates with multiplicative (matrix exponential/log) updates to preserve SPD and improve update geometry.	Covariance matrix update	Guaranteed SPD; potentially better-behaved large updates and numerical stability.	Higher computational overhead (matrix log/exp); more complex implementation.
Population-size adaptive CMA-ES (natural-gradient accuracy) [107]	Adapt population size using an estimation-accuracy criterion for the natural gradient.	Population size/sampling budget control	Improves robustness across problem difficulty; can allocate evaluations more efficiently.	Additional estimator/criteria; may introduce sensitivity to meta-parameters of the adaptation rule.
NBC-niching CMA-ES [97]	Detect niches using nearest-better clustering and run/seed CMA-ES per niche to tackle multimodality.	Restart/niching (multi-modal search control)	Handles multiple optima; preserves CMA-ES local efficiency within each niche.	Extra clustering overhead; niche detection may be sensitive in high dimensions/noisy fitness.

Table 8 (continued)

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
Evolved mean-update selection schemes (hyper-heuristic CMA-ES) [110]	Use an evolutionary/hyper-heuristic process to design/select the mean-update selection rule used inside CMA-ES.	Selection for mean update/recombination weights	Can discover selection rules tailored to a target function class; potential performance gains.	Offline training/evolution cost; risk of overfitting and reduced generality.
SA-Niching CMA-ES (SA-CMA) [96]	Self-adaptive niche radius coupled to CMA-ES step-size (replaces fixed niche radius).	Step-size adaptation (σ) + niching/peak maintenance.	Less niche-radius tuning; adapts to basin scale changes.	Euclidean niches; weaker on anisotropic/rotated basins.
Mahalanobis Niching CMA-ES (M-CMA) [96]	Use Mahalanobis distance (via niche covariance) for niche assignment; ellipsoidal/asymmetric niches.	Covariance matrix exploited as metric (C^{-1}).	Better niche separation on anisotropic landscapes; improved robustness vs Euclidean niching.	Extra computation; depends on reliable covariance estimate.
Mahalanobis + SA Niching CMA-ES (M-SA-CMA) [96]	Combine Mahalanobis niche metric with self-adaptive niche radius.	Covariance matrix + step-size adaptation inside niching.	Combines adaptive radius + shape-aware niches; strongest variant in paper.	More moving parts; overhead vs baseline niching CMA-ES.
ABSE-CMA-ES [98]	Combine CMA-ES with ABSE and SIC to estimate attraction-basin spheres for niching.	Niching/multimodal extension around CMA-ES sampling.	Good CEC2013 niching results; avoids fixed niche radius; mitigates ABSE efficiency issues.	Extra evaluations/parameters for ABSE+SIC; scalability may drop with many optima/high D.
TPE-CMA-ES [105]	Offline tune c_σ , c_1 , c_μ using TPE + Expected Improvement; run tuned CMA-ES.	Covariance update hyper-parameters (c_σ , c_1 , c_μ).	Improves convergence/quality vs default CMA-ES and self-CMA-ES on BBOB noiseless.	Two-stage overhead; risk of overfitting; can fail on some functions; harder for large function sets.

Previous Population Midpoint Fitness (PPMF) rule, which is based on the comparison of the fitness values of the current population with the midpoint of the previous population. This approach is inspired by the 1/5 success rule but is adapted to work with multiple offspring, which is typical in CMA-ES. PPMF-CMA.ES shows superior performance to CSA on both sets of benchmarks. In particular, PPMF performs well on low-dimensional problems and maintains competitive performance on higher dimensions, although its performance decreases slightly on problems of dimension 100 compared to CSA.

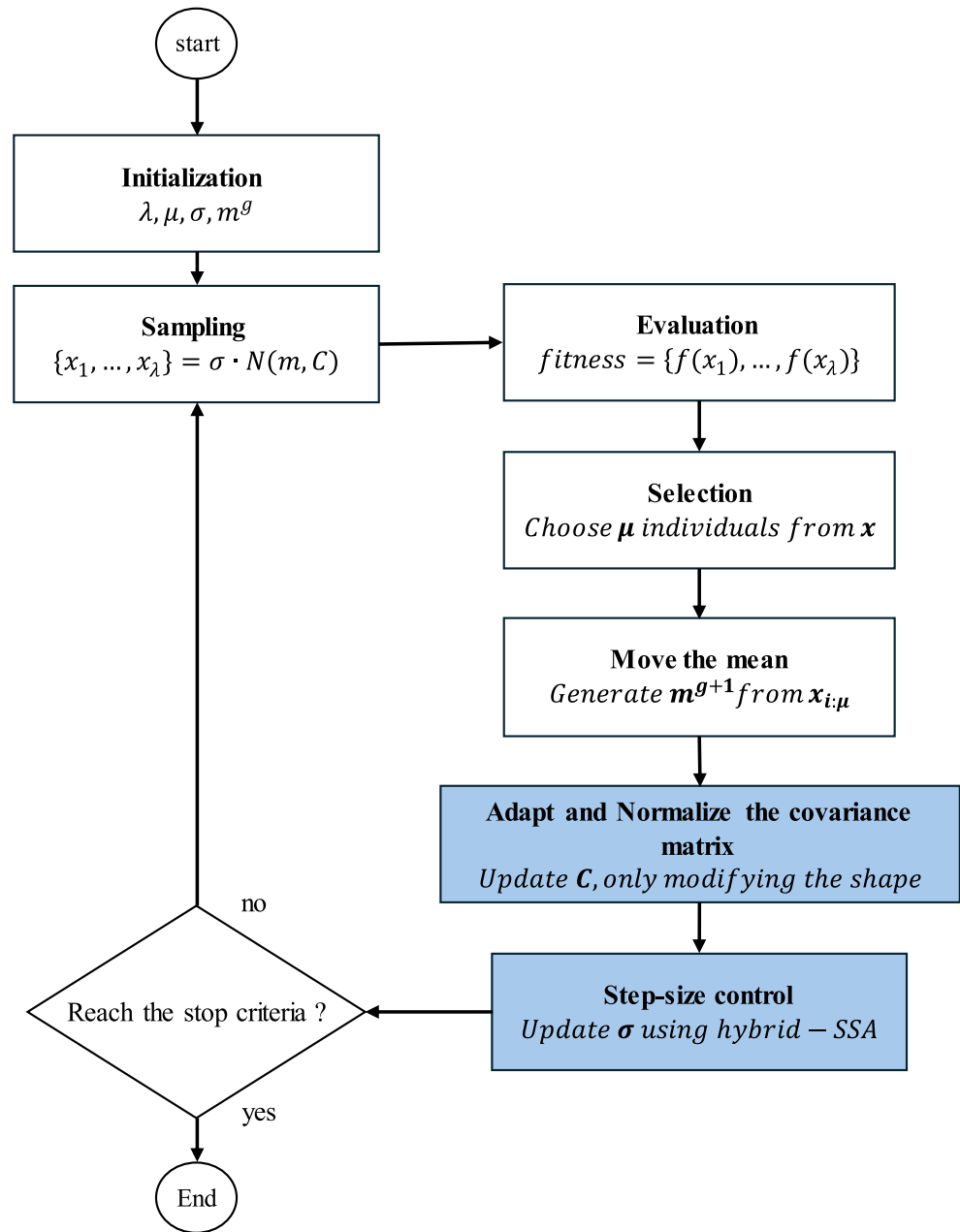
An alternative that modifies the step size more subtly is the Covariance Matrix Self-Adaptation Evolution Strategy [120] (CMSA-ES) algorithm, which combines covariance matrix adaptation with mutational self-adaptation of the step size. The proposed CMSA-ES simplifies the covariance matrix adaptation process and reduces the required strategic parameters, making it more efficient, especially in problems with large populations. CMSA-ES outperforms CMA-ES on problems with large populations, although its performance decreases slightly at higher dimensions due to the increased number of mixing matrix conditions; however, CMSA-ES outperforms with a wide margin over CMA-ES on the Rastrigin function, especially with large population sizes. This suggests that CMSA-ES is more effective in multimodal problems with many local optima.

Table 9 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection in terms of step-size modifications.

4.8.2 Restart Strategies

The most popular modification of the CMA-ES is the IPOP-CMA-ES algorithm [121], which was developed by the same author as the original method and is intended to improve the overall performance of the algorithm itself. IPOP-CMA-ES incorporates a restart mechanism in which the population size is increased, allowing the search to become more global after each restart. The core idea is simple: after each restart, the population size (λ) is multiplied by a constant factor (typically 2), thereby progressively shifting the search behavior from local exploitation toward increasingly global exploration. In each restart, the algorithm uses the default CMA-ES parameterization, sampling (λ) new offspring from a multivariate normal distribution, updating the mean, the covariance matrix, and the step-size through evolution paths and rank-based recombination mechanisms, as can be seen in Fig. 18. In unimodal functions, the performance of IPOP-CMA-ES is similar to that of the local restart strategy, confirming that CMA-ES is effective in this type of problem. However, in multimodal functions, IPOP-CMA-ES significantly outperforms the local restart strategy in 29 out

Fig. 17 FS-CMA-ES flowchart



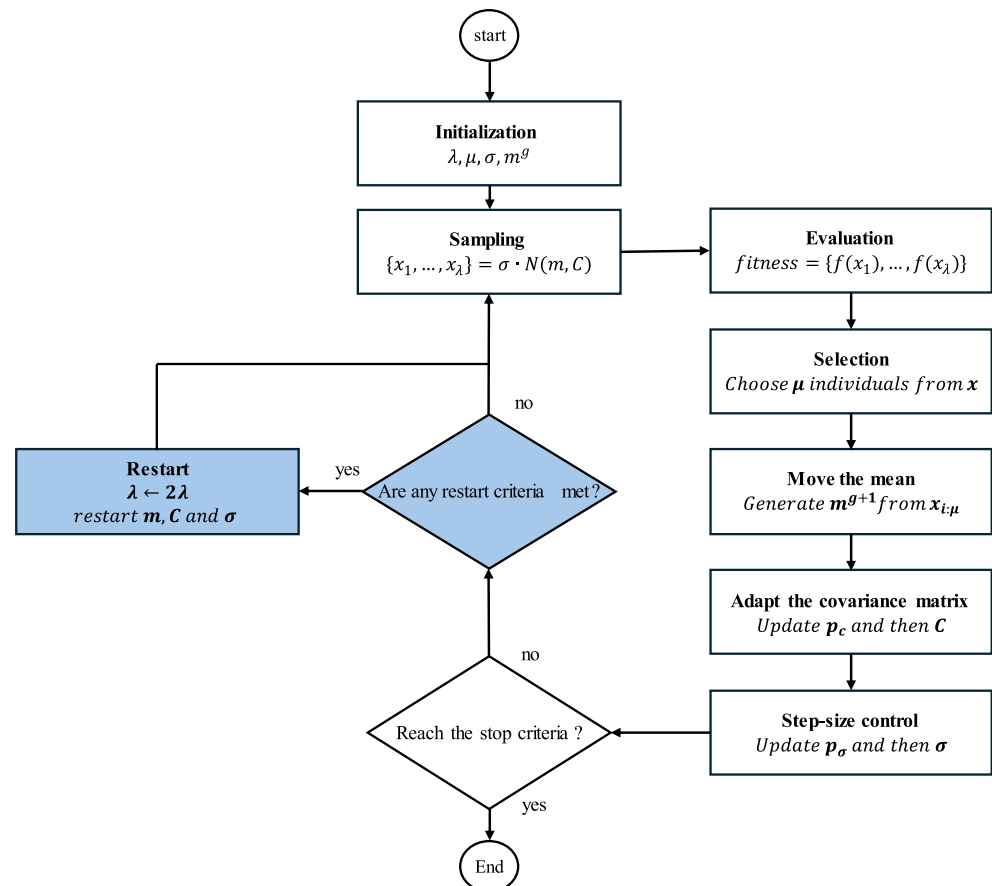
of 60 cases tested, while the local strategy is only better in 4 cases.

However, this variant was upgraded by automatically tuning its parameters, which led to the design of the iCMA-ES-tsc algorithm [122]. The authors of this variant use an automatic setup tool called Iterated F-Race to adjust the iCMA-ES parameters on a set of training functions that do not overlap with the test functions of the CEC'05 set. This is done to avoid over-fitting and obtain a fairer assessment of the improvement potential, showing that the results are stable concerning minor changes in the automatic tuning configuration, reinforcing the validity of the results obtained. Although the IPOP-CMA-ES algorithm achieves highly

superior results to the original CMA-ES, a variant of IPOP-CMA-ES named RB-IPOP-CMA-ES [123] was developed, which achieves even better results than previous versions of this variant. The RB-IPOP-CMA-ES uses the midpoint of the population as an approximation of the optimum and introduces a new restart mechanism based on the evaluation of the fitness of the midpoint. In addition, adjustments are made to the parameters and restart criteria of the original algorithm to improve its performance in multimodal optimization problems. The RB-IPOP-CMA-ES algorithm outperforms the original IPOP-CMA-ES version on most of the problems in the 2017 CEC benchmark, observing a significant improvement in high dimensionality problems.

Table 9 Summary of step-size modifications of CMA-ES

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
LTO-CMA-ES (learned step-size adaptation for CMA-ES) [118]	Replaces the hand-crafted step-size control with a learned policy; other CMA-ES settings kept at default.	Step-size adaptation (σ)	Data-driven σ control; plug-in change (keeps covariance update); can improve performance on tested suites.	Requires training phase/data; generalization risk; added implementation/compute overhead.
FS-CMA-ES (functionally specialized CMA-ES) [117]	Specializes/retunes the roles of covariance adaptation vs. step-size adaptation as a function of population size.	CMA+CSA parameterization (population-size dependent)	Targets better behavior across different population regimes; maintains CMA-ES framework.	Adds design choices tied to λ ; may increase tuning burden; benefit can be problem-dependent.
CMSA-ES [120]	Introduces a simpler covariance adaptation step closer to mutative self-adaptation.	Covariance adaptation (C) (simplified); step-size control (alternative to CSA)	Simpler adaptation/parameter set; reported superiority especially for large population sizes.	May underperform on some landscapes (e.g., Rosenbrock); loses some path-based effects.
CMA-ES+PPMF (Previous Population Midpoint Fitness) step-size rule [119]	Updates σ using comparison between current population fitness and the previous population midpoint fitness (generalized 1/5 success rule).	Step-size adaptation (σ)	Comparable to CSA and superior to MSR in reported benchmarks; uses midpoint as informative reference.	Midpoint-as-optimum proxy can fail on asymmetric landscapes; needs extra bookkeeping (previous midpoint).

Fig. 18 IPOP-CMA-ES flowchart

In [124], a variant method of the CMA-ES is proposed, focusing on a multiple restart scheme with two intertwined population strategies: one with increasing population size and one with varying small population sizes. This scheme, named BI-population CMA-ES, allows balancing global exploration and local exploitation, improving the algorithm's ability to escape local optima and find global

solutions. Within the BBOB-2009 test environment, the BI-population CMA-ES solves 23, 22, and 20 of the 24 functions in dimensions 10, 20, and 40, respectively, showing satisfactory performance on most functions but can be outperformed on smooth, regular, and moderately ill-conditioned functions, as well as on separable and multimodal functions with weak structures.

The CMA-ES has been an algorithm that has inspired not only improved versions of itself, but its operators have gone on to form new algorithms in the literature, such as the Adaptive Coordinate Descent (ACiD) [125] algorithm, which combines Adaptive Encoding (AE), a generalization of the covariance matrix adaptation mechanism used in the CMA-ES algorithm, with an adaptive coordinate descent method. The main idea of this algorithm is to use adaptive encoding to gradually transform the coordinate system so that the objective function becomes as separable as possible. Then, a coordinate descent method is applied to the transformed space. Extensive experiments are performed on the BBOB testbed, comparing ACiD with other CMA-ES variants, such as (1+1)-CMA-ES and IPOP-CMA-ES. ACiD demonstrates superior performance on several functions, up to 2 times faster than (1+1)-CMA-ES on problems such as Rosenbrock, Ellipsoid, and Discus.

The restart strategies are a set of tactics that have proven effective in CMA-ES, which is a remarkable improvement, especially in multimodal optimization functions. In [126], the restart strategy NIPOP-aCMA-ES increases the population size in problems with multiple local optima. It decreases the initial mutation step size at each restart, aiming for a smaller initial step size that allows for better exploration of the local basins of attraction. A larger population size helps explore the function's global structure.

On the other hand, the NBIPOP-aCMA-ES combines the NIPOP-aCMA-ES strategy with a random sampling of the initial step size. The choice between the two strategies is made adaptively, allocating more computational budget to the strategy that obtains better results at each point. These restart strategies improve the results obtained by previous versions of CMA-ES on the interplanetary trajectory optimization problem. The results of these restart methods have been validated on various testbeds such as CEC 2013, concluding that these alternative restart strategies show slightly better performance than the original strategies [127].

Table 10 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on restart strategies.

4.8.3 Warm Starting

Warm starting is a technique within optimization algorithms that consists of initializing an optimization algorithm with previous information obtained from a related task or a previous execution instead of starting from scratch with a random or predetermined initial configuration. This technique has been used in different ways in the implementation of metaheuristic algorithms, where when implemented in the CMA-ES, we find variants such as the CMA-ES algorithm with contextual warm starting (CMA-ES-CWS) [128], which

uses the results of the optimization of the previous context vectors to train the regression of the multivariate Gaussian process or the Warm Starting CMA-ES (WS-CMA-ES) algorithm [129], which introduces a warm starting method that initializes the CMA-ES parameters using the promising distribution of a source task, which accelerates convergence on the target task. In both strategies, it is possible to see a reduction in the time required to find optimal solutions, especially in environments with limited evaluation budgets, in addition to the fact that it allows to take advantage of previous information from related tasks, which is especially useful in practical applications where models are optimized on different datasets or tasks.

Table 11 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on warm starting strategies.

4.8.4 Surrogate Models

The surrogate models are mathematical tools that help mathematically simplify the objective functions, becoming a methodology widely used in metaheuristic algorithms. This concept has already been addressed in several modifications of the CMA-ES algorithm. However, we can find these models with different objectives in their implementation, such as the ^s*ACM-ES algorithm [130, 131], which adjusts online the model lifetime and the surrogate model hyperparameters, allowing the algorithm to automatically adjust how long a surrogate model should be used before being retrained, as well as to optimize the model parameters to improve its accuracy. ^s*ACM-ES shows a 2- to 3-fold speedup compared to CMA-ES, furthermore, on unimodal problems of dimensions ranging from 2 to 40. In multimodal problems, the algorithm also shows improvements, although less significant.

As a widely used technique, several approaches to surrogate models exist, as seen in [132], where a study on using surrogate models based on Gaussian processes (GP) to improve CMA-ES performance is presented. This research proposes two new algorithms: S-CMA-ES and DTS-CMA-ES, of which the DTS-CMA-ES (Doubly-Trained Surrogate CMA-ES) is the primary approach, which uses an evolution control based on the uncertainty of the surrogate model to select points to be evaluated with the original function. In multimodal problems with global structure, the self-adaptive version of DTS-CMA-ES outperforms other algorithms, showing a significant improvement in convergence speed, being particularly effective in multimodal problems, where different algorithms tend to get trapped in local optima. DTS-CMA-ES compares favorably with other algorithms based on CMA-ES and surrogate models, such as Imm-CMA and ACM-ES, as well as Bayesian optimizers

Table 10 Summary of restart strategies for CMA-ES

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
IPOP-CMA-ES [121]	Restart CMA-ES; double population size (λ) at each restart	Restart strategy (population size schedule)	Stronger global exploration on multimodal problems; simple mechanism	Can spend many evaluations on repeated restarts; needs sufficient budget
LR-CMA-ES (local restart baseline) [121]	Restart with constant small λ (local-focused multistart)	Restart strategy (multistart policy)	Low overhead; fast refinement when a good basin is found	Weaker global search; higher risk of repeatedly restarting in similar basins
BIPOP-CMA-ES [124]	Interlace IPOP restarts with small-population restarts using varied initial step-size (σ_0)	Restart strategy (λ and σ_0 schedules)	Robust on multimodal tasks; balances global and local restart regimes	More moving parts than IPOP; behavior depends on budget allocation choices
Auto-tuned iCMA-ES (iCMA-ES + irace) [122]	Parameter tuning (via irace) of iCMA-ES (CMA-ES with increasing population size)	Hyperparameter setting (restart + internal constants)	Improves default iCMA-ES performance on benchmark suite; systematic tuning protocol	Tuning cost and possible overfitting to chosen tuning set; tuned settings may not transfer
ACiD (Adaptive Coordinate Descent) [125]	Use CMA-ES Adaptive Encoding to learn a coordinate transform, then apply coordinate-wise dichotomy search	Covariance learning as adaptive encoding (coordinate transform)	Low-cost local optimization; can pair AE with other optimizers; preserves some invariance via learned encoding	Less effective on strongly non-separable/rotated landscapes than full CMA-ES; mainly local-search oriented
NIPOP-aCMA-ES [126]	IPOP-style restarts while decreasing initial σ_0 across restarts (plus active covariance update)	Restart strategy (λ , σ_0) + covariance update (active)	More diverse restart settings; improved robustness on multimodal problems	More hyperparameters/heuristics (σ_0 schedule); potential sensitivity to restart design
NBI-POP-aCMA-ES [126]	BIPOP-like scheme with adaptive budget allocation among restart settings (plus active covariance update)	Restart strategy (budget allocation) + covariance update (active)	Adaptive allocation reduces manual tuning; strong anytime behavior on multimodal tasks	More complex controller; performance can depend on budget allocation dynamics
IPOP-CMA-ES [127]	Baseline restart strategy: increase λ (doubling) after each restart	Restart strategy (population size schedule)	Strong baseline on multimodal benchmarks; simple and well-studied	May be outperformed by bi-population/ f_0 -varying schemes on harder multimodal sets
BIPOP-CMA-ES [127]	Interlaced restarts combining increasing λ with small-population restarts and varying σ_0	Restart strategy (λ and σ_0 schedules)	Better balance of exploration/exploitation than pure IPOP in many settings	Controller complexity; can be sensitive to overall budget and restart management
IPOP-aCMA-ES [127]	IPOP restarts + weighted active covariance matrix update	Covariance adaptation (active update) + restart strategy	Improves performance on ill-conditioned problems; competitive overall	Potentially less robust under noise; extra implementation detail vs standard update
BIPOP-aCMA-ES [127]	BIPOP restarts + weighted active covariance matrix update	Covariance adaptation (active update) + restart strategy	Strong on multimodal + ill-conditioned cases; good anytime performance	More hyperparameters/logic than IPOP; may require careful budget setting
NIPOP-aCMA-ES [127]	Alternative restart: increase λ and decrease σ_0 across restarts + active update	Restart strategy (λ , σ_0) + covariance update (active)	Broader restart coverage; competitive on multimodal sets	Heuristic schedules; may require more evaluations to realize benefits
NBI-POP-aCMA-ES [127]	Alternative restart: adaptive budget allocation among σ_0/λ regimes + active update	Restart strategy (budget allocation) + covariance update (active)	Competitive; ranked highly by solved target pairs in reported study	Budget allocation search can be inefficient if poorly matched to problem class
RB-IPOP-CMA-ES [123]	Use population midpoint as an optimum proxy; use midpoint fitness to trigger additional restarts	Selection/statistics for restart control (midpoint trigger)	Adds inexpensive information signal for restart decision; targets bound-constrained single-objective cases	Midpoint proxy may be misleading on complex landscapes; benefits may be problem-specific

such as SMAC and local search methods such as BOBYQA and fmincon.

Hansen presents in [133] his research on the use of surrogate models to improve the performance of the CMA-ES optimization algorithm in continuous search spaces, exploring the construction of a global surrogate model of linear, diagonal quadratic, or full quadratic complexity, depending

on the amount of data available. If the correlation between the actual function value and the value predicted by the model is high, the surrogate model is used; otherwise, more data are added to the model. The model evaluates candidate solutions instead of the fitness function to reduce the computational cost. This proposed approach, named lq-CMA-ES, improves CMA-ES performance in dimensions greater

Table 11 Summary of warm starting strategies for CMA-ES

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
CMA-ES-CWS [128]	Context-aware warm start: train multi-output Gaussian process regression on past (context, best-solution) pairs; initialize $m^{(0)}$ from GP posterior mean and $\sigma^{(0)}$ from posterior trace; keep $C^{(0)}=I$	Initialization/warm starting (m_0, σ_0) using contextual surrogate	Uses context information explicitly; reduces early adaptation cost; strong empirical gains on transformed benchmarks and robot-control tasks	Requires prior solved contexts (pre-optimizations); GP training/choice of kernels may be costly or sensitive; does not initialize covariance C from GP (may limit benefit when landscape structure varies)
WS-CMA-ES [129]	Warm start from source-task observations: build a γ -promising Gaussian mixture around top- γ samples; initialize CMA-ES mean/covariance by minimizing KL divergence to that mixture.	Initialization/warm starting (m_0, Σ_0) via promising-distribution approximation	Shortens adaptation phase under small evaluation budgets; leverages transferable HPO evidence; low extra runtime since prior used only at start	Can hurt when source-target similarity is low; depends on γ and α (mixture width) and on quality of source observations
WS-sep-CMA-ES [129]	Same warm-start principle as WS-CMA-ES but restrict Σ warm starting to diagonal (separable model) for faster/stabler early adaptation	Initialization + restricted covariance model (diagonal)	Lower parameter count and faster adaptation in high dimension; useful when budget is very limited	Loses rotational invariance; weaker on non-separable/rotated landscapes compared to full-covariance warm start

than or equal to 5, especially for evaluation budgets up to 10 times the problem dimension. This is attributed to the use of linear or diagonal quadratic models before having enough data for a full quadratic model, making the proposal a modification able to be competitive compared to other approaches based on surrogate models, such as Imm-CMA-ES and DTS-CMA-ES, even outperforming these methods in some cases, especially in higher dimensions.

Within the surrogate models, Zengcong Li has made several contributions that have allowed further development of the CMA-ES algorithm from different perspectives. The first of these algorithms developed was the Competitive Variable-Fidelity Surrogate-assisted CMA-ES (CVFS-CMA-ES) [134]. This algorithm introduces a variable fidelity model to improve the computational efficiency of CMA-ES. Its strategy uses information from low-fidelity (LF) and high-fidelity (HF) models to balance accuracy and computational cost. Besides improving computational efficiency by reducing costly evaluations, this algorithm incorporates a chaotic operator to enhance diversity and avoid local minima. Thanks to the effectiveness of surrogate models, Li expanded the idea of CVFS-CMA-ES by focusing on multiobjective optimization and improving the selection of promising points through clustering and knowledge extraction, resulting in Knowledge-Extraction-based Variable-Fidelity-assisted CMA-ES (KE-VFS-CMA-ES) [135]. This approach addresses problems such as multiobjective optimization by using K-means clustering to extract knowledge from low-fidelity models and implement a model management strategy based on hypervolume enhancement. Despite their advantages, these algorithms had problems with computational cost and loss of accuracy if the models did not correctly represent the problem. These questions led to the development of the Elite-driven surrogate-assisted CMA-ES (ES-CMA-ES) algorithm [136], which obtains better computational efficiency by reducing

costly evaluations using an efficient pre-selection strategy and incorporating a chaotic operator to improve diversity and avoid local minima. This algorithm improved CMA-ES efficiency with the Improved Lower Confidence Bound (ILCB) method and selection and diversification strategies; however, it does not use variable fidelity like the other two proposals, as each algorithm addresses CMA-ES problems differently.

Table 12 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection in surrogate models for CMA-ES.

4.9 Noisy Problems Optimization

In the context of an optimization function, noise is a random disturbance or variability that affects the evaluation of the objective function. This noise can arise for various reasons, such as measurement errors, uncertainty in the data, or the stochastic nature of the problem being modeled [137]. Noise introduces uncertainty in the evaluation of the objective function, which hinders the task of optimization algorithms, causing issues such as difficulty to converge, complicating the balance between exploration and exploitation, and complicating the calibration of parameters, among others [138].

To address the noise problem, several techniques have been developed and embedded into the CMA-ES algorithm to form a robust algorithm capable of dealing with such challenges. One of these proposals is the Uncertainty Handling CMA-ES (UH-CMA-ES) algorithm, which incorporates an uncertainty handling mechanism that dynamically adjusts the number of episodes used to evaluate a policy [139]. This mechanism balances learning speed and accuracy, increasing the number of evaluations only when necessary to ensure a reliable ranking of policies. The UH-CMA-ES dynamically adjusts the number of evaluations, using fewer episodes in the early stages of learning and increasing it as

Table 12 Summary of surrogate models for CMA-ES

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
KE-VFS-CMA-ES [135]	Fitness evaluation (surrogate-assisted)+covariance adaptation in a multi-objective MO-CMA-ES framework	Knowledge-extraction-based variable-fidelity surrogate: obtain LF non-dominated set, map to HF samples via K-means+space-filling; model management with MHVI-based pre-screening and oversampling	Cuts expensive HF evaluations; exploits LF+HF information; accelerates convergence in expensive multi-objective cases	Requires reliable LF/HF models and surrogate management; extra hyperparameters; overhead grows with model/training complexity
S-CMA-ES [132]	Fitness evaluation/selection (surrogate-integrated), while CMA-ES sampling & update are retained	Uses GP surrogate (with uncertainty) to rank/select candidates to reduce true evaluations (simple surrogate-assisted loop)	Improves sample-efficiency on tight budgets; leverages GP uncertainty during selection	GP training overhead; sensitivity to surrogate quality; scalability issues in higher dimensions
SurrogateCMA-ES [132]	Fitness evaluation/selection (surrogate-guided) with CMA-ES update unchanged	Surrogate-based candidate assessment combined with some true evaluations each iteration (surrogate used to guide which points get evaluated)	Reduces expensive evaluations while keeping CMA-ES update structure	Requires model management; may degrade if surrogate ranking is poor or noisy
DTS-CMA-ES (Doubly Trained Surrogate CMA-ES) [132]	Fitness evaluation pipeline (surrogate+objective) that feeds standard CMA-ES updates	Combines cheap surrogate predictions with objective evaluations every iteration; emphasizes uncertainty-aware selection for (surrogate/true) evaluation	Often faster convergence for small evaluation budgets; robust mix of surrogate and true feedback	Additional overhead and implementation complexity; GP scalability limits
AdaptiveDTS-CMA-ES [132]	Surrogate-driven evolution control (how many/which points are truly evaluated)	Adapts the degree of surrogate assistance based on online model reliability/control signals	Balances risk of surrogate bias vs. evaluation savings automatically	More moving parts (reliability thresholds, schedules); can be brittle if reliability metric is miscalibrated
lq-CMA-ES (model-assisted CMA-ES with linear/diagonal-quadratic/full quadratic surrogate) [133]	Fitness evaluation (global surrogate) with acceptance gated by rank correlation; CMA-ES update retained (incl. active CMA)	Builds a global low-order surrogate; uses Kendall- $\bar{I}_r$, rank correlation to decide surrogate usage; injects surrogate optimum into next generation	Very simple surrogate (low overhead); explicit safety check (rank correlation) reduces harmful surrogate use	Low-order models may miss strong nonlinearities; benefits depend on correlation test and data quality
ES-CMA-ES [136]	Surrogate-assisted sampling/evaluation and elite selection feeding standard covariance/step-size adaptation	ILCB acquisition uses step-size to adapt uncertainty term; model management via efficient pre-screening+competitive chaotic operator (local search/diversity)	Fewer true evaluations; adaptive exploration-exploitation control; better global search robustness on expensive problems	Extra hyperparameters; reliance on surrogate accuracy; chaotic operator may be problem-sensitive
s*ACM-ES [130]	Surrogate assistance schedule (lifelength) inside ACM-ES/CMA-ES while preserving rank-based invariance	Online self-adaptation of the surrogate model lifelength based on dimension and population size	Automatically controls how long the surrogate is trusted; improved efficiency on expensive objectives	Needs stable surrogate learning; still risks misguidance when surrogate becomes inaccurate
saACM-ES (intensive exploitation variant of s*ACM-ES) [131]	Surrogate exploitation intensity (e.g., stronger pre-selection/internal surrogate evaluations) with CMA-ES update preserved	Increases exploitation of the surrogate model (stronger pre-selection/higher surrogate usage) to boost speed-up vs. s*ACM-ES	Higher speed-up (notably on unimodal functions); retains CMA-ES mechanics while exploiting surrogate more aggressively	Higher risk of surrogate-induced bias; may require tuning of exploitation intensity; robustness depends on surrogate quality
IPOP-saACM-ES (restart variant used for benchmarking) [131]	Restart mechanism (IPOP) combined with surrogate-assisted CMA-ES exploitation schedule	Adds IPOP increasing-population restarts to saACM-ES to improve robustness across problem classes	Better global reliability across multimodal cases; inherits restart benefits with surrogate speed-up	More control parameters; restarts can waste budget if surrogate is weak or objectives are highly noisy

higher accuracy is required. The UH-CMA-ES significantly outperforms other methods, achieving a higher success rate on tests with noisy observations.

However, an improved version of the UH-CMA-ES was developed to amplify the search characteristics in noisy

environments, called Uncertainty Handling CMA-ES, for finding Robust Optima (RO-UH-CMA-ES) [140]. The RO-UH-CMA-ES approach focuses on combining uncertainty handling techniques with robustness approximation schemes based on the Multi Evaluation Model method

(MEM), allowing adjustment of evaluation accuracy throughout the optimization process and reducing noise as the algorithm approaches the optimum. The RO-UH-CMA-ES is compared with other evaluation schemes, such as the Single Evaluation Model (SEM) or the standard CMA-ES without uncertainty handling, where experiments are performed on six benchmark problems, showing that the RO-UH-CMA-ES outperforms the other methods, especially on unimodal problems with significant discontinuities near the optimum.

To improve the selection of individuals in noisy environments, where the evaluation of the objective function is subject to uncertainty, an approach named KL-KG (named after the union between the Knowledge Gradient and the Kullback-Leibler divergence) proposes a sequential sampling scheme for noisy optimization using the CMA-ES algorithm as a basis [141]. In this approach, an adapted version of the Knowledge Gradient (KG) method is proposed to minimize the Kullback-Leibler divergence between the distribution of selected individuals and the distribution that would be obtained if the best individuals were chosen in the absence of noise, allowing a more informed selection of individuals, improving the quality of the distribution used to generate the next generation of offspring. KL-KG is especially effective in high-noise environments, where the uncertainty in evaluating the objective function is higher because as the noise level increases, KL-KG maintains superior performance compared to the other methods.

The population size is a critical parameter in CMA-ES since a larger size can improve performance in multimodal and/or noisy functions. However, a smaller size is preferable to speed up optimization. Because of this, the Population Size Adaptation CMA-ES (PSA-CMA-ES) algorithm was developed, which proposes a population size adaptation mechanism based on the estimated accuracy of the update of the normal distribution parameters [142]. It is evaluated on noiseless and noisy benchmark functions and compared with existing strategies. PSA-CMA-ES outperforms CMAES-APOP and pcCMSA-ES because while these algorithms have difficulties handling functions with noise and multimodality simultaneously, PSA-CMA-ES can effectively address both challenges.

The Learning Rate Adaptation CMA-ES (LRA-CMA-ES) [143] algorithm was designed to determine whether the CMA-ES with a predetermined population size can solve multimodal and noisy problems without hyperparameter tuning. The LRA-CMA-ES implements a mechanism that adapts the learning rate (η) to maintain a constant signal-to-noise ratio (SNR), allowing the CMA-ES to solve multimodal and noisy problems without manually adjusting η . In problems with additive noise, the LRA-CMA-ES continued to improve the quality of solutions even in the presence of

intense noise. In contrast, the CMA-ES with fixed η stopped improving before reaching the global optimum.

Although the proposals seen above are methods for dealing with noise in a generic landscape, there are various types of noise in a search space. The variant of CMA-ES called Reevaluation Adaptation CMA-ES (RA-CMA-ES) focuses on improving CMA-ES performance in noisy environments, particularly under multiplicative noise [144]. The RA-CMA-ES variant dynamically adapts the number of solution reevaluations based on the estimated correlation of two update directions. The method combines learning rate adaptation and reevaluation to improve the accuracy of correlation estimation. Experiments were performed on test functions with multiplicative and additive noise, comparing the performance of RA-CMA-ES with other CMA-ES variants, such as UH-CMA-ES, PSA-CMA-ES, and LRA-CMA-ES, where RA-CMA-ES proved to be superior in most test functions with multiplicative noise, especially in cases of intense noise.

Table 13 shows the comparative analysis between all CMA-ES variants proposed in the articles in this subsection on optimization of problems with noise.

4.10 Safe Optimization

Safe optimization is a special type of constrained optimization that seeks to minimize an objective function by avoiding evaluating “unsafe” solutions, i.e., solutions that violate certain safety constraints [145]. This problem is common in medical and control engineering applications, where evaluating unsafe solutions can have serious consequences.

The Safe CMA-ES algorithm [146] was developed to achieve both safety and efficiency in safe optimization using a combination of Gaussian process regression (GPR) to estimate the Lipschitz constants of the safety functions and a projection mechanism to ensure that the generated solutions are within a safe region. In addition, the algorithm adjusts the distribution of initial parameters using safe solutions (called safe seeds). Safe CMA-ES managed to significantly reduce the number of unsafe evaluations compared to other methods, such as the traditional CMA-ES and the violation avoidance method, as well as effective on multimodal and noisy problems, maintaining a low number of unsafe evaluations while finding optimal solutions.

4.11 Summary of Studies

This subsection compiles all studies related to modifications and variants of the CMA-ES algorithm, which can be viewed in condensed and categorized form in Table 14.

Table 13 Summary of CMA-ES modifications for noisy problems

Algorithm	Main CMA-ES Component	Key Modification	Strengths	Weaknesses
LRA-CMA-ES (CMA-ES with Learning Rate Adaptation) [143]	Learning rates (distribution parameter updates)	Adapt CMA-ES learning rate online to maintain an approximately constant signal-to-noise ratio (SNR), targeting multimodal/noisy cases while keeping default λ .	Reduces manual tuning; improves robustness in multimodal/noisy settings without increasing population size.	Adds adaptation logic and SNR estimation; benefits may be problem-dependent and can add overhead/instability if SNR is misestimated.
RA-CMA-ES (Reevaluation Adaptation CMA-ES) [144]	Fitness evaluation under noise (reevaluation)	Compute two update directions using half of the evaluation budget and adapt the number of reevaluations using the estimated correlation between those directions; designed for multiplicative noise.	Targets multiplicative noise explicitly; adapts reevaluation effort; can improve performance under multiplicative noise while staying competitive under additive noise.	Requires extra evaluations for reevaluation; correlation estimation can be noisy; specialization to noise types may limit gains on noiseless tasks.
Sequential-sampling CMA-ES (KG/KL-KG based allocation) [141]	Selection under noise (adaptive sampling allocation)	Sequentially allocate samples to candidates to maximally inform $(\mu/\mu, \lambda)$ selection; use Knowledge Gradient with KL-divergence objective to reduce selection errors due to noise.	More efficient use of evaluation budget under noise; reduces misranking and improves selection quality.	Additional computation/modeling for allocation criterion; may depend on noise assumptions and increases implementation complexity.
UH-CMA-ES (Uncertainty Handling CMA-ES) [139]	Noise/uncertainty handling in evaluation & selection	Augment CMA-ES with adaptive uncertainty handling by allocating additional evaluations (episodes) when the ranking is uncertain, suitable for stochastic RL returns.	Improves robustness to stochastic RL evaluations; focuses reevaluations where selection uncertainty is high.	Can be evaluation-expensive (more episodes); requires choices for uncertainty/racing criteria and may slow down when noise is high.
Robust UH-CMA-ES (UH-CMA-ES modified for robust fitness estimation) [140]	Noise handling adapted to robust fitness (effective fitness)	Recast robust optimization as optimization with uncertain/noisy objective (robust fitness via stochastic approximation) and modify UH-CMA-ES to handle approximations of effective fitness.	Leverages proven noisy-optimization machinery for robustness; directly targets robust optima via uncertainty handling.	Requires repeated sampling to estimate robust fitness; added evaluation cost and sensitivity to how robustness/noise is modeled.
PSA-CMA-ES [142]	Population size (λ)	Adapt population size online based on an estimated accuracy of the update of the normal distribution parameters (aim: keep λ small but increase when needed, e.g., multimodal/noisy).	Reduces manual λ tuning; can balance speed vs. robustness by expanding λ only when necessary.	May introduce oscillatory/slow dynamics in λ ; extra bookkeeping for accuracy estimation; can increase evaluations when λ grows.

4.12 Recent Advances

Since its introduction in the 1990s, CMA-ES has evolved from a general-purpose continuous black-box optimizer into a modular paradigm in which extensions can be interpreted as targeted responses to regime-induced bottlenecks. Recent contributions from 2024 to 2026 reinforce this view and fit naturally within the taxonomy adopted in this survey: some advances refine internal mechanisms while preserving the core CMA-ES loop, whereas others introduce regime-driven extensions addressing multimodality, noise, expensive evaluations, and large-scale scalability. In contrast to earlier “single-knob” variants, several recent proposals increasingly adopt a systems-oriented design that combines complementary mechanisms—diversity preservation, stabilization of update signals, ranking reliability, or automated structural choices—into coordinated pipelines [147].

Within the problem-driven extensions for multimodal-ity, Kbp-LaF-CMAES frames multimodal optimization as a coordination problem between basin discovery and basin exploitation through CMA-ES instances. Its novelty is the joint use of a leader–follower coevolution structure to

maintain diversity and mitigate comparison bias, together with a knowledge-based perturbation restart mechanism whose perturbation magnitude is informed by an estimated basin size [148]. This makes the restart component more structured than purely random perturbations and explicitly targets stagnation and insufficient cross-basin exploration that often limit restart-only strategies. At the same time, the approach illustrates a familiar trade-off in multimodal orchestration: the “knowledge acquisition” phase for estimating basin size can be evaluation-intensive and may become prohibitive under tight budgets or expensive objectives, and the effectiveness of perturbation remains sensitive to whether it is large enough to leave a basin without degrading refinement efficiency.

In the problem-driven family for noisy/stochastic objectives, recent work further emphasizes a theme already highlighted in this survey: because CMA-ES is driven by ranks, robustness under noise often depends on protecting ranking reliability rather than simply increasing resampling. AS-CMA proposes an adaptive sampling supplement that allocates measurement time per candidate based on predicted sorting difficulty, explicitly leveraging the speed–accuracy

Table 14 A summary of CMA-ES modifications

Area of improvement	Studies
Reducing Computational Complexity	Igel et al. [29], Knight and Lunacek [31], Ros and Hansen [32], Suttorp and Hansen [30], Akimoto and Hansen [33, 34], Beyer and Sendhoff [35], Mao et al. [37], Ajani et al. [38].
Large-Scale Optimization	Shir and Bäck [40], Shir et al. [42], Liu and Tang [43], Loshchilov [44, 46], Tong and Yuan [51], Jin et al. [47], Li et al. [48], Yamaguchi et al. [50], Shimizu and Toyoda [49].
Reducing Function Accesses	Au and Leung [53], Kruisselbrink et al. [54], Huang et al. [55], Bouzarkouna et al. [56], Pitra et al. [57].
Constrained Optimization	Kramer et al. [60], Arnold and Hansen [61], de Melo and Iacca [62, 63], Kumar et al. [64], Atamna et al. [65], Dufossé and Hansen [66], Biedrzycki [67].
Multi-Objective Optimization	Igel et al. [70, 71], Voß et al. [72, 73], Krimpmann et al. [74], Rodrigues et al. [75], Limmer and Fey [76], Touré et al. [77], Chen et al. [78], Krause et al. [79], Gharafi et al. [80].
Mixed-Variable Optimization	Hansen [84], Brys et al. [85], Karmelita and Pawlak [86], Hamano et al. [87], Uchida et al. [88].
Parallel Optimization	Müller et al. [91], Hakkarinen et al. [92].
Noisy Problems Optimization	Heidrich-Meisner and Igel [139], Kruisselbrink et al. [140], Groves and Branke [141], Nishida and Akimoto [142], Nomura et al. [143], Uchida et al. [144].
Safe Optimization	Uchida et al. [146].
Step-Size Adaptation for performance improvement	Akimoto et al. [117], Shala et al. [118], Warchulski and Arabas [119], Beyer and Sendhoff [120].
Restart Strategies for performance improvement	Auger and Hansen [121], Liao et al. [122], Biedrzycki [123], Hansen [124], Loshchilov et al. [125, 126], Loshchilov [127].
Warm Starting for performance improvement	Sekino et al. [128], Nomura et al. [129].
Surrogate Models for performance improvement	Loshchilov et al. [130, 131], Hansen [133], Li et al. [134–136], Bajer et al. [132]
Other mechanisms for performance improvement	Auger et al. [93], Jastrebski and Arnold [94], Arnold and Hansen [95], Shir et al. [96], Preuss [97], Xu et al. [98], Kundu et al. [99], Caraffini et al. [100], Loshchilov et al. [101], Krause and Glasmachers [103], Andersson et al. [104], Zhao and Li [105], Teytaud [106], Nishida and Akimoto [107], Drugan [108], Abdolmaleki et al. [109], Richter et al. [110], Benhamou et al. [111], Varelas et al. [112], Akimoto and Hansen [113], Hamalainen et al. [114], Fontaine et al. [115], Friess et al. [116].

trade-off mediated by measurement time while keeping the CMA-ES update structure intact [149]. This perspective is particularly well matched to experimental and robotic contexts where evaluations have non-negligible overhead and where switching costs can be substantial. Its scope, however, is naturally tied to settings in which noise meaningfully depends on measurement time and where an appropriate noise–time model can be obtained; when noise is not time-dependent or is dominated by other sources, the benefit of time allocation may be limited.

Still within robustness-oriented refinements of internal dynamics, LRA-CMA-ES-LED targets the low effective dimensionality regime by estimating direction-wise effectiveness along covariance eigenvectors and reducing the influence of redundant directions on learning-rate adaptation [150]. This is aligned with your taxonomy’s view that step-size and learning-rate control constitute a major axis of robustness, especially where instability, premature contraction, or ineffective progress signals can dominate outcomes. The approach also highlights that improving robustness often requires stabilizing update signals rather than adding new operators, although the practical impact remains dependent on how accurately effectiveness can be estimated under the available budget and noise conditions.

For surrogate-assisted CMA-ES under expensive evaluations, Rank-based Linear-Quadratic Surrogate Assisted

CMA-ES addresses a long-standing tension between surrogate efficiency and invariance: CMA-ES is invariant to strictly increasing transformations because it relies on ranks, but value-based surrogate regression can break this property. By building the surrogate on rank-transformed data and exploring model selection via cross-validation, the method makes invariance preservation an explicit design goal in surrogate construction [151]. This direction aligns tightly with your survey’s view of surrogate assistance as model-management rather than simple evaluation saving, where preserving ranking reliability is often the decisive factor. At the same time, the work illustrates a nuanced trade-off: strict invariance can come with performance costs relative to stronger but less invariant surrogates, and robust improvements may require blending rank- and value-based information, which in turn partially relaxes invariance and increases design complexity.

Within computational scaling strategies for large-scale optimization, LCC-CMAES builds on cooperative coevolution but shifts attention to an increasingly relevant bottleneck: decomposition strategy selection is often rule-based and expert-dependent and may not generalize. By framing strategy scheduling as a learning problem and using reinforcement learning to select among decomposition strategies online, the method explicitly targets scalability without requiring hand-crafted decomposition rules [152]. This

represents a concrete step toward automated structural decision-making in large-scale CMA-ES pipelines and resonates with the broader trend toward “adaptive orchestration” rather than fixed strategy design. However, as with many learning-based hybrids, the added benefit depends on generalization of the learned scheduler across problem classes and budgets, and introduces additional configuration choices (state representation, reward, training protocol) that may increase implementation and tuning burden compared to purely hand-designed CC variants.

Finally, in application-facing hybridizations for complex engineering design, A2-CMA-ES exemplifies a pipeline-style hybrid in which novelty arises from coordinating multiple stabilizers and diversity mechanisms rather than from a single operator. It integrates adaptive archiving to incorporate historical information into the mean update, a step-size adjustment strategy combining nonlinear suppression and reward amplification, a stagnation-triggered multiplicative population growth mechanism as a soft restart, covariance condition-number monitoring with regularization for numerical stability, and a directional mining module based on a probabilistic local model for anisotropic refinement [147]. This design directly targets failure modes that appear repeatedly across the CMA-ES literature—premature convergence, diversity loss, and convergence-rate attenuation—and supports its choices with comparative evidence across benchmark and engineering contexts. The same integration, however, underscores the recurring survey-level trade-off between regime-specific gains and increased system complexity: multi-component architectures can introduce overhead and additional design choices, which may affect computational efficiency and reproducibility unless accompanied by careful ablation and sensitivity reporting.

5 CMA-ES Hybridizations

The hybridizations combine two or more techniques to take advantage of each other’s strengths and compensate for their weaknesses. Hybridizations are necessary because individual metaheuristic algorithms have inherent limitations, such as an imbalance between exploration and exploitation, null scalability, and a low convergence speed [153].

Hybridizations overcome these limitations by combining the strengths of different approaches, as they often strike a balance between exploration and exploitation, resulting in better quality solutions and faster convergence. In addition, the hybridization of algorithms allows a wide range of problems to be adapted, from continuous to combinatorial optimization, and these methodologies are generally more robust to variability in problem structure, such as the presence of multiple local optima or nonlinear constraints [154].

5.1 CMA-ES and PSO Hybridizations

Within the developed hybridizations of the CMA-ES, Particle Swarm Optimization (PSO) is one of the most used algorithms when combining different search strategies. The first version of hybridization between PSO and CMA-ES is the PS-CMA-ES algorithm [155], which has as its primary objective to improve performance in optimization problems with multi-funnel landscapes, i.e., problems with multiple “funnels” or local optimization regions, as well as to deal with noisy problems. PS-CMA-ES combines the best of both approaches, using multiple CMA-ES instances acting as particles in a swarm, communicating with each other to improve global exploration and search direction. When PS-CMA-ES is subjected to test functions, the algorithm performs well in higher dimensions. However, its performance in unimodal noiseless functions is comparable or slightly inferior to other CMA-ES variants, having a computational cost similar to that of standard CMA-ES.

The Epsilon Adaptive Mutation Particle Swarm Optimization - Covariance Matrix Adaptation (EAPSO-CMA) algorithm [156] was developed to solve constrained optimization problems (COP) in continuous spaces, which combines a well-known variant of PSO called EAPSO with the advantages offered by CMA-ES. This combination improves the algorithm’s capability to optimize the objective function after finding feasible regions. EAPSO is used to locate feasible regions, while CMA-ES is used to refine solutions. The combination of EAPSO and CMA-ES proved effective, successfully solving all CEC2010 benchmark problems regarding constraint satisfaction.

Finally, to improve the exploration and exploitation capability in global optimization problems, the Hybrid of Time-Windows PSO, PSO, and CMA-ES (HTPC) algorithm [157] was developed, which is a combination of the TW-PSO, PSO, and CMA-ES algorithms. The TW-PSO is an improvement of the traditional PSO that introduces a time-window mechanism to reduce the influence of historical positions that may lead individuals to fall into local optima. The combination of these three algorithms in HTPC leverages the strengths of each: the global exploration of TW-PSO, the local exploitation capability of CMA-ES, and the social interaction of PSO. On average, HTPC achieved 10-digit accuracy on approximately 300,000 objective function evaluations for various optimization problems; however, there are situations where HTPC failed to achieve the desired accuracy on problems, suggesting that there is room for improvement in the local exploitability of the algorithm.

5.2 CMA-ES and DE Hybridizations

The first hybridization between the Differential Evolution (DE) [158] algorithm and the CMA-ES is proposed in [159], where the Differential Covariance Matrix Adaptation Evolutionary Algorithm (DCMA-EA) is presented. The DCMA-EA integrates the DE difference vector-based mutation scheme with the covariance matrix adaptation of CMA-ES. The idea is to combine the strengths of both algorithms to achieve a better balance between exploration and exploitation in the search space. The DCMA-EA shows statistically superior performance on 17 of the 20 test functions performed in dimensions 30 and 50, compared to the original DE and CMA-ES, in addition to other variants such as the SaDE [160], JADE [161], IPOP-CMA-ES [121], and DMS-PSO [162] algorithms. In particular, the hybrid algorithm is especially effective on multi-modal, rotated, and noisy functions.

In addition to the DCMA-EA, more hybridizations of the CMA-ES with the original DE have been developed, such as the DECMSA algorithm [163], which may look similar to the DCMA-EA but have different approaches. While DCMA-EA is a variant of CMA-ES that incorporates DE operators, DECMSA is a variant of DE that integrates the covariance self-adaptation proposed by CMA-ES. The DECMSA methodology includes a local search strategy enhanced with the Quasi-Newton BFGS method when the population appears to converge, in addition to using a new DE/current-to-better/1 mutation scheme, which guides the search using a Gaussian distribution based on the covariance matrix. DECMSA was compared with popular DE and CMA-ES variants on 10-, 30-, and 50-dimensional continuous optimization problems, where the results show that DECMSA outperforms DE variants and is competitive with more advanced CMA-ES variants, such as IPOP-CMA-ES and BIPOP-CMA-ES.

The LSHADE-SPACMA algorithm [164] was developed as a combination of the LSHADE-SPA and a modified version of the CMA-ES to create an algorithm with high stability and robustness in multimodal and hybrid optimization problems. The LSHADE-SPA is a modified version of the LSHADE algorithm, which in turn is one of the most popular modifications of the DE. It stands out for its adaptability and efficiency by reducing the population size at crucial moments of the algorithm. A semi-parametric adaptation (SPA) approach is proposed for the F and Cr parameters of the DE algorithm. In the first half of the function evaluations, only Cr is fitted using the LSHADE approach, while F is randomly generated from a uniform distribution. In the second half, F is fitted using a Cauchy distribution, while Cr is kept fixed. The LSHADE-SPACMA was compared with the original LSHADE algorithm on 30 test functions of

the CEC 2017 benchmark. The results show that LSHADE-SPACMA outperforms LSHADE and LSHADE-SPA in most functions, especially in high-dimensionality problems.

The jSO_CMA-ES_LBFGS algorithm [165], which combines the CMA-ES, DE, and LBFGS (Limited-Memory Broyden-Fletcher-Goldfarb-Shanno) methods, was developed to find solutions to complex optimization problems where traditional algorithms are not practical. The jSO_CMA-ES_LBFGS algorithm uses jSO, a DE variant, as the global search operator. At the same time, CMA-ES is activated when the population stagnates, generating reliable initial solutions for the LBFGS local search operator. Statistical test results confirm that jSO_CMA-ES_LBFGS is significantly better than other compared algorithms such as Sandgre, GeneAS, and LSHADE-cnEpSin in most cases, especially on 30- and 50-dimensional problems.

5.3 CMA-ES and Cooperative-Coevolution Hybridizations

Cooperative Coevolution (CC) is a framework used in metaheuristic algorithms to address large-scale optimization problems [166]. This approach is based on the idea of “divide and conquer”, where a high-dimensional optimization problem is decomposed into smaller, tractable subproblems, which are then solved cooperatively.

The Two-Stage Cooperative Coevolution CMA-ES (TSCC-CMAES) [167] is a cooperative-coevolution-based algorithm that combines two stages of variable clustering: differential analysis (DA) and differential clustering (DG2), intending to improve efficiency in resource allocation and detection of interactions between variables. The algorithm combines the covariance matrix adaptive evolution strategy (CMA-ES) for solving non-separable subproblems and with sep-CMA-ES for separable subproblems. However, the cooperative-coevolution concept can be applied in different ways, as can be seen with the Cooperative Coevolutionary CMA-ES With Landscape-Aware Grouping (CC-CMAES-LAG) algorithm [168], which can decompose optimization problems into more tractable subproblems in the presence of multiplicative noise. CC-CMAES-LAG uses two key features to group variables: the convergence time of variables (t_c) and the cumulative path (p_c), which are less sensitive to noise and allow the identification of non-separable subcomponents more effectively.

Although these two hybrid approaches may be similar, they have several differences, such as the development objective, where the TSCC-CMAES aims to cover more optimization problems. At the same time, the CC-CMAES-LAG focuses on being efficient in noisy and high-dimensional environments.

5.4 Other CMA-ES Hybridizations

The BIPOP-aCMA-ES algorithm is a variant that, due to its performance, has been used for several hybridizations. In this case, we can find three different algorithms in the study carried out in [169], these being the ones named below:

- BIPOP-saACM-ES-k: A version of the CMA-ES with self-adaptive surrogate model assistance that increases the size of the population used to optimize the surrogate model.
- BIPOP-aCMA-STEP: A hybrid between the BIPOP-aCMA-ES and the STEP algorithm (A coordinate linear search method).
- HCMA: a hybrid between the BIPOP-saACM-ES-k, the STEP algorithm, and the NEWUOA algorithm (a derivative-free optimization algorithm).

Each of the aforementioned hybridizations was designed for a different purpose, with BIPOP-saACM-ES-k improving performance on ill-conditioned unimodal functions, BIPOP-aCMA-STEP improving performance on separable functions, and HCMA achieving superior performance on continuous optimization problems.

Other methodologies have been used to develop new search strategies, such as the Efficient Global Optimization (EGO) algorithm, which results in the EGO-CMA algorithm when combined with the CMA-ES [170]. The EGO-CMA uses the EGO algorithm to fill the design space efficiently, employing Gaussian processes. When it is detected that EGO no longer significantly improves the search space, the CMA-ES strategy is sequentially implemented to converge to the optimum. The EGO-CMA algorithm performs better than EGO and CMA-ES in most tested cases, especially on test functions from 2 to 10 dimensions.

In [171], a hybridization between a variant of the Multi-Objective Evolutionary Algorithm based on Decomposition (MOEA/D) and the CMA-ES is presented to obtain a fusion capable of solving complex multi-objective problems. From this fusion, a version of MOEA/D-CMA and an improved variant MOEA/D-CMA+I are proposed, where an injection mechanism of solutions from neighboring subproblems is introduced to update the covariance matrices of CMA-ES. By performing an analysis where the performance of MOEA/D-CMA+I is compared against MOEA/D, MOEA/D-DE, and MOEA/D-CMA in multi-objective optimization benchmarks, it is possible to visualize how MOEA/D-CMA+I shows significant improvements in most cases.

The HMO-CMA-ES algorithm [172] was designed as a hybridization based on its modifications of the CMA-ES, combining several techniques in an improved version.

HMO-CMA-ES combines different components of the following algorithms: BOBYQA for a fast start on the Pareto front approximation, ss-MO-CMA-ES to obtain increasing population size, restart-CMA-ES to obtain restarts on different scaled objectives, and IPOP-MO-CMA-ES with restarts. Combining different optimization components allows HMO-CMA-ES to be robust against various problems, making it a versatile algorithm for multi-objective optimization, possessing high scalability and performing well in different search space dimensions.

The Cell-like Evolutionary Tissue Membrane System (CETMS) algorithm [173] is a methodology that combines the CMA-ES algorithm with Tissue Membrane Systems (TMS) to solve real-parameter numerical optimization problems. The CETMS is based on the structure of tissue membrane systems, where each cell contains a CMA-ES algorithm that evolves candidate solutions. In addition, this proposal introduces a cell-to-cell communication mechanism that allows the sharing information about the best solutions found. The algorithm performs well in multimodal problems, although it does not outperform other algorithms, such as L-SHADE. However, CETMS proves scalable while maintaining good performance in high dimensions, although the computational cost increases with the number of dimensions.

Univariate sampling effectively solves non-separable multimodal problems because this technique efficiently reduces the search space and focuses on promising areas. The Hybrid Sampling-Evolution Strategy (HS-ES) algorithm [174] combines the advantages of univariate sampling with the outstanding unimodal function performance of CMA-ES. HS-ES uses a cascade model that combines univariate sampling and CMA-ES in different stages. First, univariate sampling is used to find an improved initial solution, then CMA-ES is applied to refine the solution, and finally, univariate sampling is reapplied with some fixed variables. HS-ES was evaluated on 29 test functions from the CEC-2018 set, which include unimodal, multimodal, hybrid, and compositional problems, where the results show that HS-ES is effective in solving problems of different dimensions.

The Grey Wolf Optimizer (GWO) algorithm has been one of the state-of-the-art algorithms with the greatest impact within metaheuristics, so its combination with the CMA-ES aims to overcome its limitations, such as tending to fall into local optima when facing complex optimization problems [175]. This hybridization, called CMAGWO, uses the global search capability of the GWO in the first stage to explore the solution space widely and locate the region where the optimal solution is found. In the second stage of CMAGWO, three instances of CMA-ES, starting from the leading wolf positions (α , β , δ) obtained in the first stage, are used to perform a refined local search in parallel. Each

Table 15 A summary of CMA-ES hybridizations

Hybridization	Studies
PSO and CMA-ES (PS-CMA)	Muller et al. [155]
EAPSO and CMA-ES (EAPSO-CMA)	Reza et al. [156]
TW-PSO, PSO and CMA-ES (HTPC)	Xu et al. [157]
DE and CMA-ES (DCMA-EA and DECMSA)	Ghosh et al. [159], He and Zhou [163]
LSHADE-SPA and CMA-ES (LSHADE-SPACMA)	Mohamed et al. [164]
jSO, LBFSGS and CMA-ES (jSO_CMA-ES_LBFSGS)	Zhao et al. [165]
TSCC and CMA-ES (TSCC-CMAES)	Irawan et al. [167]
CC and CMA-ES (CC-CMAES-LAG)	Wu et al. [168]
STEP and BIPOP-aCMA-ES (BIPOP-aCMA-STEP)	Loshchilov et al. [169]
STEP, NEWUOA, and BIPOP-aACM-ES-k	Loshchilov et al. [169]
EGO and CMA-ES (EGO-CMA)	Mohammadi et al. [170]
MOEA/D and CMA-ES (MOEA/D-CMA)	Zapotecas-Martínez et al. [171]
BOBYQA, ss-MO-CMA-ES, restart-CMA-ES and IPOP-MO-CMA-ES (HMO-CMA-ES)	Loshchilov and Glasmachers [172]
TMS and CMA-ES (CETMS)	Liu and Fan [173]
US and CMA-ES (HS-ES)	Zhang and Shi [174]
GWO and CMA-ES (CMAGWO)	Zhao et al. [175]
BBO and CMA-ES (HBBO-CMA)	Zhao et al. [176]
BSO and CMA-ES (BSO-CMA-ES)	Qu et al. [177]
PBIL and CMA-ES (CMA-PBIL)	Konieczka et al. [178]
SFS and CMA-ES (CMASFS)	Paçacı et al. [179]

CMA-ES instance has a different step size to avoid overlaps in the search. CMAGWO shows faster and more accurate convergence than other algorithms such as GWO, CMA-ES, MVO, WOA, PSOGWO, and DEGWO. Besides, the algorithm acquires robustness, avoiding falling into local optima and maintaining high diversity in the first stages of the search.

The HBBO-CMA [176] is a hybrid biogeography-based optimization (BBO) algorithm with an improved mutation operator and CMA-ES to solve non-separable problems. The HBBO-CMA incorporates a rotation-invariant migration operator to reduce the dependence of the BBO on the coordinate system and control population diversity. The HBBO-CMA algorithm was evaluated on the CEC-2017 test suite, demonstrating its superiority over other BBO variants and comparative algorithms such as CMA-ES, BSA, IWO, and BBO, obtaining better results in terms of accuracy and convergence speed on 10- and 30-dimensional problems.

In [177], a combination of the CMA-ES with the Brain Storm Optimizer (BSO) is presented, resulting in the BSO-CMA-ES. This approach makes multiple CMA-ES train collaboratively, increasing population diversity in multi-modal optimization problems and showing superior results to various modifications of CMA-ES.

The CMA-PBIL [178] is a hybrid algorithm between the Population-Based Incremental Learning (PBIL) algorithm and the CMA-ES, introducing a covariance matrix to describe the probability distribution of the population, which allows taking into account the correlations between variables. In problems with high correlation between items, CMA-PBIL converges faster than PBIL, reducing the

number of evaluations of the objective function needed to reach the optimal solution. However, on problems with low to moderate correlation, CMA-PBIL shows no significant degradation in performance compared to PBIL, indicating that the modification does not harm the algorithm in cases where correlations are not critical.

The CMASFS [179] hybrid methodology combines the Stochastic Fractal Search Algorithm (SFS) and the CMA-ES. The CMASFS is based on sharing information between the SFS and the CMA-ES, which allows for a better exploration of the search space and higher accuracy in the convergence to optimal solutions. The Gaussian Walk equations in the FSS diffusion process were modified to create a more dynamic and efficient algorithm. In addition, a probabilistic mechanism is introduced to decide when to run each of the algorithms (SFS or CMA-ES) depending on the quality of the generated solutions and the diversity of the population. CMASFS obtains the best mean error values on most CEC2017 test functions, especially hybrid and composition functions, indicating that the algorithm can handle complex and high-dimensional optimization problems.

5.5 Summary of CMA-ES Hybridizations

This subsection compiles all studies related to CMA-ES algorithm hybridizations, which can be viewed in condensed and categorized form in Table 15.

6 Conclusions and Future Work

This survey provides a structured synthesis of CMA-ES variants and hybridizations through a hierarchical taxonomy that links algorithmic modifications to the regimes and bottlenecks they address. Beyond summarizing the literature, we distill consensus observations and cross-family trade-offs that support more principled selection and development of CMA-ES extensions. In particular, the reviewed evidence indicates that mechanism-preserving refinements are often the most transferable across benchmarks. In contrast, regime-driven hybrids can yield stronger gains in specific settings at the cost of increased configuration burden and reduced reproducibility. Accordingly, many of the most consistently competitive proposals preserve the core CMA-ES paradigm—rank-based selection coupled with distribution adaptation—while introducing targeted changes that reduce information loss or improve sample efficiency. In this sense, high-performing variants tend to be “mechanism-preserving,” retaining invariance and stability while focusing modifications on a single regime-induced bottleneck (expensive evaluations, high dimensionality, stochasticity, or feasibility constraints). This observation is supported by the comparative synthesis summarized in Tables 8 and 12, where many competitive approaches frequently highlighted preserve the rank-based selection and distribution-adaptation core while introducing targeted changes—often through performance-oriented refinements (Table 8) or surrogate-assisted mechanisms that maintain the CMA-ES update structure (Table 12).

A central trend is the primacy of covariance adaptation, together with increasing specialization in how covariance (or its proxies) is represented and updated. Active and multiplicative updates, diagonal acceleration, sparse precision learning, and large-population covariance schemes aim to accelerate metric learning and improve conditioning. However, they expose a recurring trade-off between expressivity and scalability. Full covariance remains advantageous when rotational structure is essential, whereas diagonal, low-rank, or sparse structures become compelling when effective interactions are limited or when memory and time constraints dominate. Closely related, step-size and learning-rate control remains a dominant axis for robustness in multimodal and noisy regimes, where premature contraction and unstable progress can be decisive failure modes.

Surrogate-assisted CMA-ES has evolved from saving evaluations into model-management strategies that control when and how surrogate information influences ranking and updates. Local versus global surrogates, reliability gating via rank-correlation or uncertainty criteria, adaptive surrogate lifetimes, aggressive pre-screening, and variable-fidelity pipelines—also in multiobjective settings—reflect

a shift toward treating surrogates as first-class components that interact with selection pressure, restart logic, and evaluation allocation. A similar perspective appears in noise handling and reinforcement-learning deployments, where uncertainty handling, sequential sampling, and adaptive reevaluation policies emphasize that the “fitness” driving CMA-ES is often a stochastic estimate whose accuracy can be controlled—at non-negligible cost—to reduce misranking and stabilize distribution updates.

A further observation is that modern CMA-ES research is increasingly application-facing, where hybridizations are dictated by problem structure rather than algorithmic elegance alone. Constraint handling via augmented Lagrangian formulations, penalties, surrogate constraints, and repair or projection operators suggests that feasibility mechanisms often determine overall success in real-world settings. In multimodal optimization, niching, clustering, and restart portfolios (including increasing-population and bi-population strategies) improve robustness by orchestrating multiple search regimes rather than relying on a single adaptation rule. For discrete and mixed-integer problems, margin-based mechanisms and integer-aware operators help preserve distribution learning while avoiding premature fixation and loss of exploration in non-continuous coordinates.

From a comparative standpoint, the literature suggests regime-driven preferences: (1) in high-dimensional settings, diagonal, low-rank, sparse, or limited-memory covariance models are often favored over full covariance once costs dominate; (2) in expensive optimization, surrogate assistance and warm-start strategies are most beneficial when model-management explicitly protects ranking reliability; (3) under noise, uncertainty handling and reevaluation policies tend to outperform naive resampling under fixed budgets by reducing misranking; (4) in constrained problems, feasibility-driven sampling or repair and AL or penalty frameworks are frequently effective but sensitive to feasibility metrics and reporting practices; (5) for multimodal objectives, restart portfolios and niching mechanisms commonly improve robustness; and (6) for multiobjective and mixed-variable cases, specialized MO-CMA-ES and encoding-based methods are typically necessary, albeit with stronger scenario dependence and tuning overhead.

A critical perspective emerging from the survey is that outcomes within some of the most active families remain heterogeneous and occasionally difficult to reproduce, particularly for surrogate-assisted and constraint-driven hybrids. In surrogate-assisted CMA-ES, gains are often contingent on model-management decisions that preserve ranking reliability; when surrogate influence becomes overly aggressive, misranking and negative transfer can occur, improving early progress but deteriorating later-stage convergence or cross-benchmark robustness. Likewise, in

constraint-handling and other application-facing hybrids, reported success is frequently shaped by feasibility mechanisms and reporting choices, including how constraint-evaluation budgets are accounted for, which violation metrics are used, and whether feasibility rates are tracked consistently. Across both families, multiple interacting components are commonly introduced without ablation or sensitivity analyses, making it difficult to attribute improvements to a single mechanism and underscoring the need for more standardized, reproducible evaluation protocols.

From a real-world engineering perspective, these trends translate into regime-driven choices that depend on application context. In simulation-based design problems common in aerospace, structures, and electromagnetics, surrogate-assisted CMA-ES and warm-starting are typically most beneficial when surrogate influence is gated by reliability criteria and periodically refreshed with true evaluations. In robotics and control, where objectives are often noisy and may involve stochastic rollouts, uncertainty handling and controlled reevaluation policies are often decisive for preventing misranking and stabilizing distribution updates. In constrained engineering design and parameter identification tasks, AL or penalty-based variants, combined with feasibility-aware sampling or repair, often drive practical success. In contrast, large-scale calibration problems commonly favor structured covariance representations once full covariance becomes prohibitive. In mixed-variable engineering design, margin-based encodings and integer-aware operators are typically required to preserve distribution learning without premature fixation. These mappings motivate the practitioner-oriented decision guide proposed in this survey and underscore the importance of reporting budgets, feasibility or violation metrics, restart policies, and surrogate or noise-handling rules for reproducible deployment.

While this survey covers a broad spectrum of CMA-ES variants and hybridizations, it is inevitably shaped by the scope and reporting practices of the available literature. First, the review is inherently biased toward continuous black-box optimization and benchmark-driven evidence, with domain-specific industrial studies and proprietary implementations likely underrepresented. Second, cross-paper comparability remains limited in constrained, noisy, and surrogate-assisted settings due to heterogeneous evaluation budgets, feasibility or violation metrics, and reporting choices and implementation details, which may affect the strength of aggregate conclusions. These limitations motivate the emphasis on standardized reporting and reproducible evaluation protocols discussed in the conclusions and future directions.

Future research is likely to be most impactful where it provides unifying principles and safer, more automated adaptation without sacrificing the algorithm's invariance-driven

strengths. To prioritize actionable directions, we highlight three high-impact avenues. First, scalable covariance structure learning is expected to remain central, since much of the surveyed progress in high-dimensional regimes already relies on diagonal, low-rank, sparse, or limited-memory proxies; the next step is to develop diagnostics that infer when each structure is appropriate and to enable online structure selection while preserving positive definiteness and, when possible, invariance properties. Second, principled reliability control under uncertainty appears increasingly unifying across families, because surrogate assistance, noise handling, and reinforcement-learning deployments repeatedly hinge on preventing misranking; future work should therefore formalize reliability measures that jointly govern surrogate gating, reevaluation, and budget allocation so that performance gains do not come at the cost of instability or negative transfer. Third, automated and reproducible CMA-ES design is likely to shape how new hybrids are adopted in practice, as the survey highlights growing configuration burden and module interactions; progress here will depend on modular configuration frameworks with conservative fallbacks, along with ablation-driven and sensitivity-aware evaluation protocols that make increasingly complex variants comparable and reproducible. Beyond these priorities, portfolio-level adaptation that dynamically selects among restart regimes, covariance models, and surrogate policies remains a natural extension toward CMA-ES systems that are both computationally efficient and broadly reliable across heterogeneous optimization landscapes.

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Declarations

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